KCSE 2026 PREDICTION TRIALS

MATHEMATICS

KCSE 10 TOP PREDICTIONS (MATHEMATICS PAPER 1&2)

A SERIES OF TOP PREDICTION QUESTIONS PREPARED BY A PANEL OF TOP KNEC EXAMINERS!

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KENYA EDUCATORS CONSULTANCY

KCSE TOP PREDICTION TRIAL 1 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1 (Alt A)

Time 2 ½ hours

Name ……………………………………………………………………… **Adm Number**……………………………..

Candidate's Signature ……………………………………………………….**Date** ……………………………………

INSTRUCTIONS TO CANDIDATES

1. Write your name, index number and class in the spaces provided above.
2. The paper contains two sections: **Section I** and **Section II**.
3. This paper contains **14 PRINTED** pages make sure all **PAGES ARE PRINTED** and **NON IS MISSING**
4. Answer **ALL** the questions in **Section I** and **ANY FIVE** questions from **Section II**.
5. All working and answers must be written on the question paper in the spaces provided below each question.
6. Marks may be awarded for correct working even if the answer is wrong.
7. Negligent and slovenly work will be penalized.
8. Non-programmable silent electronic calculators and mathematical tables are allowed for use.

For examiners use only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

SECTION I (50 MARKS)

Answer ALL the questions in this Section in the spaces provided

1. Evaluate without using calculator; (3 marks)

$$\frac{-2(3-5)+7\div -2+4}{5\times -3 \, of \, 2-6}$$

2. Find the acute angle formed between the lines y + 2x = 5 and 2y - 6x + 8 = 0. (3 marks)

3. A water tank 10m high stands on a tower. Mrs. Kamau observes the angles of elevation of the top and bottom of the tank to be 40° and 25° respectively. Calculate the height of the tower. (4 marks)

4. Evaluate without using mathematical tables or a calculator. (3 marks)

$$100\left(\frac{\sqrt{8.41}}{\sqrt[3]{195.112}}\right)$$

5. Given that the matrix $\begin{pmatrix} 3 & 5 \\ (x+2) & x \end{pmatrix}$ is singular, find the value of x. (2 marks)

6. The exchange rates for a certain year were as follows:

	Buying (Kshs)	Selling (Kshs)
Chinese Yuan	12.34	12.38
1 US Dollar	80.24	80.44

A Kenyan businessman had 100,000 dollars which he converted into Kenyan shillings. He spends 5 million Kenyan shillings to import goods from China. How much is his balance in Chinese Yuan. (3 marks)

7. Find the value of y which satisfies the equation.
$2^{2y} - 3(2^{y+1}) + 8 = 0$ (3 marks)

8. Solve $\frac{2(2 - x)}{5} < 3x + 2 \leq \frac{x}{2} + 9$, and state the integral values of x that satisfy the inequalities. (3 marks)

9. On one side of line AB below, use a ruler and a pair of compasses only to construct the locus of a point P such that $\angle APB = 67.5°$. (3 marks)

10. Using tables of cubes, square roots and reciprocals, evaluate; (4 marks)

$$\frac{5}{\sqrt{0.876}}-(23.59)^3$$

11. Given that $P_1(6, 4)$ is the image of $P(1, -2)$ under an enlargement with scale factor ½. Find the centre of enlargement. (3 marks)

12. Simplify completely; $\frac{3x^3-12x}{2x^2+10x+12}$ (3 marks)

13. The masses of two similar containers are 23.2g and 185.6g. If the surface area of the smaller container is $40cm^2$, find the surface area of the larger one. (3 marks)

14. Find the value of y in the equation

$$2 - \log(3x + 2) = \log 25 - \log(x - 1)$$ (3 marks)

15. Find the equation of the normal to the curve;

$$y = \frac{2x^3 + 4x^2 - 3x - 6}{x + 2}$$ at the point y = 5. (4 marks)

16. Calculate the surface area of a solid hemisphere of radius 7cm, giving your answer to 3 significant figures. (3 marks)

SECTION II (50 MARKS)
Answer only FIVE questions from this section in the spaces provided

17. On a certain day, a matatu left Nakuru for Nairobi at 8am travelling at 80km/hr while a personal car left Nairobi for Nakuru at 8.15am, travelling at 100km/hr. Given that the distance between Nairobi and Nakuru is 200km;

a) Calculate;

i) the time when the two vehicles met. (3 marks)

ii) the distance of Nakuru from their meeting point. (2 marks)

b) At the meeting point the matatu stopped for 30 minutes and continued to Nairobi, while the personal car broke down and the owner took a taxi back to Nairobi, 20 minutes after the matatu left. If the taxi moves at 150km/h, calculate;

i) the time the taxi caught up with the matatu. (2 marks)

ii) the difference of their arrival in Nairobi. (2 marks)

18. Four points ABCD are positioned in a garden such that point B is 200m on a bearing of 063° from point A and 320° and 300m from point C. Point D is due North of C and 250m from B.

a) Using a scale of 1cm to represent 50m, show the positions of the four points. (4 marks)

b) From your diagram in (a) above, find;

i) the distance of D from C.

ii) the true bearing of D from B.

iii) the distance of A from C.

iv) the compass bearing of A from D. (6 marks)

19. P, Q, R and S are vertices of a parallelogram. Find the co-ordinates of S, given that P(-1, 1), Q(2, 2 and R(3, 5) (3 marks)

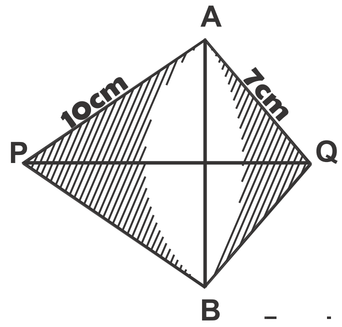
b) i) Given that **OT** = **i** + 3**j**, show that P, R and T are collinear. (3 marks)

ii) State the ratio OT:TR (1 mark)

c) Calculate the magnitude of PR to 2 significant figures. (2 marks)

d) Find vector QS in terms of **i** and **j**. (1 mark)

20. The figure below shows two intersecting circles with centres P and Q and radii 10cm and 7cm respectively.



Given that the distance between the centres is 15cm, calculate;
a) the length of the common chord AB. 4 marks)

b) the area of the shaded region. (6 marks)

21. A particle moves in a straight line with a velocity Vm/s given by

$$V = t^3 - 4t^2 + 5t$$

If at $t = 0$, $S = 4m$

i) Find the distance covered at $t = 2s$. (4 marks)

ii) Find the acceleration of the particle at $t = 3s$. (3 marks)

iii) Find the maximum velocity attained. (3 marks)

22. The table below represents marks scored by 50 students in a test.

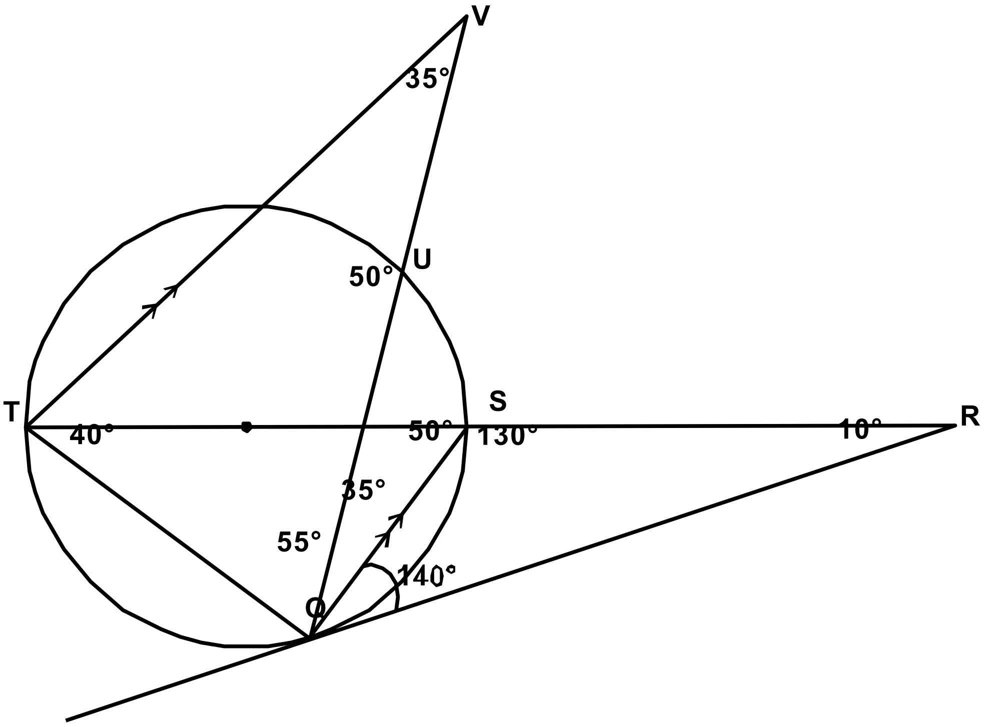
Marks	$0 \leq x < 10$	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$	$40 \leq x < 50$	$50 \leq x < 60$	$60 \leq x < 80$
No. of students	4	7	10	6	9	12	2

a) Calculate the mean mark. (3 marks)

b) Calculate the median (3 marks)

c) On the grid provided, draw a frequency polygon to represent the data. (4 marks)

23. In the figure below, PQR is a tangent to the circle at Q. TS is a diameter and TSR and QUV are straight lines. QS is parallel to TV. Angles SQR = 40° and TQR = 55°



a) Find the following angles, giving reasons in each case;

i) ∠QTS (2 marks)

ii) ∠QRS (2 marks)

iii) ∠QVT (2 marks)

iv) ∠QUT (2 marks)

b) Given that QR = 8cm and SR = 4cm, find the radius of the circle. (2 marks)

24. Triangle ABC has vertices A(1, 1), B(3, 1) and C(1, 3)

a) On the grid provided, plot triangle ABC. (1 mark)

b) Triangle ABC is transformed onto $A^1B^1C^1$ by a translation vector $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$. State the co-ordinates of A^1, B^1 and C^1 and plot it on the graph. (2 marks)

c) $A^1B^1C^1$ is reflected along the line x= 0 onto $A^{11}B^{11}C^{11}$. Obtain the co-ordinates of $A^{11}B^{11}C^{11}$ and plot them on the graph. (2 marks)

d) $A^{11}B^{11}C^{11}$ undergoes an enlargement scale factor -1 and centre origin onto $A^{111}B^{111}C^{111}$. Obtain the co-ordinates of the image $A^{111}B^{111}C^{111}$. (2 marks)

e) $A^{111}B^{111}C^{111}$ undergoes a rotation of 120° about (1, -2). Obtain the co-ordinates of the final image $A^{iv}B^{iv}C^{iv}$. (3 marks)

KCSE TOP PREDICTION TRIAL 1 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ……………………………………………………………… **Adm Number**……………………………..

Candidate's Signature ……………………………………………….**Date** ………………………………………

INSTRUCTION TO CANDIDATE'S:

(a) *Write your name, admission number and class in the spaces provided at the top of this page.*
(b) *Sign and write the date of examination in spaces provided above.*
(c) *This paper consists of **TWO** sections: **Section I** and **Section II**.*
(d) *Answer **ALL** the questions in **Section I** and any **five** questions from Section **II**.*
(e) ***Show all the steps in your calculation, giving your answer at each stage in the spaces provided below each question**.*
(f) *Marks may be given for correct working even if the answer is wrong.*
(g) ***Non-programmable** silent electronic calculators and **KNEC** Mathematical tables may be used, except where stated otherwise.*

FOR EXAMINER'S USE ONLY:

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

SECTION II

17	18	19	20	21	22	23	24	TOTAL

Grand Total	

SECTION A (50 MARKS)

Answer all questions in this section in the spaces provided

1. Use completing square method to solve $2x^2 - 4x - 5=0$ (3 marks)

2. Find the radius and the centre of a circle whose equation is $2x^2 + 2y^2 - 6x + 10y + 9 = 0$. (3 marks)

3. A bag contains 10balls of which 3 are red, 5 are white and 2 are green. Another bag contains 12balls of which 4 are red, 3 are white and 5 are green. A bag is chosen at random and then a ball chosen at random from the bag. Find the probability that the ball so chosen is red. (3 Marks)

4. (a) Expand and simplify the binominal expression $(1 + \frac{1}{2}x)^6$ (2 marks)

(b) Use the expansion up to the fourth term to evaluate $(1.05)^6$ to 2 decimal places. (2 Marks)

5. The latitude and longitude of two stations **P** and **Q** are (47°N, 25°W) and (47°N, 70°W) respectively. Calculate the distance in nautical miles between **P** and **Q** along the latitude 47°N. (3 Marks)

6. Solve the equation $4 \sin x = 5 - 4 \cos^2 x$ for $0° \leq x \leq 360°$ (4 Marks)

7. T is a transformation represented by the matrix $\begin{pmatrix} 5x & 2 \\ x & -3 \end{pmatrix}$. Under T, a square of area 10cm^2 is mapped onto a square 110cm^2. Find the values of *x*. (3 Marks)

8. The marks obtained by 10 students in a math test were:- 25, 24, 22, 23, 20**,** 26, 21, 23, 22 and 24. Calculate the Standard deviation. (3 Marks)

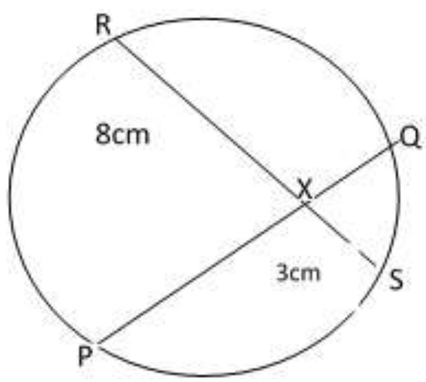
9. Brand **A** tea costing Kshs.80 per kg is mixed with Brand **B** tea costing Kshs.100 per kg such that the mixture is sold at Kshs.114 making a profit of 20%. Find the ratio of **A:B** (3 Marks)

10. **B** varies partly as the square of **M** and partly as the inverse of **N**. **B,M** and **N** are such that when **M**=2, **N**= ½ , **B**=96 while when **M**= 3 , **N**=2, **B** = 46. Write an expression for **B** in terms of **M** and **N.** (3 Marks)

11. The cash price of a T.V set is Ksh.26,000. Linda bought the set on hire purchase terms by paying a deposit of Ksh.6,000 and the balance by 24 equal monthly installments of Kshs.1,045.30. Find the compound rate of interest per year. (3 Marks)

12. Evaluate without using tables or calculators. (3 Marks)
    Log (3x + 8) - 3 log2 = log (x – 4)

13. In the circle below chords **PQ** and **RS** interest intersect at **X** .Given that **RX**=8cm, **XS**=3cm and **PQ** =10cm.Calculate **PX** (3Marks)



14. Given that the temperature of fluid A = 2.7°C, B = 9.8°C and C= 4.2^0, find the percentage error in calculation $\frac{A+B}{C}$ (3 marks)

15. Simplify the expression $\dfrac{3x^2 - 4xy + y^2}{9x^2 - y^2}$ (3 Marks)

16. Simplify giving your answer in rationalised form (3 marks)

$$\frac{Cos60^0}{Cos60^0 - Sin60^0}$$

SECTION 11 (50 MARKS)

Answer ONLY FIVE questions in this section in the spaces provided.

17. The table below shows income tax rates.

Monthly income in the Kenya shillings	**Tax rates percentage (%) in each shilling**
Up to 10164	10
From 10165 to 19740	15
From 19741 to 29320	20
From 29321 to 37040	25
From 37041 and above	30

In one year, Mr. Kamau's monthly earning was as follows:

Basic salary 20,000

House allowance 12,000

Medical allowance 2,808

Transport allowance 1,764

a) Calculate:

i. Mr. Kamau's taxable income. (2 marks)

ii. Tax charged on Mr. Kamau's earning (4 marks)

b) Mr. Kamau was entitled to the following tax reliefs:

i. Monthly personal relief at the rate of 15% of the premium paid. If Mr. Kamau paid a monthly premium of Kshs.2,500 and cooperative shares of Kshs.5,000 per month, calculate his net salary. (4 marks)

18. a) Triangle XYZ has the vertices X(2,-1) Y(4,-1) and Z(4,2) is mapped onto triangle $X^/Y^/Z^/$ by transformation matrix $T_1=\begin{pmatrix}1 & -3\\0 & 1\end{pmatrix}$. Determine the coordinates of $X^/Y^/Z^/$. (2 Marks)

b) Another triangle $X^{//}$(2,10) $Y^{//}$(2,14) and $Z^{//}$ (-4,-4) is the image of $X^/Y^/Z^/$ after transformation T_2. Determine the matrix of transformation T_2. (3 Marks)

c) Find the single matrix of transformation T^{-1} that maps $X^{//}Y^{//}Z^{//}$ onto XYZ. (3 Marks)

d) Given that the area of triangle XYZ is 15 cm^2, Find the area of the triangle $X^{//}Y^{//}Z^{//}$. (2 Marks)

19. The second, third and fourteenth terms of arithmetic progression are the three consecutive terms of a geometric progression. The 10^{th} term of the arithmetic progression is 18. Find:
    a) The first term and the common difference of the progression. (4 marks)

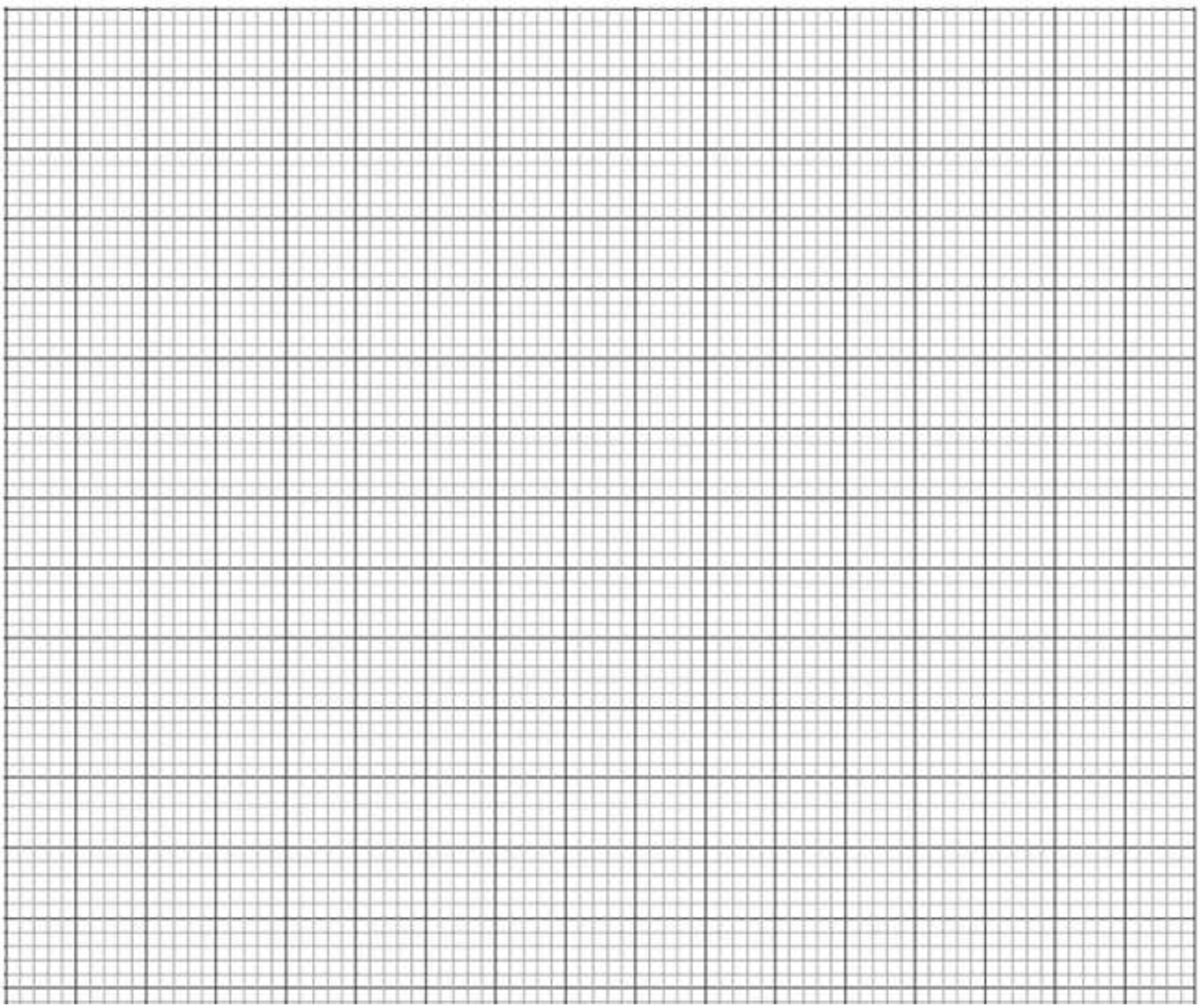
    b) The sum of the first 10 terms of the Arithmetic progression. (2 marks)

    c) The sixth term f the G.P. (4 marks)

20. a) Complete the table below for y = sin 2x and y = sin 2x + 30^{O} giving values to 2d.p. (2 marks)

x	0	15	30	45	60	75	90	105	120	135	150	165	180
sin 2x	0	0.5		1		0.5	0	-0.5				-0.5	
sin (2x + 30)		0.866	1		0.5	0			-1.0		-0.5		0.5

b) Draw the graph of y = sin 2x and y = sin (2x + 30^{O}). On the same axis. Take 1cm to represent 15^{O} on the horizontal axis and 4cm to represent one unit on the vertical axis. (5 marks)

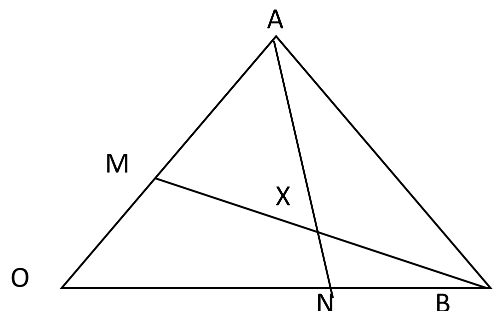


c) Use the graph to solve sin (2x + 30^{O}) - sin 2x = 0 (1 mark)

d) Determine the transformation which maps sin 2x onto sin $(2x + 30^{O})$ (1 mark)

e) State the period of y = sin)2x + 30^{O}) (1 mark)

21. In the figure below OAB is a triangle in which M divides OA in the ratio 2:3 and N divides OB in the ratio 4:1. AN and BM intersects at X



(a) Given that $\underset{\sim}{OA}$ = a and $\underset{\sim}{OB}$ = b, express in terms of $\underset{\sim}{a}$ and b

(i) $\underset{\sim}{AN}$ (1 Mark)

(ii) $\underset{\sim}{BM}$ (1 Mark)

(iii) $\underset{\sim}{AB}$ (1 Mark)

(b) If $\underset{\sim}{AX}$ = s$\underset{\sim}{AN}$ and $\underset{\sim}{BX}$ = t$\underset{\sim}{BM}$, where **s** and **t** are constants, write two expressions for OX in terms of **a**, **b**, **s** and **t**. Find the value of **s** and **t** hence write OX in term of **a** and **b.** (7 Marks)

22. The table below shows the masses of Form Four students in a school.

Mass (kg)	40-44	45 - 49	50-54	55- 59	60-64	65-69	70-74
No. of	3	30	29	33	13	1	1

Using an assumed mean of 57, calculate :

a) the mean (3 marks)

b) the standard deviation (4 marks)

c) The lower quartile. (3 marks)

23. Using a ruler and a pair of compass only.

a) Construct a triangle PQR such that PR=7.5 cm, PQ=5.0 cm and $< QPR = 60^0$. (3 Marks)

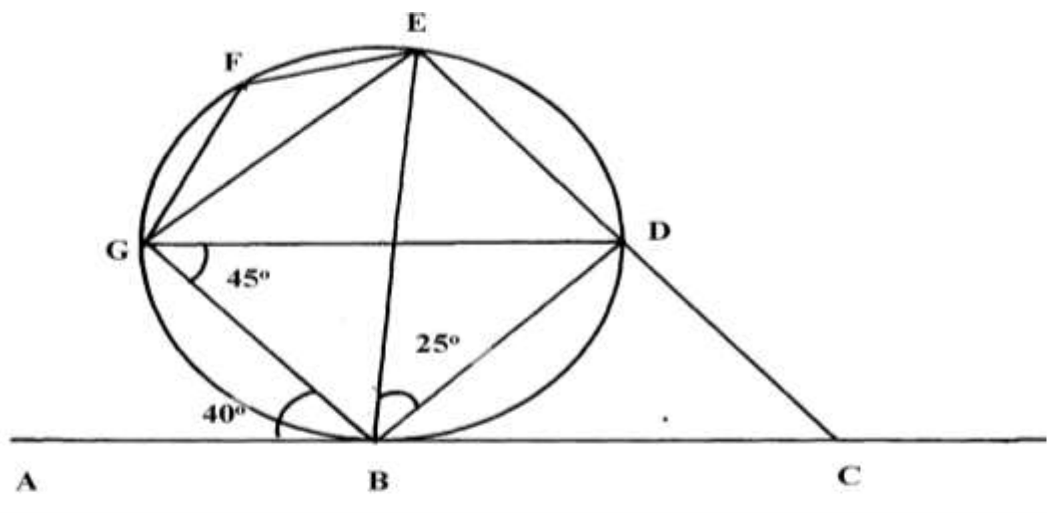
b) Construct the locus T of points which are equidistant from a point L and passes through the vertices P, Q and R. (2 Marks)

c) Locate the locus S on T such that it is equidistant from sides PQ and QR of the triangle. (2 Marks)

d) Locate the locus of points G enclosed by PQ and QS such that QG$< 3.5cm$. (2 Marks)

e) Measure SL. (1 Mark)

24. In the figure below ABC is a tangent to the circle at B. Given that <ABG=40º, <BGD=45º, and <DBE=25º as shown below.



Find the sizes of the following angles giving reasons in each case:

a) ∠BDG (2 Marks)

b)∠DGE (2 Marks)

c)∠EFG (2 Marks)

d)∠CBD (2 Marks)

e)∠BCD (2 Marks)

KCSE TOP PREDICTION TRIAL 2 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ……………………………………………………..……………… Adm Number…………………………..

Candidate's Signature ………………………..……………………….Date ……………………………………

Instructions to Candidates

(a) Write your name and Admission number in the spaces provided above.
*(b) This paper consists of two sections: **Section** I and **Section II.***
*(c) Answer **ALL** questions in **section I** and **ANY** five questions in **section II.***
(d) All answers and workings must be written on the question paper in the spaces provided below each question.

FOR EXAMINER'S USE ONLY

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

17	18	19	20	21	22	23	24	TOTAL

SECTION I (50 MARKS)

Answer **ALL** questions in this section

1. Without using a calculator, evaluate; (3 marks)

$$\frac{\frac{1}{2}\,of\ 3\frac{1}{2}+1\frac{1}{2}\left(2\frac{1}{2}-\frac{2}{3}\right)}{\frac{3}{4}\,of\ 2\frac{1}{2}\div\frac{1}{2}}$$

2. Solve for x in the equation. (3 marks)

$$2^{(2x-1)}\ \text{x}\ 16^{(2x-1)}=1$$

3. A metallic solid cone has a base radius of 6.4 cm and slant height 15.8 cm. If the density of the metal is 7.9 g/cm^3, calculate its mass in kg.(Take $\pi=\frac{22}{7}$) (3 marks)

4. Members of a group decided to raise K£100 towards a charity by contributing equal amount. Five of them were unable to contribute. The rest had, therefore, to pay K£1 more each to raise the same amount. How many members were in the group originally? (3 marks)

5. The LCM of three numbers is 1512 and their GCD IS 6. If two of the numbers are 54 and 72, find the least third possible number. (3 marks)

6. The exterior angle of a regular polygon is equal to one-third of interior angle. Calculate the number of sides of the polygon. (3 marks)

7. Find the reciprocal of 2.234.Hence use tables to evaluate$\left(\frac{5}{2.234}\right)^3$ to 4 significant figures

(3 marks)

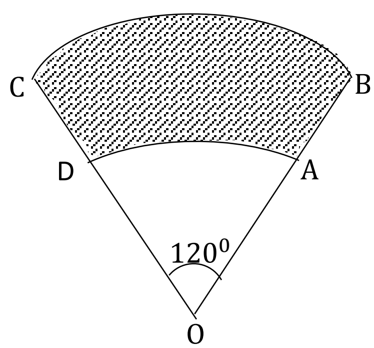
8. The image of A (-2,5) under a transformation T is A' (2,2) B' (9, -5) is the image of B under the same translation T determine the coordinates of B. (3 marks)

9. Six men take 28 days working for 10 hours a day to pack 4480 parcels. How many more men working 8 hours a day will be required to pack 2560 parcels in 4 days. (3 marks)

10. From a point A, the student observed the angle of elevation to the top of the building to be 30^0, after walking from point A to point B 20m towards the foot of the building in an horizontal ground, he observed the angle of depression from the top of the building to be 72^0, find the height of the building correct to 2 decimal places. (4 marks)

11. A windscreen wiper of a car sweeps through an angle of 120^0. The shaded region in the figure below represents the area swept clean by the blade of the wiper AB. If OA = 14cm and OB = 21cm. Taking л = $\frac{22}{7}$, find the area of the glass swept clean to 1d.p

(3 marks)



12. Use matrix method to solve the equation; $\begin{array}{r} 4x - 5y = 13 \\ -2y + 3x = 8 \end{array}$ (4 marks)

13. Simplify completely. (3 marks)

$$\frac{3x^2 - 48}{3x^2 - 24x + 48}$$

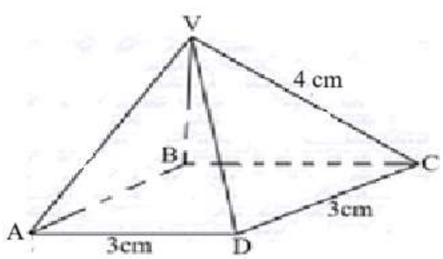
14. Use a ruler and a pair of compasses only to construct a triangle ABC in which AB = 4.3cm, BC = 3.9cm and angle ABC is 135^0. Measure AC. (3 marks)

15. A Kenyan bank bought and sold foreign currencies on two different days as shown below.

		Buying (In Ksh)	Selling (In Ksh)
12/28/2024	1 Sterling pound (£)	130.10	130.54
12/29/2024	South African Rand	9.52	9.58

A businessman arrived in Kenya on 12/28/2024 with £ 50000 Sterling pound. He changed the whole amount to Kenya shillings. While in Kenya, he spent 80% of the money and changed the balance to South African Rand before leaving for South Africa on 12/29/2024. Determine, to the nearest Rand, the amount he obtained. (3 marks)

16. The figure below shows a solid square based pyramid, draw a net of the solid (3 marks)



SECTION II: (50 MARKS)

*Answer any **FIVE** questions from this section*

17. A solid Q is made up of a hemi spherical bottom, cylindrical middle and conical top. The height of the cylinder is 14m and the height of the cone is 5m, the common radius of both figures is 2.1m. (Take $\pi=\frac{22}{7}$)

(a) Find the slant height of the cone correct to 2 decimal places. (2 marks)

(b) Find the total surface area of the solid Q correct to 4 significant figures (4 marks)

(c) Find the volume of solid Q to the nearest metres. (4 marks)

18. (a) The equation of a straight line L_1 is given by $6x + 3y - 12 = 0$. Another line L_2 is perpendicular to L_1 at $(m, 13)$. Line L_2 also passes through point $K(3,5)$.

(i) Find the value of m. (3 marks)

(ii) Find the equation of L_2 in the form of $ax + by + c = 0$, where a, b and c are integers. (2 marks)

(b) The equation of the base AB of an isosceles triangle ABC is $y = -2$ and the equation of side BC is $y + 2x = 6$. If points A and C are $(-6, -2)$ and $(-1, 6)$ respectively, Find;

(i) The coordinates of B (2 marks)

(ii) The equation of side AC. (3 marks)

19. A curve is represented by the function y = $\frac{1}{3}x^3 + x^2 - 3x + 2$

(a) Find the gradient function of the curve (1 mark)

(b) Determine the turning points on the curve (4 marks)

(c) Determine the nature of the turning points of the curve (3 marks)

(d) Sketch the curve of y = $\frac{1}{3}x^3 + x^2 - 3x + 2$ (2 marks)

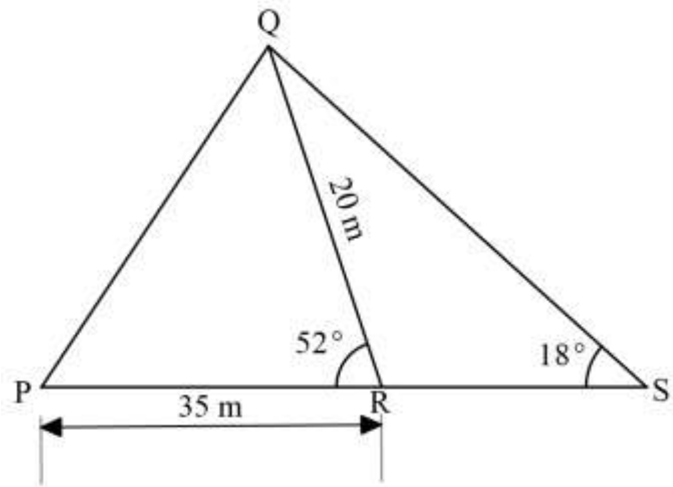
20. (a) A triangle has vertices A (1,2), B (7,2) and C (5,4). Draw triangle ABC on the Cartesian place (1 mark)

(b) A’ (2, -1), B’ (2, -7) and C’ (4, -5) is the image of ABC under a transformation T. Draw triangle A’B’C’ on the same grid and describe T fully. (3 marks).

(c) Draw triangle A” B”C” the image of triangle A’B’C; under a reflection in the line Y=X. State the coordinates of the vertices. (3 marks)

(d) Triangle A’”B’”C’” is the image of triangle A”B”C” under an enlargement centre (-5,0) and linear scale factor of -1, and hence draw triangle A’”B’”C’” and state its coordinates. (3 marks)

21. The figure below represents two neighboring plots with QR as their common boundary.



Find to 2 decimal places,

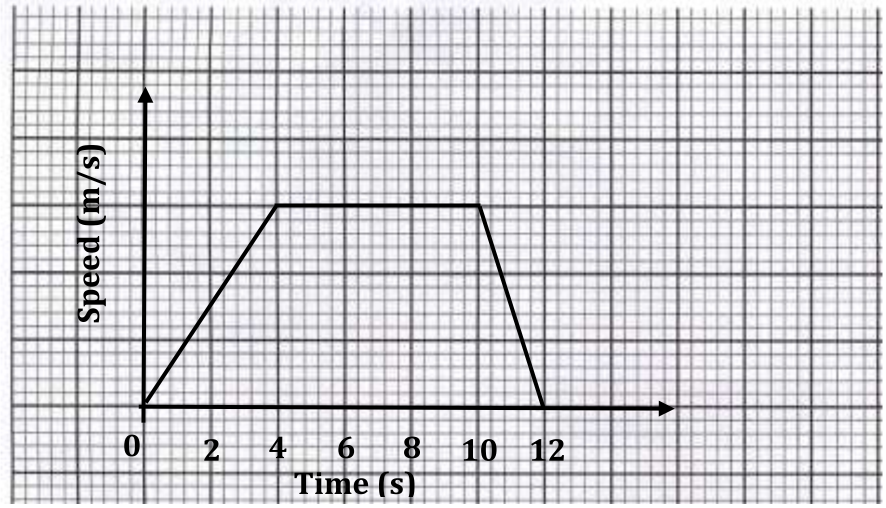
(a) The length of boundary PQ. (3 marks)

(b) The angle RQS. (1 mark)

(c) The length of boundary RS. (3 marks)

(d) Area of triangle QRS. (3 marks)

22. (a) The graph shows the speed of a car during an interval of 12 seconds.



The Distance travelled in the first 4 seconds is 40m.

Find:

(i) the maximum speed reached. (2 marks)

(ii) the average speed for the whole journey. (2 marks)

(b) Wafula left Webuye at 8.00 am towards Mumias at an average speed of 90km/hr. Nekesa also left Webuye at 8.21 am towards Mumias along the same road of an average speed of 97km/hr.

Determine:

(i) the time Nekesa caught up with Wafula. (3 marks)

(ii) the distance from Webuye when Nekesa caught up with Wafula. (3 marks)

23. The table below show marks obtained by 60 students in mathematics exam.

MARKS	30- 34	35-39	40 -44	45 -49	50 -54	55 -59	60-64
NO. OF STUDENTS	3	6	5	12	8	Y	7

(a) Find the value of Y in the above table. (2 marks)

(b) State the modal class. (1 mark)

(c) Calculate the mean mark of the students (3 marks)

(d) On the grid provided draw a histogram and a frequency polygon on the same axes to represent the above data. (4 marks)

24. (a) Complete the table below for the function $y = x^2 - 3x - 4$ (2 marks)

x	-1	0	1	2	3	4
y						

Use the trapezium rule with 5 strips to estimate the area bounded by the curve $y = x^2 - 3x - 4$, x − axis and the lines $x = -1$ and $x = 4$. (3 marks)

(b) Use integration method to calculate the exact area in (b) above (3 marks)

(c) Determine the percentage error in estimating the area by trapezium rule. (2 marks)

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KCSE TOP PREDICTION TRIAL 2 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ……………………………………………..…………………… Adm Number…………………………..

Candidate's Signature ………………………..…………………….Date ……………………………………

Instructions to Candidates

1. Write your **name, Admission number and class** in the spaces provided.
2. The paper contains **TWO sections**: **Section I and II**
3. Answer **ALL** questions in section I and **STRICTLY ANY FIVE** questions from section II.

For Examiner's use only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

17	18	19	20	21	22	23	24	TOTAL

GRAND TOTAL

SECTION I (50 MARKS)

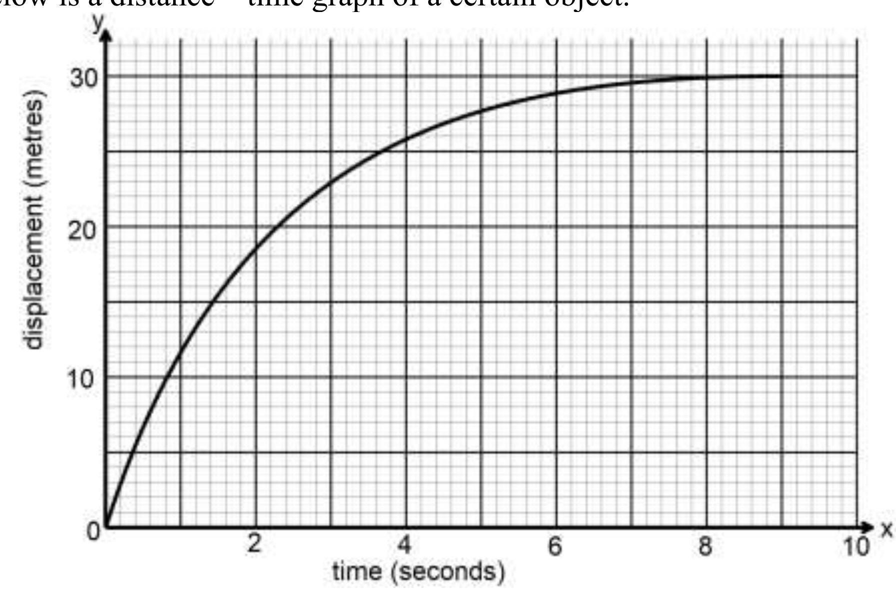
1.Given that $A = \pi(R - r)(R + r)$. Make R the subject of the formula. [3 marks]

2. In a transformation. An object with an area of 6 cm^2 is mapped onto an image whose area is 36 cm^2. Given that the matrix of the transformation is $\begin{pmatrix} 4 & x-1 \\ 2 & x \end{pmatrix}$, find the value of x. [3 marks]

3. A quantity P varies partly as n and partly as the square of n. When $P = -3, n = -1$ and when $P = 18, n = 2$. Find P when $n = 1$ [3 marks]

4. A businessman obtained a loan of sh.450,000 from a bank to buy a matatu valued at the same amount. The bank charges interest at 20% per annum compounded quarterly. Calculate the total amount of money the businessman paid to clear the loan in $1\frac{1}{2}$years to the nearest shillings. [3 marks]

5. The figure below is a distance – time graph of a certain object.



From the graph, determine the average velocity of the object between t = 1 and t = 5 seconds. [2 marks]

6. Given that $2\log m^2 + \log\sqrt{m} = a\log m$, find the value of a. [3 marks]

7. Given that $\frac{x}{23}\boldsymbol{i} + \frac{6}{23}\boldsymbol{j} - \frac{x-19}{23}\boldsymbol{k}$ is a unit vector, determine two possible values of x. [3 marks]

8. The equation of a circle is $2x^2 + 2y^2 + 12x - 20y - 4 = 0$. Determine the coordinates of the centre of the circle and state its radius. [3 marks]

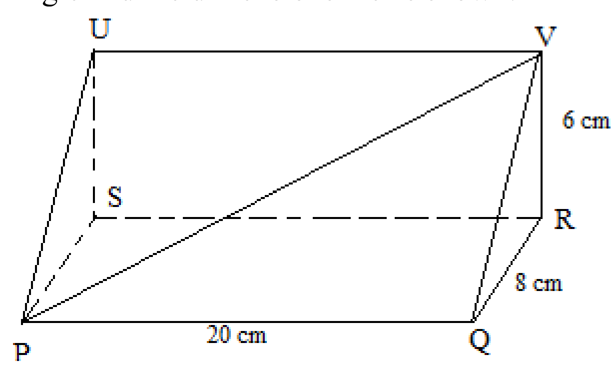
9. (i) Expand and simplify $(1-3x)^5$ up to the term in x^3 [1 mark]

(ii) Hence use your expansion to estimate $(0.97)^5$ correct to 4 d.p. [2 marks]

10. Without using a calculator or mathematical tables, evaluate; [3 marks]

$$\frac{\sin 45° + \cos 30°}{\tan 30°}$$

11. The diagram below represents a wedge whose cross – section is a right angle triangle. PQVU is a rectangle and the dimensions are as shown.

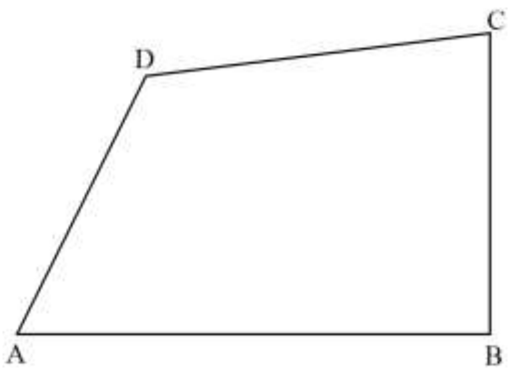


Calculate to 2 decimal places:-

a) The length of PV [2 marks]

b) The size of the angle between PV and plane PQRS. [2 marks]

12. The figure below shows a ranch ABCD drawn to a scale of 1: 10,000. A security light tower is to be installed in the ranch such that it meets the following conditions:



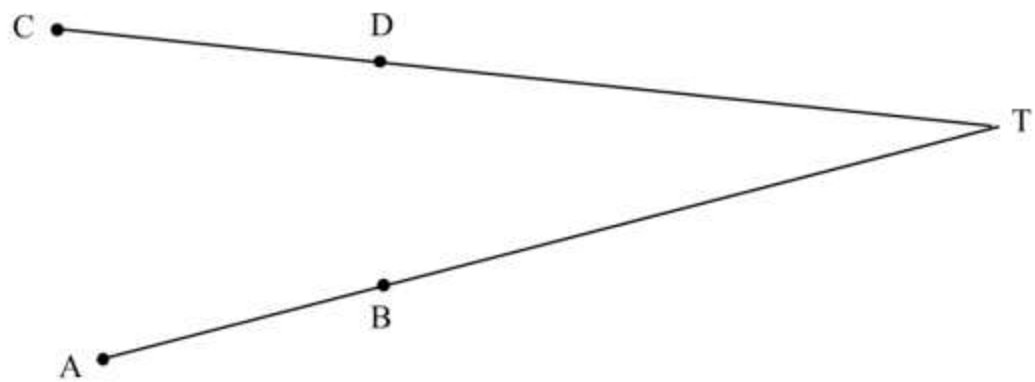
- It is nearer to B than it is to A
- It is at least 200 metres from the point C
- It is further from the line AD than it is to CD

Shade the possible region in the ranch where the tower would be installed to meet the above conditions. [4 marks]

13. The velocity of a particle after **t** seconds is given by $V = t^2 - 4t + 4$. Find displacement of the particles during the third second [3 marks]

14. Working together two taps A and B can fill a tank in 6 hours. By itself, tap A can fill the tank in 8 hrs. Tap A and B are opened at the same time and after running for 2 hours, an outlet tap which can drain the full tank by itself in 12 hours is opened. How much longer will it take the tank to be filled? [3 marks]

15. In the figure below, AB and CD are chords of a circle that intersect externally at point T. Using a ruler and a pair of compasses only, construct the circle hence construct a tangent from point T to the circle at E. Measure ET. [4 marks]



16. Determine the period and phase angle for the function [3 marks]

$y = 6\sin(\frac{x}{2} - 90^{\circ})$

SECTION II (50 MARKS)

17. In a certain year income tax for all the income earned was charged at the rate shown below.

Monthly taxable pay in KShs	**Rate of tax % in each KShs**
1 -9,680	10%
9,681-18,800	15%
18,801-27,920	20%
27,921-37,040	25%
Over 37,040	30%

Mr. mwangi earned a basic salary of KShs. 32,000 and a house allowance of KShs. 10,000, Medical allowance of KShs 4,000 and a non-taxable risk allowance of ksh 8,000 per month. He claimed a tax relief of KSh 1056 per month.

(a) Calculate his total taxable income per month. [2 marks]

(b) Calculate her PAYE In that month . [5 marks]

(c) Other than tax, the following deductions are made:

A service charge of Sh. 2028
Health insurance fund Sh. 3, 000
A co-operative loan of Sh. 12,000
Burial benevolent fund of KSh. 900

Calculate his net monthly income from his employment. [3 marks]

18. A bag contains 8 blue, 3 green and x red marbles. The marbles are similar except for the colour. Two marbles are picked from the bag, one after the other and without replacement. The probability that the two marbles picked are both blue is $\frac{7}{30}$

    (a) Find the value of x [4 marks]

    (b) Represent the information given using a tree diagram [2 marks]

    (c) Using the diagram in (b) above, find:

    (i) The Probability that the two marbles picked are of the same colour [2 marks]

    (ii) The probability atleast a red marble is picked [2 marks]

19. A geometric progression (G.P.) is such that the product of its first three terms is 8,000.

(a) Taking the first term as a and the common ratio as r , express r in terms of a . [2 marks]

(b) The sum of the first three terms in (a) above is 78.

(i) Determine the first term and the common ratio of two possible sequences. [4 marks]

(ii) Hence write the first 4 terms of the two sequences. [2 marks]

(c) Find the product of the 8th term of the two sequences. [2 marks]

20. Two towns on the earth's surface are such that $A(75°N, 100°E)$and $B(75°N, 80°W)$. A pilot can fly from A to B along the parallel of latitude, or along the great circle through the North

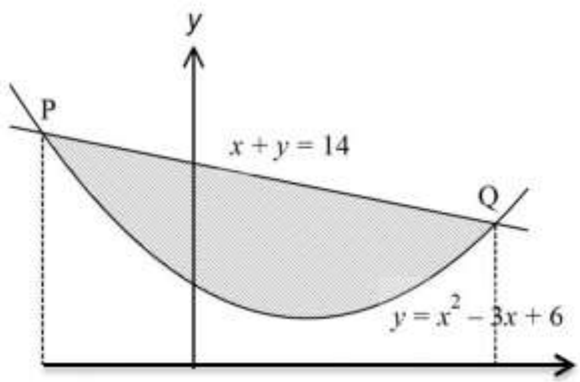
a) Determine the difference in distances of the two routes. Give the answer to the nearest kilometre. Use $\pi = \frac{22}{7}$ $and\ R = 6370km$ [5 marks]

b) At 3.20 p.m., the pilot had to fly from B due west at 300 knots for 2 hours to town C Determine:

(i) the longitude of town C to the nearest whole number [3 marks]

(ii) the local time at C when he landed. Give the answer in 24-hour clock system. [2 marks]

21. The diagram shows a sketch of a curve $y = x^2 - 3x + 6$ intersecting line $x + y = 14$ at points P and Q



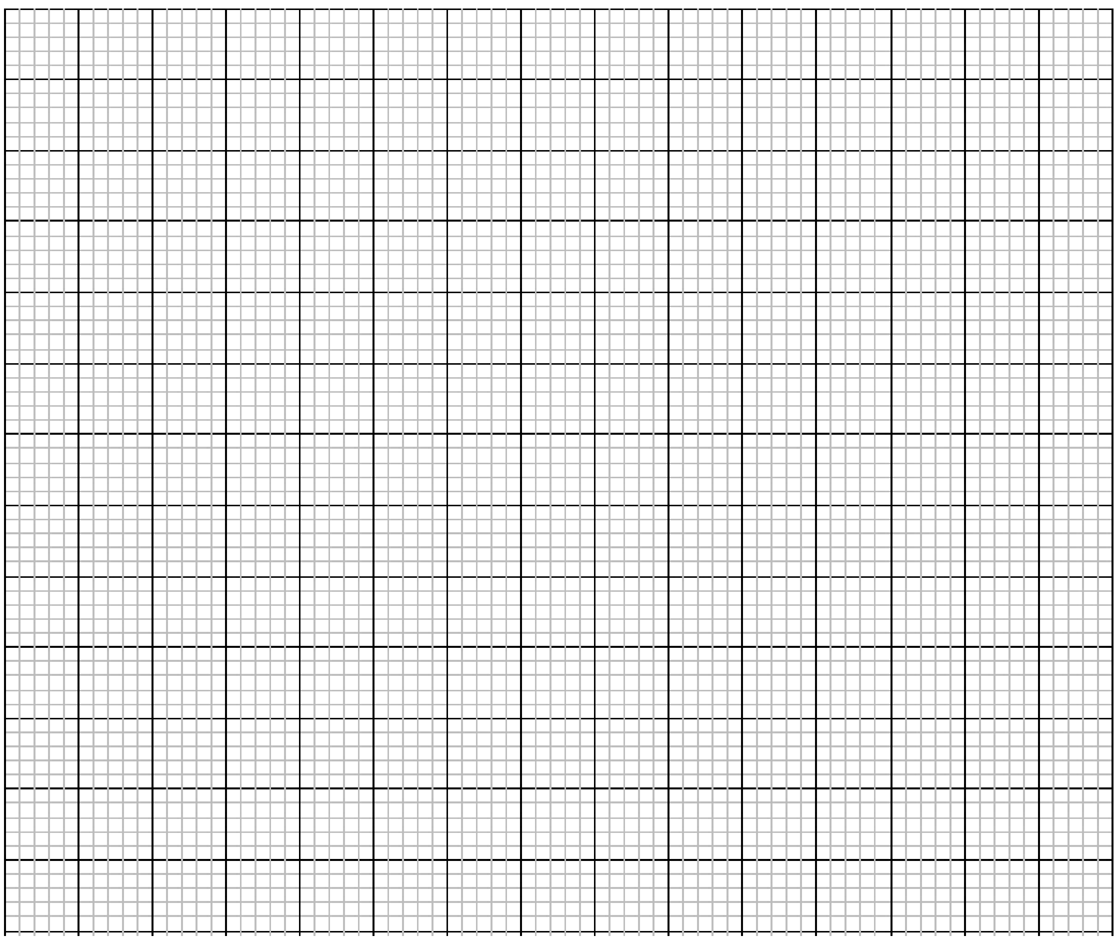
(a) Find the coordinates of points P and Q. [4 marks]

(b) Calculate the exact area of the shaded region. [6 marks]

22. A bus company runs a fleet of two types of buses operating between Nairobi and Nyeri. Type A bus has a capacity to take 70 passengers and $2000kg$ luggage. Type B can carry 50 passengers and $3000kg$ of luggage. On a certain day, at least 500 passengers with at most $35000kg$ of luggage have to be transported. The company could only use a maximum of 15 buses altogether.

a) If the company uses x buses of type A and y buses of type B write down all the inequalities satisfying the given conditions. [4 marks]

b) Represent the inequalities in part (a) above on the grid provided. [4 marks]



c) If the cost of running one bus of type A is Kshs. 7200 and that of running one bus of type B Kshs. 6000. Find the maximum cost of running the buses. [2 marks]

23. The heights of 100 maize plants were measured to the nearest centimeter and the results recorded in the table below

$Height\ (cm)$	$Frequency\ f$	x	d	d^2	fd	fd^2	cf
$25-29$	5	27	-3	9			5
$30-34$	12	32	-2	4			17
$35-39$	18	37	-1	1			35
$40-44$	30	42	0	0			65
$45-49$	17	47	1	1			82
$50-54$	11	52	2	4			93
$55-59$	7	57	3	9			100

(a) Complete the table [2 marks]

(b) Calculate to 2d.p

(i) The mean [2 marks]

(ii) The standard deviation [2 marks]

(c) Using the data above plot an ogive and use it to find the 30^{th} percentile [4 marks]

24. The vertices of a triangle PQR are $P(2,4)$, $Q(8,5)$ and $R(x, y)$. A transformation matrix **V** maps triangle PQR onto triangle $P'Q'R'$whose vertices are $P'(6,6), Q'(2,13)$ and $R'(10,14)$.

(a) Find the;

(i) Matrix **V**. [3 marks]

(ii) The coordinate of R [2 marks]

(b) Triangle $P''Q''R''$ the image of triangle $P'Q'R'$ under a stretch y-axis invariant and $P'(6,6)$ is mapped onto $P''(-18,6)$. Determine the ;

(i) The stretch matrix U [2 marks]

(ii) The single matrix which maps $P''Q''R''$ onto PQR [3 marks]

KCSE TOP PREDICTION TRIAL 3 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1 (Alt A)

Time 2 ½ hours

Name ………………………………………………......……………… Adm Number…………………………..

Candidate's Signature ……………………..………………………….Date ……………………………………

Instructions to candidates

a) *Write your name and Index number in the spaces provided above.*
b) *Write your class, date of examination and sign in the spaces provided above.*
c) *This paper consists of **two** sections; **Section I** and **Section II**.*
d) *Answer all the questions in **Section I** and only **five** questions from **Section II**.**Show all the steps in your calculations, giving your answers at each stage in the space***
e) ***s provided below each question.***
f) *Marks may be given for correct working even if the answer is wrong.*
g) ***Non – programmable** silent electronic calculators **and** KNEC Mathematical tables may be used, except where stated otherwise.*

For Examiner's Use Only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand Total

SECTION I (50 marks)

*Answer **all** the questions in this section*

1 Without using a calculator, evaluate $\left(2.4\times10^{-5}\right)\div\left(9.6\times10^{-3}\right)$ giving your answer in standard form.

(2 marks)

2. Mr. Owino spends $\frac{1}{4}$ of his salary on school fees. He spends $\frac{2}{3}$ of the remainder on food and a fifth of what is left on transport. He saves the balance. In certain month he saved Sh. 3400. What was his salary?

(3 marks)

3. Some information about six numbers is provided as below:

- Lowest number 37
- The range 24
- The mode 43
- The median 46
- One number is a multiple of 11

Find the mean of the six numbers. (3 marks)

4. A train passes through a station at a speed of 108 km/h. The length of the station is 120 m. The train takes 7 seconds to completely pass through the station. Calculate the length of the train. (3 marks)

5. A point Q lies on the line PR below such that PQ:QR = 4:1. PT and PQ are two sides of a quadrilateral.

(a) Using a ruler, a pair of compasses and a set square, locate the point Q. (1 mark)

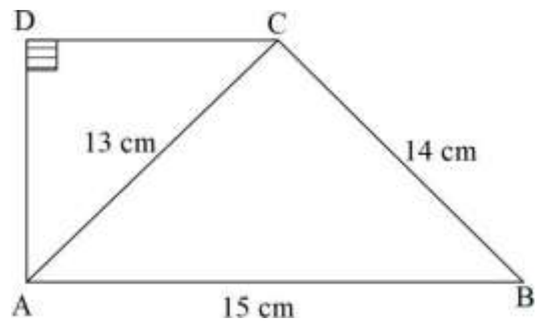
(b) Using a ruler and a pair of compasses only, construct a quadrilateral PQST such that angle PQS = 75˚ and angle QST is a right angle. Measure ST. (3 marks)

6. Find the value of x in the equation $81^{x-1} \times \frac{1}{9^{x+2}} = \sqrt{27}$. (3 marks)

7. A peg is 8 metres from the foot of a building. Two children Ali and Mary can spot the peg from the first floor and second floor at angles of depression 42° and 60° respectively. Calculate the height of the second floor from the first floor correct to 2 decimal places. (3 marks)

8. The dimensions of a rectangle are $(2x+1)$cm by $(x-1)$cm. The area of the rectangle is 29 cm^2 greater than the area of a square of side x cm. Find the perimeter of the rectangle. (4 marks)

9. In the trapezium ABCD, AB = 15 cm, AC = 13 cm and BC = 14 cm. The size of angle ADC = 90°.

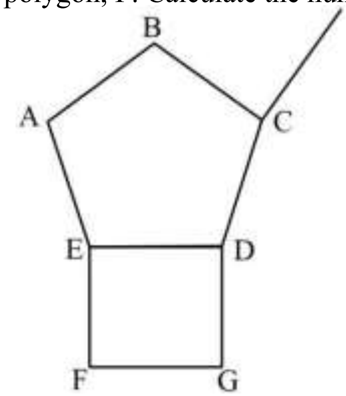


Calculate the area of the trapezium. (4 marks)

10. A water tank is in the shape of a frustum of a cone. The internal radii of the base and top of the vessel are 63 cm and 126 cm respectively and the internal height of the container is 280 cm. Calculate the radius of the water surface in the container when it is filled to a height of 140 cm. (3 marks)

11. Four bells ring simultaneously at starting and an interval of 6 seconds, 12 seconds, 15 seconds and 20 seconds respectively. How many times will they ring together between Monday 11.30 p.m. and Tuesday 2.30 a.m. (3 marks)

12. The figure below shows a regular pentagon, ABCDE, and a square, EDFG. The lines CD and DG are two sides of another regular polygon, P. Calculate the number of sides of polygon P. (3 marks)



13. Use the exchange rates below to answer this question.

	Buying	Selling
1 US dollar	63.00	63.20
1 UK £	125.30	125.95

A tourist arriving in Kenya from Britain had 9600 UK Sterling pounds (£). He converted the pounds to Kenya shillings at a commission of 5%. While in Kenya, he spent ¾ of this money. He changed the balance to US dollars after his stay. If he was not charged any commission for this last transaction, calculate to the nearest US dollars, the amount he received. (3 marks)

14. Solve the following inequality and show your solution on a number line. (3mks)

$$4x - 3 \leq \frac{1}{2}(x + 8) < x + 5$$

15. A measuring cylinder is filled with 110 ml of water. The mass of the empty measuring cylinder is 238.6 grams. When an object is immersed in the cylinder, the mass of the cylinder increases to 642.3 grams and the volume of water rises to 165 ml. calculate the density of the object in g/cm^3. (3 marks)

16. Simplify $\frac{6ax - 3ay - 4bx + 2by}{8ay + 2by - 16ax -}$ (3marks)

SECTION II (50 marks)

*Answer **only five** questions in this section in the spaces provided*

17. A salesman is paid a basic salary of Ksh 30000 and commissions for selling computers as follows:

    5% for sales up to Ksh 100 000

    6% for sales from Ksh 100 001 to Ksh 250 000

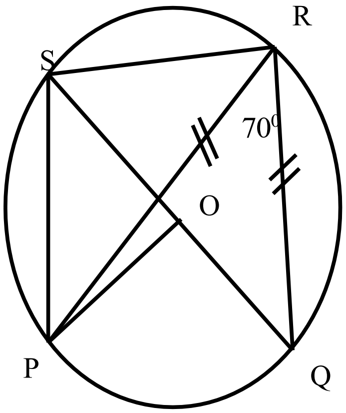
    8% for sales in excess of Ksh 250 000

    (a) In one month, the salesman sold goods worth Ksh 370 000. Determine his total earnings that month. (3 marks)

    (b) In the month that followed, the salesman earned a total of Ksh 57 600 in salary and commissions. How much sales did he make? (4 marks)

    (c) The marked price of each computer is Ksh 42 000. If the basic salary of the salesman is increased by 10%, and the computers are sold at 7% discount, the salesman would earn a total of Ksh 73 872 by selling x computers. Find x. (3 marks)

18. The figure below shows a circle center O, PQRS is a cyclic quadrilateral and QOS is a straight line.



Giving reasons for your answers, find the size of

(a) Angle PRS (2 Marks)

(b) Angle POQ (2 Marks)

(c) Angle QSR (3 Marks)

(d) Reflex angle POS (3 Marks)

19. ABCD is a rhombus in which AC is a major diagonal. The coordinates of the vertices A and C are $(-3,-1)$ and $(5,7)$ respectively.

(a) Find:

(i) The x intercept of the line AC. (3 marks)

(ii) The equation of the minor diagonal BD. (3 marks)

(b) Given that the equation of the side AB is $3y - x = 0$, find the coordinates of D. (4 marks)

20. The distance between towns A and B is 360km. A minibus left town A at 8.15 a.m. and traveled towards town B at an average speed of 90km/hr. A matatu left town B two and a third hours later on the same day and travelled towards A at average speed of 110km/hr.

(a) (i) At what time did the two vehicles meet? (4 Marks)

(ii) How far from A did the two vehicles meet? (2 Marks)

(b) A motorist started from his home at 10.30 a.m. on the same day as the matatu and travelled at an average speed of 100km/h. He arrives at B at the same time as the minibus. Calculate the distance from A to his house. (4 Marks)

21 (a) Given that $\mathbf{A} = \begin{pmatrix} 5 & 3 \\ 12 & 7 \end{pmatrix}$ show that $\mathbf{A}^2 - 12\mathbf{A} - \mathbf{I}$ is a null matrix. (3 marks)

(b) A teacher ordered for 12 calculators and 10 revision books costing Ksh 22 500. Due to financial constraint, he was advised to cut the budget to Ksh 20 100 by reducing the number of calculators by two.

(i) Form and simplify two linear equations to represent this information. (1 mark)

(ii) Use matrix method to find the cost of each item. (4 marks)

(c) If the teacher bought 5 calculators and 7 revision books, represent the number of items bought in a 1×2 matrix hence find the total cost of the items. (2 marks)

22. The table below shows marks scored by students in a class.

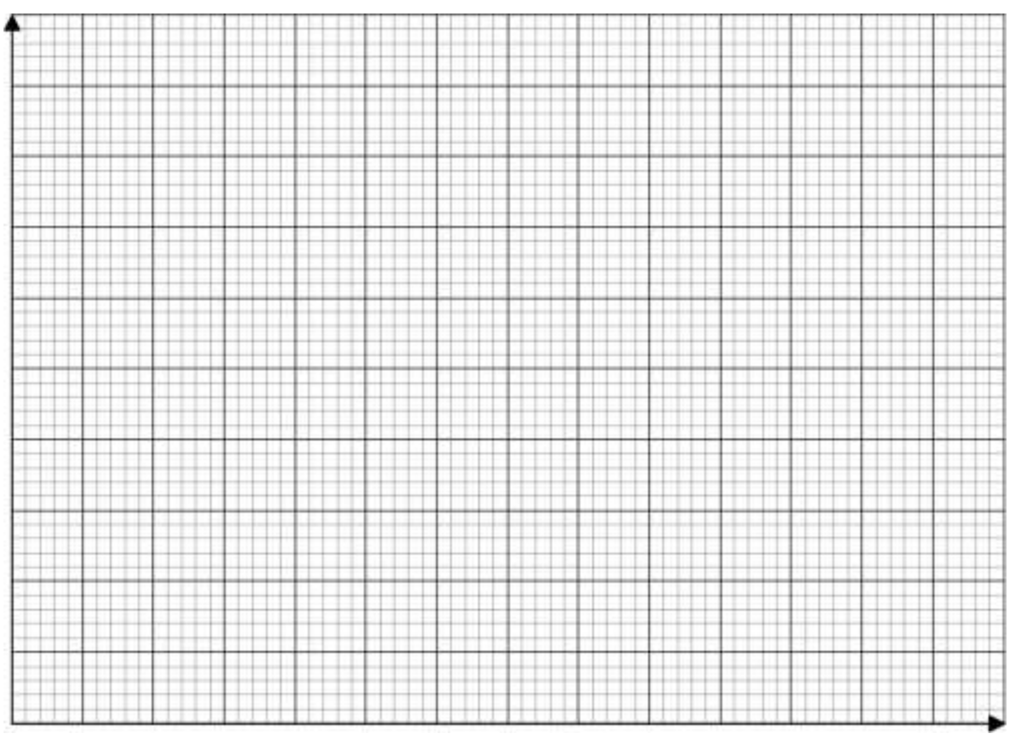
Marks	10 – 14	15 – 24	25 – 39	40 – 44	45 – 54	55 – 69
Frequency	11	8	9	4	6	t

(a) Given that the mean mark is 35.7, find:

(i) The value of t. (3 marks)

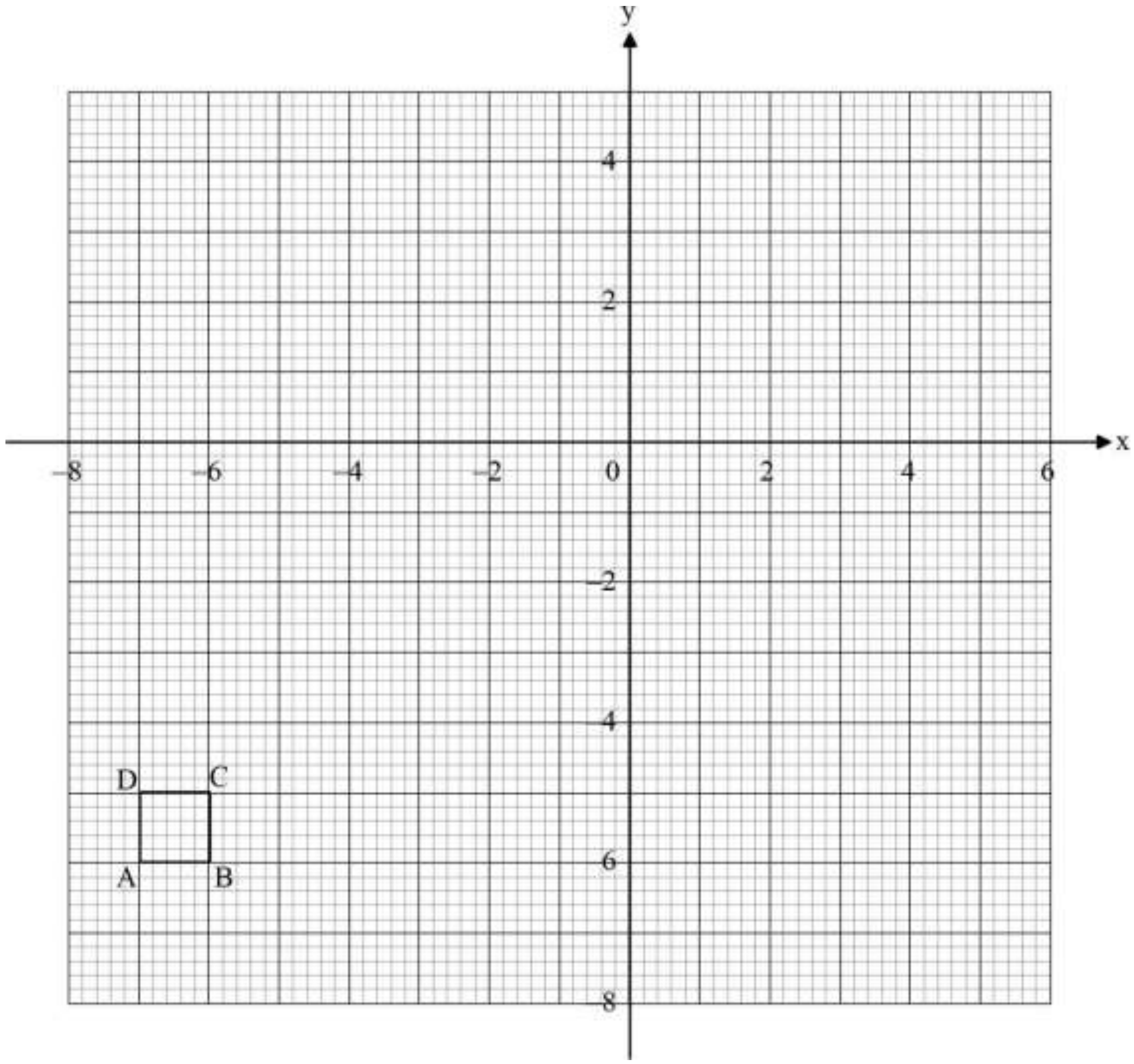
(ii) The class interval for the modal class. (1 mark)

(b) On the grid provided below, draw a histogram to represent the information in the table. (3 marks)



(c) Using the histogram in (b) above, calculate the median mark. (3 marks)

23. A square ABCD with vertices at $A(-7,-6)$, $B(-6,-6)$, $C(-6,-5)$ and $D(-7,-5)$ is drawn on the grid below. $A'B'C'D'$ is the image of ABCD under enlargement center $(-7,-7)$ and scale factor 3.



(a) Draw ABCD on the same axes. (3 marks)

(b) ABCD is the image of ABCD under reflection in the line $y = 0$. Draw ABCD and state its vertices. (2 marks)

(c) ABCD is the image of ABCD under $+90°$ turnabout center $(-1,-2)$.

(i) State the transformation that maps ABCD onto ABCD. (1 mark)

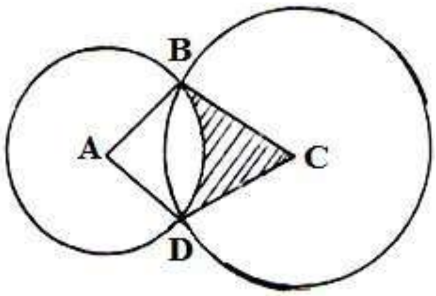
(ii) On the same axes, draw and label the vertices of the square ABCD (2 marks)

(d) State any two pairs of the squares that are:

(i) Directly congruent. (1 mark)

(ii) Oppositely congruent. (1 mark)

24. In the diagram below, two circles, centers A and C and radii 7cm and 24cm respectively intersect at B and D. AC = 25cm.



(a) Show that angle ABC = 90^0. (3 Marks)

(b) Calculate

(i) the size of obtuse angle BAD (3 Marks)

(ii) the area of the shaded part (4 Marks)

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KCSE TOP PREDICTION TRIAL 3 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ……………………………………………………….…………… **Adm Number**…………………………..

Candidate's Signature ……………………..……………………….**Date** ……………………………………

Instructions to Candidates

(a) *Write your name and index number in the spaces provided above.*
(b) *Sign and write the date of the examination in the spaces provided above.*
(c) *This paper consists of **TWO** sections: **Section I** and **Section II**.*
(d) *Answer **ALL** the questions in **Section I** and only **Five** from **Section II**.*
(e) *All answers and working must be written on the question paper in the spaces provided below each question.*
(f) ***Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.***

For Examiner's Use Only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand Total	

SECTION I

Answer ALL questions in the spaces provided

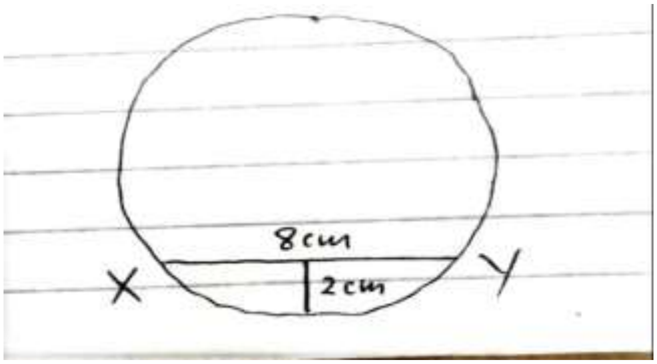
1) Wambugu bought two types of rice, Pishori at Ksh. 160 per two kilogram and mixed with Falcon costing Ksh. 140 per kilogram. In what ratio should he mix the two brands to sell a kilogram of the mixture at Ksh. 124.80 making 20% profit (3mks)

2) a) Expand $\left(1 - \frac{1}{2}x\right)^5$ (1mk)

b) Use the expansion up to x^3 in (a) above to evaluate $(0.8)^5$ correct to 4 d.p (2mks)

3) A rectangular plot of land has length of 125.8m and an exact width of 80m. Find the percentage error in calculating the area of the plot correct to 3 d.p (3mks)

4) Chord XY is of length 8cm and the maximum distance between the chord and the lower part of the circle is 2cm. determine the diameter of the circle (3mks)



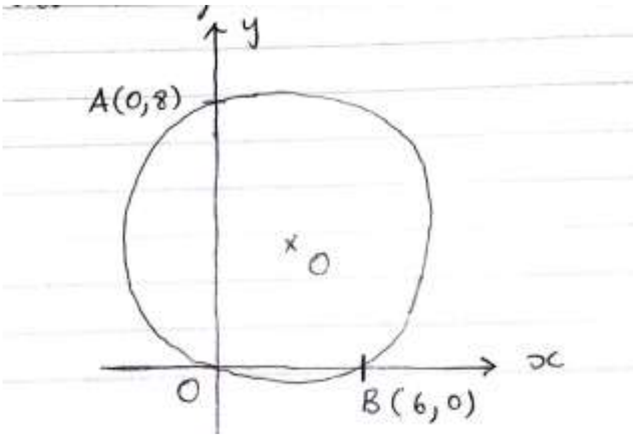
5) The position vectors of points P and Q are -2i + 3j – 4k and 4i – 2k respectively. A point M divides PQ externally in the ratio 2:1. Determine the co-ordinate of point M (3mks)

6) Solve for n in $\log_{10}(6n + 4) - \log_{10}(2n - 8) = 2$ (3mks)

7) Simplify the following giving your answer in the simplest form possible (3mks)

$$\frac{5}{\sqrt{7}+\sqrt{5}} - \frac{3}{\sqrt{6}+\sqrt{3}}$$

8) In the figure below, 0 is the centre of the circle and AB passes through the centre of the circle



a) Determine the centre and radius of the circle (2mks)

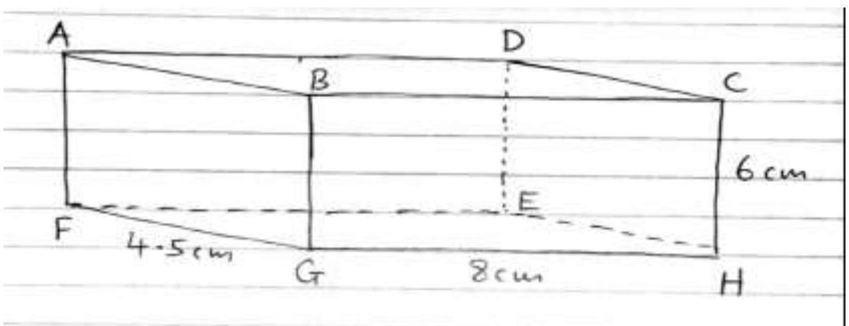
b) Express the equation of the circle in the form $x^2 + y^2 + ax + by + c = 0$ where a, b and c are constants (2mks)

9) Using completing square method, solve the equation (3mks)

$$2x^2 + 5x - 12 = 0$$

10) The probability of a student passing a mathematics exam is 60%. This probability increases by 15% on the subsequent attempt after failing the previous time. Calculate the probability of a student passing on the third attempt (3mks)

11) The diagram below represents a cuboid ABCDEFGH in which FG = 4.5cm, GH = 8cm and HC = 6cm



Calculate the size of the angle between the lines AB and FH (3mks)

12) The gradient function of a curve is given by $\frac{dy}{dx} = 2x - 4$ if the curve passes through the point (2, - 1), find its equation (3mks)

13) The period of $y = -3\sin(px - 60)^0$ is 60^0

a) Find the value of P (1mk)

b) State its amplitude (1mk)

e) State its phase angle (1mk)

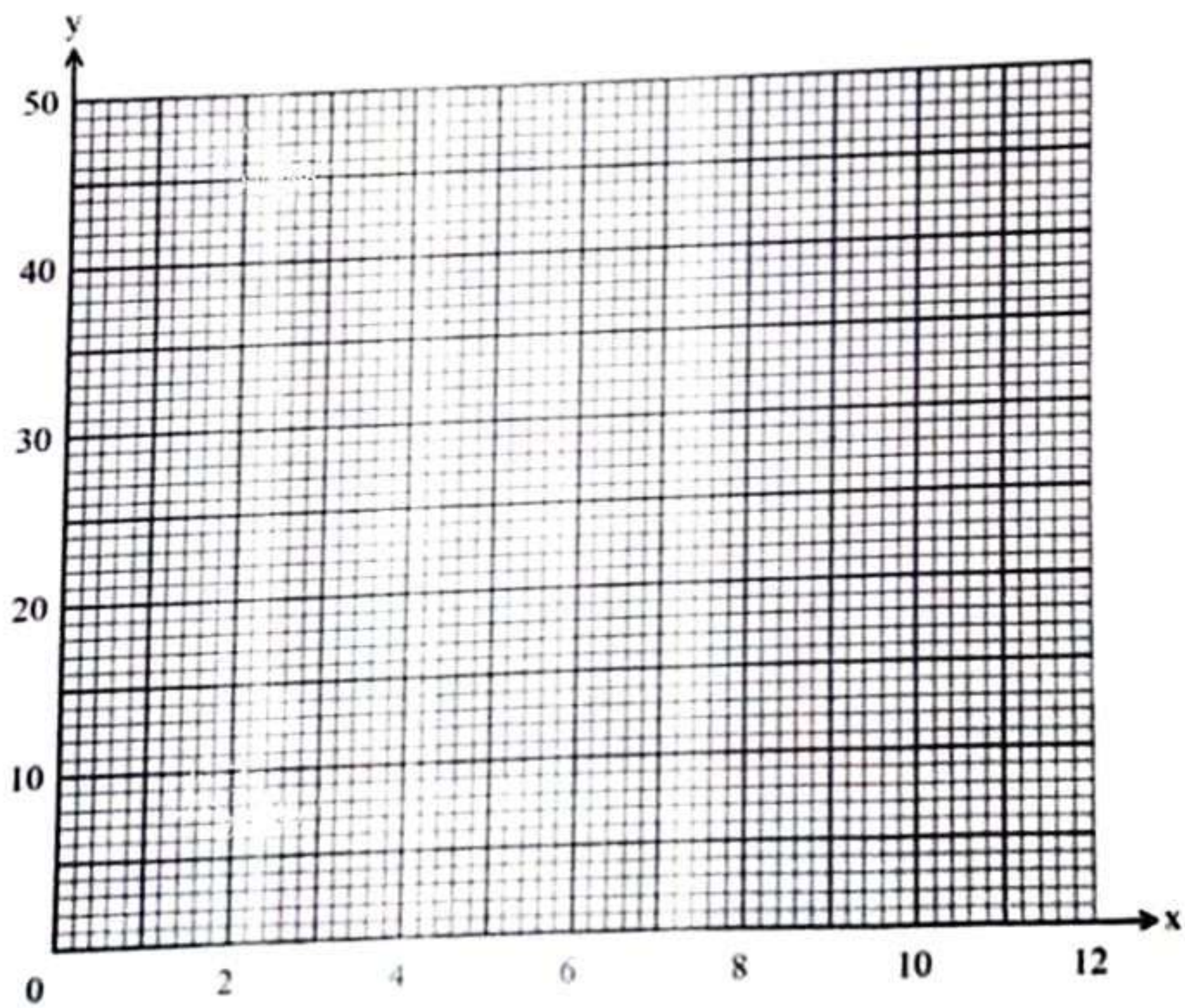
14) A car was valued at Ksh. 900,000. It depreciated at 10% p.a Find after how long its value depreciated to Ksh. 590,590 (3mks)

15) Two taps P and Q, when opened at the same time can fill a tank in 3 hours 36 minutes. Tap p working alone takes 3 hours longer than tap Q to fill the same tank. How many hours does it take tap P to fill the tank. (3mks)

16) Data was collected from an experiment involving two variables x and y and was recorded in the table below

X	1	2	3	4	5	6	7	8	8	10
y	9	12	16	20	24	8	31	35	39	42

On the grid provided below, draw the line of best fit hence find the equation connecting the variable and y (4mks)



SECTION II (50MKS)

Answer any FIVE questions in the spaces provided

17) The table below shows income tax rates in a certain year

Monthly Taxable income	Tax rate
0-14200	10%
14201 – 26700	15%
26701 – 39200	20%
39201 – 51700	25%
51701 and above	30%

Ojwera is a civil servant and earns a basic salary of sh. 34,800, medical allowance of sh. 2400, house allowance of sh. 10,000, commuter allowance of sh. 7,200 and non-taxable allowance of sh. 12,000. He has a personal tax relief of sh. 1,020 and also gets relief equivalent to 10% of the premium paid. If he pays monthly insurance premium of sh. 2,700. Calculate:

a) Ojwera's monthly taxable income (2mks)

b) Find the net tax paid by Ojwera per month (5mks)

c) If Ojwera pays a loan of sh. 8000 every month, affordable housing levy of 1.5% of taxable income and Social Health Authority fee of sh. 1805. Find his net monthly pay (3mks)

18) Angle A (1,11), B 92, 6) C (4, 10) is mapped on to Triangle Al (10, 4) B 1 (5, 3) C 1 (9, 1) by transformation M

a. Plot triangles ABC and $A^1B^1C^1$ on the grid provided (1mks)

b. Describe transformation M fully (3mks)

c. Triangle $A^1B^1C^1$ is further transformed to $A^{ll}\,B^{ll}\,C^{ll}$ by a transformation

$$N = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

i. Determine the co-ordinates of $A^{ll}\,B^{ll}\,C^{ll}$ (2mks)

ii. Plot the triangle $A^{ll}\,B^{ll}\,C^{ll}$ (2mks)

iii. Describe transformation N fully (2mks)

19) Points A (25^0N, 138^0W), B (70^0N, 138^0W) and C (70^0N, 42^0E) are on the earth's surface. Given that R = 6370km and $\pi = \frac{22}{7}$

a) Find in kilometres

i) The distance between A and B (2mks)

ii) The shortest distance between B and C (3mks)

b) A dream liner left point C on Tuesday 1530 hours for point A via B at average speed of 800km/h. It took 2hours at B for passengers to disembark then proceeded to point A using the shortest route in each case. Calculate to the nearest minute

i) The local time and day at A when the liner left C (2mks)

ii) The day and time when the liver arrived at A (3mks)

20) The date below shows length of leaves in millimeters

Length	25-34	35-44	45-54	55-64	65-74	75-84	85-94
Number of leaves	9	12	55	68	32	18	6

a) Using assumed mean of 59.5, calculate:

i)The mean length (3mks)

ii) The variance (3mks)

iii) The standard deviation (1mk)

b) Find the percentage of leaves that has a length of 69.5mm and above (3mks)

21) The second third and ninth terms of an arithmetic progression (AP) form the first three terms of a geometric progression (GP)

a) Find the common ratio of the GP (4mks)

b) Given that the sum of the first ten terms of the AP is 1110, find:

i)The first term and the common difference of the AP (3mks)

ii) The number of terms of the AP that will give a sum of at least 2400 (3mks)

22) Use a ruler and a pair of compasses only in this question

a) Construct a quadrilateral ABCD such that AB = 8cm AD = 5cm, DC = 9cm, BC = 7cm and angle BAD = 120^0 (4mks)

b) On the same diagram, construct the locus:

i) L_1 of points equidistant from A and B (1mks)

ii) L_2 of points equidistant from BC and DC to intersect L_1 at E. Measure DE (2mks)

c) By shading, identify the possible region R within the equilateral ABCD such that CR < 4 cm and angle ARB ≥ 90 (4mks)

23) Three quantities A, B and C are such that A varies directly as the square root of B and inversely as the square of C

a) Given that A = 4 when B = 64 and C = 5, find

i) The law connecting A, B and C (4mks)

ii) A when B = 16 and C = 10 (2mks)

b) If B is increased by 44% and C decreased by 20%, find the percentage change in A (4mks)

24) In an interschool mathematics contest, schools can register teams in junior and senior categories. Information on number of students and the participation fee per team in each category is given in the table below:

	Junior category	Senior category
No. of students per team	6	4
Participation fees per team	Ksh. 2000	Ksh. 3000

The organizing committee projected to register x junior teams and y senior teams.

For the contest to take place, the following conditions must be satisfied:

i)At least two junior teams must be registered

ii) The number of senior teams must be more than half the number of junior teams

iii) The total number of participating students from the two categories must not exceed 48

iv) The total amount of money raised from the participation fee must be more than Ksh. 12000

a) Write down inequalities in x and y that satisfy the conditions (4mks)

b) Represent the inequalities in (a) on the grid provided below (4mks)

c) The organizing committee expected to make a profit of Ksh. 200 for every junior team and Ksh. 500 for every senior team that participated. Using a search line, determine the number of teams of each category that should be registered in order to maximize the profit (2mks

KCSE TOP PREDICTION TRIAL 4 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ……………………………………………………………………… **Adm Number**…………………………..

Candidate's Signature ……………………..………………………….**Date** ……………………………………

Instructions to candidates

a) Write your name and admission in the space provided above.

b) Sign and write the date of examination in the spaces provided above

c) This paper consists of two sections: Section Ⅰ and section Ⅱ.

d) Answer all the questions in section Ⅰ and only five questions from Section Ⅱ.

e) Show all the steps in your calculations, giving your answers at each stage in the spaces provided below each question.

For examiner's use only Section Ⅰ

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	total

Section Ⅱ

17	18	19	20	21	22	23	24	total

1. Evaluate without using a calculator. (3mks)

$$\frac{\frac{5}{6}\ of\ (4\frac{1}{3}-3\frac{5}{6})}{\frac{5}{12}\times\frac{3}{25}+1\frac{5}{9}\div 2\frac{1}{3}}$$

2. The internal dimensions of a container at the port are 1380cm by 240cm by 300cm. The container is filled with large cubical boxes of equal sizes. Calculate the least number of the cubical boxes in the container. (3mks)

3. Two polygons sides differ by 1. If the sum of interior angles of the two polygons is 1620, find the number of sides of each polygon and name the polygons. (3mks)

4. A boat is at point P, a distance of 100km from the bottom of a hill. The angle of elevation of the top of the hill is 30^o from P. The boat sails straight towards the hill to a point Q from where the angle of elevation to the top of the hill is now 60^o. Calculate the distance PQ. (3mks)

5. Solve the equation. (3mks)

$5_{(2x+1)} - 3(5_{(x+1)}) + 10 = 0$

6. A Kenyan bank buys and sells foreign currencies at the exchange rates shown below.

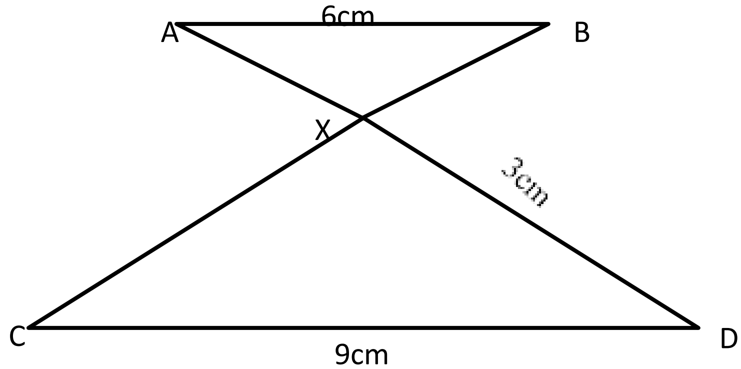
	Buying (ksh)	Selling (ksh)
1 Euro	147.86	148.00
1 US Dollar	74.22	74.50

An American arrived in Kenya with 20,000 Euros. He converted all the Euros to Kenyan Shillings at the bank. He spent Ksh. 2,512,000 while in Kenya and converted the remaining Kenyan shillings into US Dollars at the bank. Find the amount in Dollars that he received. (3mks)

7. Given that

$4p - 3q = \begin{pmatrix} 10 \\ 5 \end{pmatrix}$ and $p + 2q = \begin{pmatrix} -14 \\ 15 \end{pmatrix}$, find **q** (4mks)

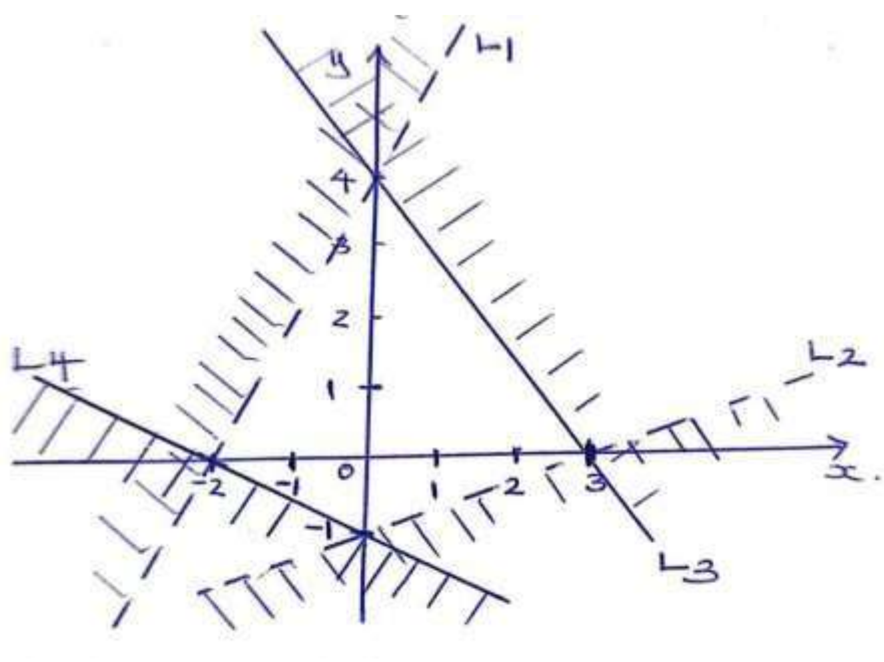
8. In the figure below lines AB and CD are parallel. AD and CB intersect at X. AB = 6cm, CD = 9cm and DX = 3cm.



a) Calculate the length of AX (2mks)

b) The area of △ ABX is 6cm^2. Calculate the area of △CXD. (2mks)

9. Write down the four inequalities that satisfy the unshaded region a shown below.



10. Given that X is an acute angle and Sin (x -20) = Cos (3x-50), find the value of x. (2mks)

11. Simplify: $\dfrac{x^2 + x - 4xy\ 4y}{4yx - x^2y + 4y^2 - xy}$ (3mks)

12. The ratio of the cost of commodity **x** to the commodity y is 2:3 and the ratio of the cost of commodity **y** to that of commodity Z is 6:1. If the total cost of the commodities is sh.
2,200, express the cost of commodity Z as a percentage of commodity **y**.
(3mks)

13. The equation of the line L_1 is $2y - 5x - 8 = 0$ and L_2 passes through the points (-5, 0) and is perpendicular to $L_{1.}$ Find;

a) the gradient of line L_1 (1mk)

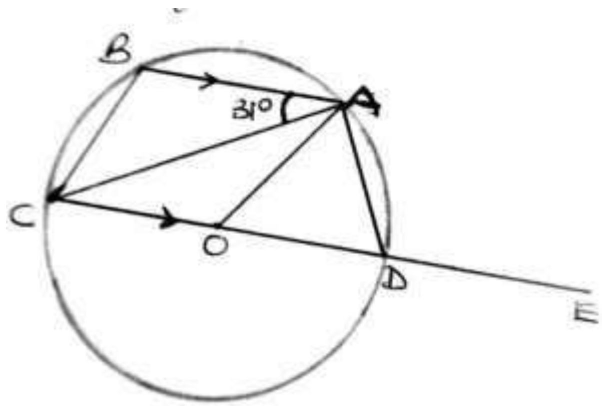
b) the equation of L_2 leaving it in double intercept form. (3mks)

14. Use reciprocal tables to find the value of $(0.325)^{-1}$ hence evaluate

$$\frac{\sqrt[3]{0.000125}}{0.325}$$

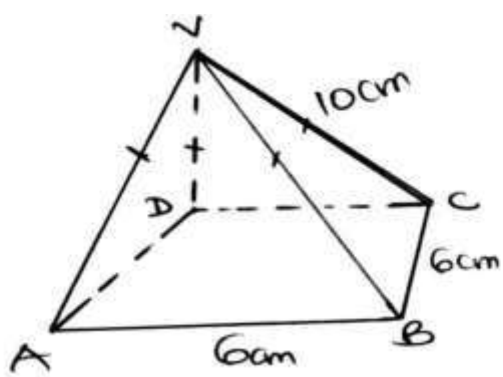
Giving your answer to 4 significant figures. (3mks)

15. In the figure below O is the center of the circle and BA is parallel to CD.



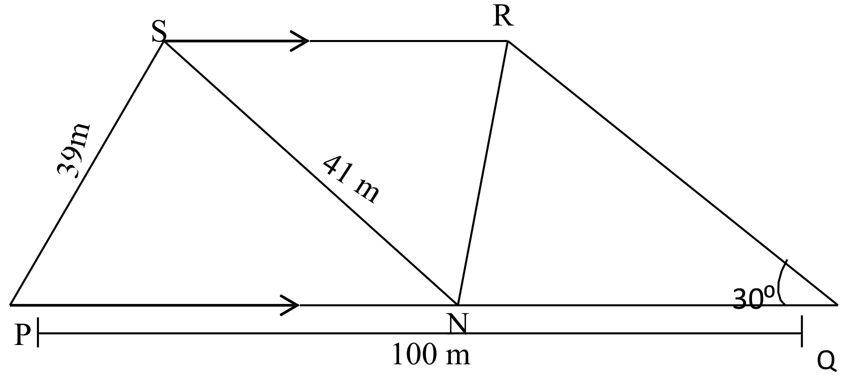
If $< BAC = 31°$, Find the size of $< CBA$. (2mks)

16. Draw the net of the solid below and calculate the surface area. (3mks)



SECTION II

17. In the figure below lines PQ is parallel to line RS point N is the Midpoint of PQ. Given that PQ = 100cm, PS = 39m, NS = 41m and PQR = 30^o



a) Calculate the area of triangle PNS and hence the perpendicular distance between the two parallel sides. (4mks)

b) Calculate the length of

i). QR (2mks)

ii). NR (2mks)

c) Calculate the size of the obtuse angle QNR (2mks)

18. The following are masses of 25 students in form 4 class.

49, 51, 50, 60, 55, 45, 56, 51, 58, 59, 40, 54, 44 44,
42, 59, 50, 62, 46, 43, 57, 56, 52, 43, 41

a) Prepare a frequency distribution table with a uniform class size starting with the class 40 – 43 (4mks)

Estimate the median mass. (3mks)

Draw a histogram for the data. (3mks)

19. (a) Determine:

The inverse of the matrix A (1mk)

$$\begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}$$

(b) Peter bought 20 bags of oranges and 15 bags of mangoes for a total of ksh. 9,500. Wilson bought 30 bags of oranges and 20 bags of mangoes for a total of ksh. 13,500. If the price of a bag of oranges is X and that of mangoes is Y.

i). Form two simplified equations to represent the information above. (2mks)

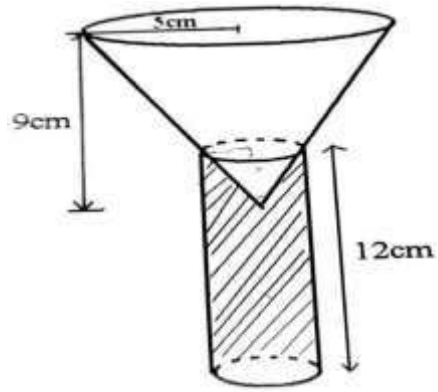
ii). Hence use the matrix A^{-1}above to find the price of one bag of each item.

(4mks)

(c) The price of each bag of orange increased by 10% and that of mangoes reduced by 10%. The businessmen (Peter and Wilson) bought as many oranges and as many mangoes as they bought earlier.

Find by matrix method, the total cost of oranges and mangoes that the businessmen bought after the percentage change. (3mks)

20. Part of the inverted open cone of base radius 5cm and height 9cm is inverted into water in a cylindrical tank of diameter of 6 cm and height of 12 cm full of water as shown;



a) Calculate the volume of water left in the tank. (6mks)

b) Calculate the surface area of the cone not submerged in water. (4mks)

21. Three islands P, Q, R and S are on an ocean such that island Q is 400Km on a bearing of 030^0 from island P. Island R is 520 km and a bearing of S60^{0}E from island Q. A port S is sighted 750km due South of island Q.

a) Taking a scale of 1cm to represent 100km, give a scale drawing showing the relative positions of P, Q, R and S. (4mks)

b) Use the scale drawing to find the compass bearing of:

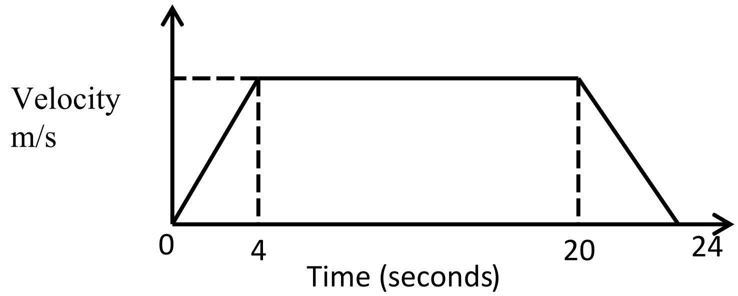
i). Island R from island P (1mk)

ii). Port S from island R (1mk)

c) Find the distance between island P and R (2 marks)

d) A warship T is such that it is equidistant from the islands P, S, and R. by construction locate the position of T. (2 marks)

22. (a) The figure below is a velocity time graph for a car.



i). If the total distance travelled by the car was 1,600M find the maximum velocity. (2 marks)

ii). Calculate the deceleration of the car. (2 marks)

(b) A car left Nairobi towards Eldoret at 7:12a.m. At average speed of 90km/h. At 8:22 a.m. a bus left Eldoret for Nairobi at an average speed of 72km/h. The distance between the two towns is 348km. calculate:

i). The time when the two vehicles met. (4 marks)

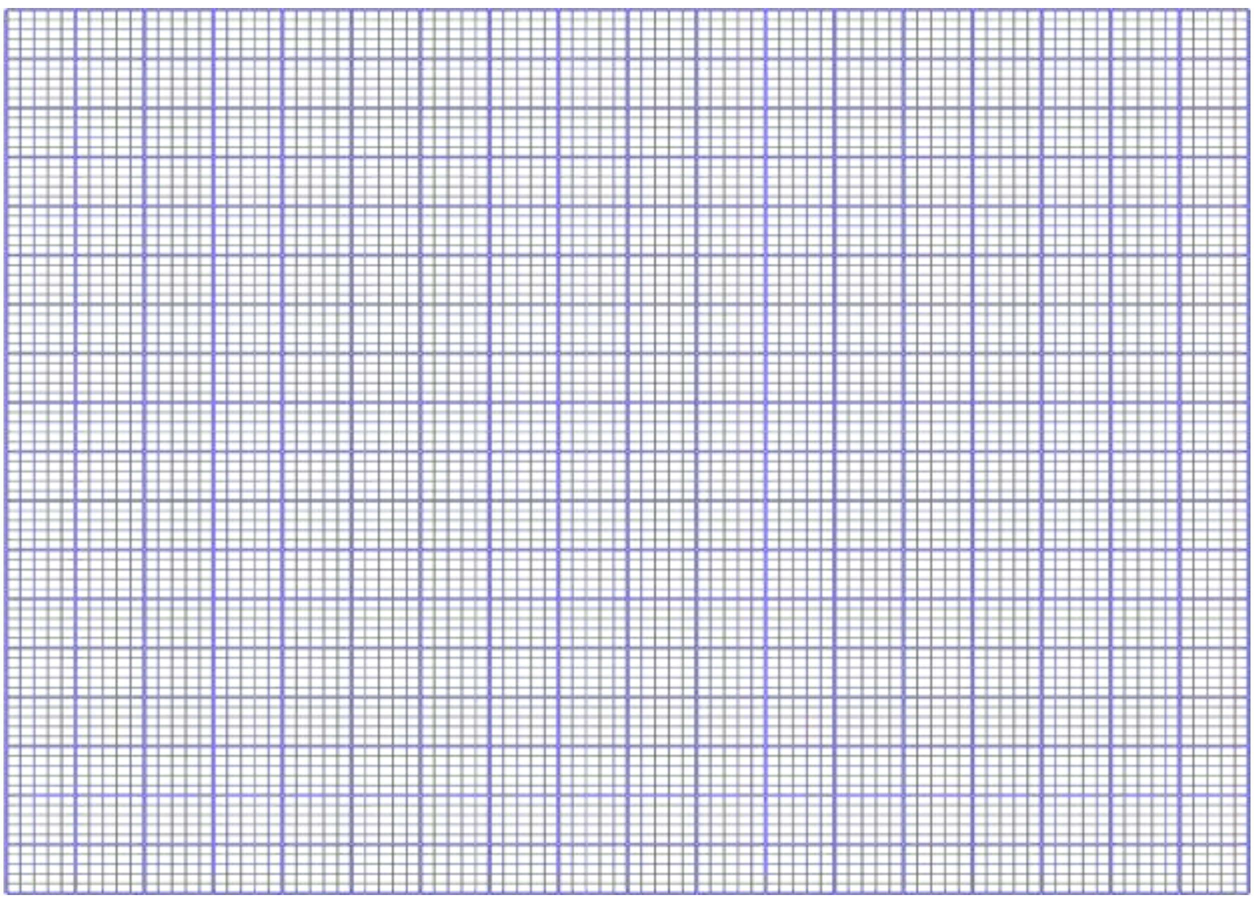
ii). The distance from Nairobi to the meeting place. (2 marks)

23. given that $y = 7 - 3x - 2x^2$ for $-4 \leq x \leq 3$ (2 marks)

a) complete the table below

X	-4	-3	-2	-1	0	1	2	3
Y	7	7		7	7	7		7
-2x	12	9		3	0	-3		-9
$-2x^2$	-32	-18		-2	0	-2		-18
y	-13	-2		8	7	2		-20

b) draw the graph $y = 7 - 3x - 2x^2$ for $-4 \leq x \leq 3$ (3 marks)



c) Use the graph to find the roots of the equation.

i). $-2x^2 - 3x + 7 = 0$ (2 marks)

ii). $-2x^2 - 4x + 9 = 0$ (3 marks)

24. a) i) Find the coordinates of the stationary points on the curve $y = x^3 – 3x + 2$. (3 marks)

ii) For each stationary point determine whether it is a minimum or a maximum. (4 marks)

b) In the space provided, sketch the graph of the function $y = x^3 -3x + 2$. (3marks)

KCSE TOP PREDICTION TRIAL 4 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ……………………………………………………………………… **Adm Number**…………………………..

Candidate's Signature ………………………..………………………….**Date** ……………………………………

INSTRUCTIONS TO CANDIDATES

1. Write your name, index number and class in the spaces provided above.
2. The paper contains two sections: **Section I** and **Section II**.
3. This paper contains **14 PRINTED** pages make sure all **PAGES ARE PRINTED** and **NON IS MISSING**
4. Answer **ALL** the questions in **Section I** and **ANY FIVE** questions from **Section II**.
5. All working and answers must be written on the question paper in the spaces provided below each question.

For examiners use only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand Total	

Section 1 (50 Marks)

1. Simplify: **(3marks)**

$$\frac{5+3\sqrt{3}}{5-3\sqrt{3}}$$

2. Give that P = $\begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ and Q = $\begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$, Find PQ hence solve the equation (**3 marks)**

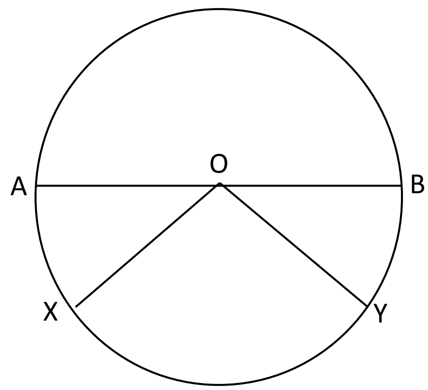
2x – 3y = 5
-x + 2y = -3

3. The gradient function of a curve is given by $\frac{dy}{dx} = x^2 - 8x + 2$
If the curve passes through the point (0, 2) find its equation **(3marks)**

4. Njau borrowed sh 500,000 from a bank which charged interest at 20% p.a compounded quarterly. If he repaid the loan after 2½ years, calculate the total amount paid

(3marks)

5. In the figure below XY= 8cm and O is the centre of the circle



Determine the area of the circle if angle AOX=15^0 **(3marks)**

6. A student at a certain college has 60% chance of passing an examination at the first attempt. Each time a student fails and repeats the examination his chances of passing are increased by 15%. Calculate the probability that a student in the college passes an examination at the third attempt. **(4marks)**

7. The product of three consecutive positive even numbers is 960 more than the product of the largest and square of the smallest. Find the numbers. **(3marks)**

8. Expand and simplify $(1-3x)^5$ up to the term in x^3. Hence use your expansion to estimate $(0.97)^5$ correct to 4 decimal places. **(4marks)**

9. A ship leavers an island (50N, 450E) and sails due west for 120 hours to another island. The average speed of the ship is 27 knots. Find the position of the second island. **(3marks)**

10. A variable y varies as the square of x and inversely as the square root of z. Find the percentage change in y when x is increase by 5% and z reduced by 19%. **(3marks)**

11. Solve the equation $2 \sin(4x-30)^0 = -1$ for the range $0^0 \leq x \leq 180^0$ **(3marks)**

12. Find the area bounded by the curve $y = x^2$ and the lines $x = -3$ and $x = 2$ using the mid ordinate rule. (Use five strips). **(3marks)**

13. Under a shear with x-axis invariant, the point A (4, 3) is mapped onto A' (10, 3). Find the coordinates of B' under this shear if point B is (2, 5) **(4marks)**

14. The figure below shows a rectangular sheet of dimensions 20cm by 15cm. equal squares of side x cm are cut from two adjacent corners and the resulting flaps bent to form the tray of a shovel. Find the value of x for which the volume of the tray is maximum. **(4marks)**

15. If $V^2 = \sqrt{\frac{1+C^2}{R^2}} + \frac{r}{3}$, make C the subject of the formula. **(3marks)**

16. The coordinates of P and Q are (4, -1,3) respectively. A point T divides PQ externally in the ratio 9.2. Find the coordinates of T. **(3marks)**

SECTION II (50 MARKS)

Answer any five questions in this section.

17. The table below shows the times in seconds taken by 50 entrants in the hearts of 100 metres free style swimming competition.

Time in seconds	41.0-41.8	41.9-42.7	42.8-43.6	43.7-44.5	44.6-45.4	45.5-46.3	46.4-47.2
Frequency	0 3	5	8	12	13	7	2

a) Draw accumulative frequency curve **(4marks)**

b) Use your graph to estimate

(i) The median **(2marks)**

(ii) The second and sixth deciles **(2marks)**

(iii) The number of swimmers who recorded between 42.05s and 46.05 s. **(2marks)**

18. A certain number of people agreed to contribute equally to buy a book worth sh 1200 for a school library. Five people pulled out so the others agreed to contribute an extra sh 10 each. Their contribution enabled them to buy a book worth sh 200 more than they originally expected.

a) If the original number of people was x write an expression of how much each was to contribute. **(1mark)**

(b) Write down two expressions of how much each contributed after the five people pulled out. **(2marks)**

(c) Calculate how many people made the contribution. **(5marks)**

(d) State how much each contributed. **(2marks)**

19. Plot the points the triangle ABC whose coordinates are A (-6, 5), B (-4, 1) and C (3, 2).
    a) Given that A (-6, 5) is mapped onto A^1(-6, -4) by a shear with y-axis invariant;
        (i) Draw triangle A^1 B^1 C^1 only one hits the target the image of triangle ABC under this shear. **(3marks)**

(ii) Determine the matrix representing this shear. **(2marks)**

b) Triangle $A^1B^1C^1$ is mapped onto $A^{11}B^{11}C^{11}$by a transformation defined by the matrix

(i) Draw triangle $A^{11}B^{11}C^{11}$. **(3marks)**

(ii) Describe fully a single transformation that maps ABC onto $A^{11}B^{11}C^{11}$. **(2marks)**

20. A new tailoring business makes two types of garments A and B. The number of type A garments made are x and that of type B y. Each garment A requires 2½ metres of material while each garment B requires 2 metres of material daily for the production of both garments but produce less than 80 garment s of type A and more than 60 garments of type B. in addition the ratio of production of y to x must be less than 5:3. **(3marks)**

(a) Write down 4 inequalities satisfying the above information **(3 marks)**

(b) Represent the inequalities on the same graph. **(4 marks)**

(c) If the business makes a profit of sh. 7.50 on garment A and a profit of sh 5.00 on garment B. How many garments of each type must the business produce in order to make maximum profit?
State the profit **(2 marks)**

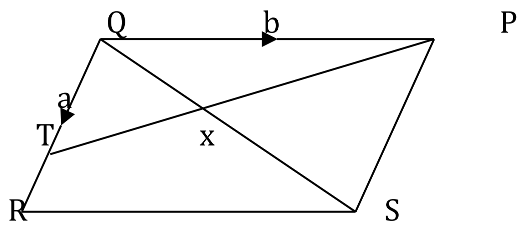
21. Without using a protractor, construct line XY = 8cm

   a) Construct the locus of all points R such that $\angle XRY = 60°$ (4 marks)

b) Construct the locus of all points P such that the area of triangle XPY is 16 cm^2.(3 marks)

c) Mark all the points where the locus of R intersects the locus of P with the letters A,B,C and D (3 marks)

22. In the figure below, QT = a and QP = b.



(a) Express the vector PT in terms of a and b. (1mark)

(b) If PX = kPT, express QX in terms of a, b and k, where k is a scalar. (3marks)

(c) If QR = 3a and RS = 2b, write down an expression for QS in terms of a and b(1mark)

(d) If QX = tQS, use your result in (b) and (c) to find the value of k and t. (4marks)

(e) Find the ratio PX : XT. (1mark)

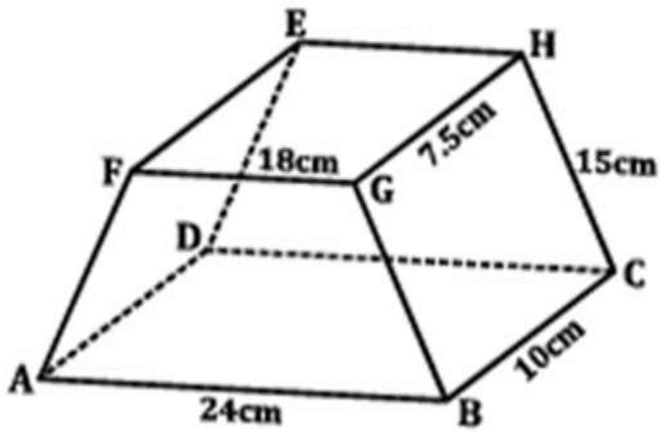
23. The 2^{nd}, 7^{th} and 27^{th} term of an AP are the first three consecutive terms of a GP. The 13^{th} term of the AP is 38. Determine

(a) The first term and the common difference of the AP. **(4marks)**

(b) The sum of the first 40 terms of the arithmetic series. **(3marks)**

c) The sum of the first 6 terms of the geometric series. **(2marks)**

24. The figure below shows a frustrum ABCDEFGH of a right pyramid AB = 24cm BC = 10cm, FG = 18cm, GH = 7.5cm and AF = BG = CH = DE = 15cm



Find

(a) The height of the pyramid. **(2 marks)**

(b) (i) The triangle between AH and the base ABCD. **(3 marks)**

(ii) The angle between the planes ABGH and ABCD. **(3 marks)**

(i) The angle between the planes ABHE and ABCD. **(2 marks)**

KCSE TOP PREDICTION TRIAL 5 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ………………………………………………………………………… Adm Number…………………………..

Candidate's Signature ………………………..……………………….Date ……………………………………

INSTRUCTIONS TO THE CANDIDATES

a) **Write your *name* and *Admn number* in the spaces provided above.**
b) **Sign and write *date* of examination in the spaces provided above.**
c) **This paper consists of two sections; *Section I* and *Section II.***
d) **Answer *All questions* in *Section I* and *only Five questions* from *section II***
e) ***All* answers and working *must* be written on the question paper in the spaces provided below each question.**
f) **Show all the steps in your calculations giving answers at each stage in the spaces provided below each question.**
g) **Non-programmable silent electronic calculators and KNEC Mathematical tables may be used except where stated otherwise.**

For examiner's use only.

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

GRAND TOTA

(Answer all the questions in the spaces provided)

1. Without using a calculator, evaluate

$$\frac{36 - 8 \times -4 - 15 \div - 3}{3 \times -3 + -8\,[6 - (^{-}2)]}$$ (3mks)

2. The G.C.D of two numbers is 12 and their L.C.M is 240. If one of the numbers is 60, find the other number. (2mks)

3. Agnes paid rent which was $\frac{1}{10}$ of her net salary. She used $\frac{1}{2}$ of the remaining amount to make a down payment for a plot. She gave her mother sh. 2500 and did shopping worth sh. 7500 for herself. She saved the remaining which was sh. 12,500. How much was the down payment that she made? (4 marks)

4. Elvis exchanged Ksh.600,000 to Sterling pounds. After settling the bills worth £1200, he

changed the balance to Euros. He then purchased goods worthy 200 Euros. Using the exchange rates below, calculate his balance in Kenyan shillings. (4mks)

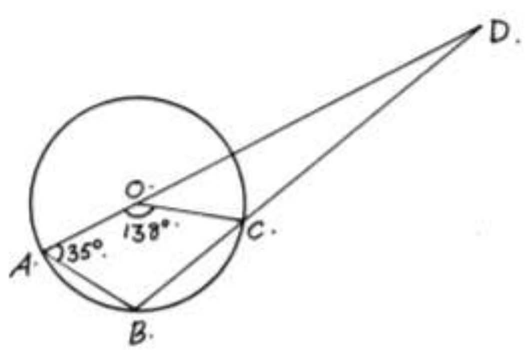
	Buying (Ksh)	Selling (Ksh)
1 Sterling pound	114.20	114.50
1 Euro	101.20	101.30

5. Each exterior angle of a regular polygon is a fifth of the interior angle.

a) Find the size of the exterior angle (2mks)

b) Find the number of sides of the polygon (1mk)

6. In the figure below, O is the center of the circle. AOD and BCD are straight lines. Angle AOC = 138^0 and angle OAB=35^0. Determine the size of angle ADB. (3mks)

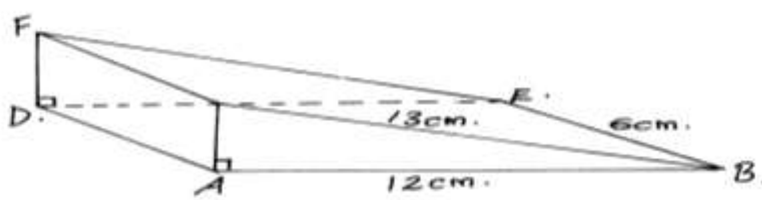


7. An observer stationed 20m away from a tall building finds that the angle of elevation of the top of the building is 68^0 and the angle of depression of its foot is 50^0. Calculate the height of the building. (3mks)

8. Simplify the expression (3mks)

$$\frac{P^2 - 4m^2}{2m^2 - 7mp + 3p^2}$$

9. a) Sketch the net of a wedge in the following figure.



b) Calculate the surface area of the net drawn above. (3mks)

10 Express the inequalities $\frac{1}{3}x-4\leq 7+2x\leq 4+\frac{1}{4}x$ in the form $p\leq x\leq q$ hence state the integral values (3mks)

11. Two similar cylindrical containers are such that the capacity of the larger container is 5 litres and that of the smaller is 320 millilitres. If the base area of the larger container is 0.25 m^2. Find the base area of the smaller container. (3mks)

12. Solve for x in the following equation (3mks)

$4^x(8^{x-1}) = \text{Tan } 45^0$

13. Use tables of logarithms to evaluate (3mks)

$$\frac{4.28\,X\,0.01677}{tan\ 20^O}$$

14. A solid block in the shape of a cylinder has a height of 14cm and weighs 22kg. If it is made of material of density 5g/cm^3, find the radius of the cylinder. Take $\pi = \frac{22}{7}$ (3mks)

15 Use the reciprocal, square and square-root tables to evaluate to 4 significant figures the expression.

$$\sqrt{\frac{1}{24.56} + 4.346^2}$$ (4mks)

16. Without using a calculator or tables, find the value of y given that $y = (a+b)(x-c)^2$ and $a = 5$, $b = 6$, $x = -3$ and $c = 2$. (3mks)

SECTION II

*(Answer **ANY FIVE** questions in the spaces provided)*

17. (a) A straight line L, whose equation is $3y - 2x = -2$ meets the x-axis at R. Determine the co-ordinates of R. (2 mks)

b) A second line L_2 is perpendicular to L_1 at R. Find the equation of L_2 in the form $y = mx + c$, where m and c are constants. (3 mks)

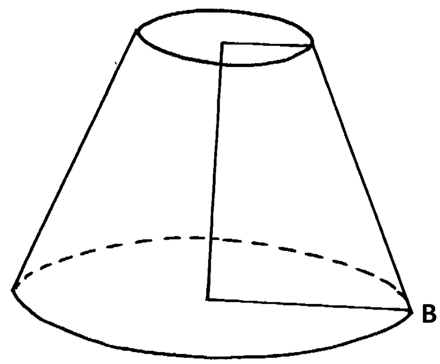
(c) A third line L_3 passes through (-4,1) and is parallel to L_2 Find:

(i) the equation of L_3 in the form $y = mx + c$, where m and c are constants

(2 mks)

(ii) the co-ordinates of point S, at which L intersects L (3 mks)

18 The diagram below represents a solid conical frustrum .Given that the height of the frustrum is 4cm, the radius of the base is 4.9cm and slant height of the frustrum is 2.1 cm.



a) Determine the height of the chopped off cone and hence the height of the bigger cone. (2mks)

b) Calculate the surface area of the solid. (4mks)

c) Calculate the volume of the solid. (4mks)

19. A matatu left Ravine for Torongo 51km away at an average speed of 48km/h at 7.00am. At 7.30am a Boda boda left Torongo for Ravine travelling along the same route at an average speed of 60km/h.Find:

(a) The time when Boda boda meet the matatu (3mks)

(b) How far from Ravine did the Boda boda meet the matatu (3mks)

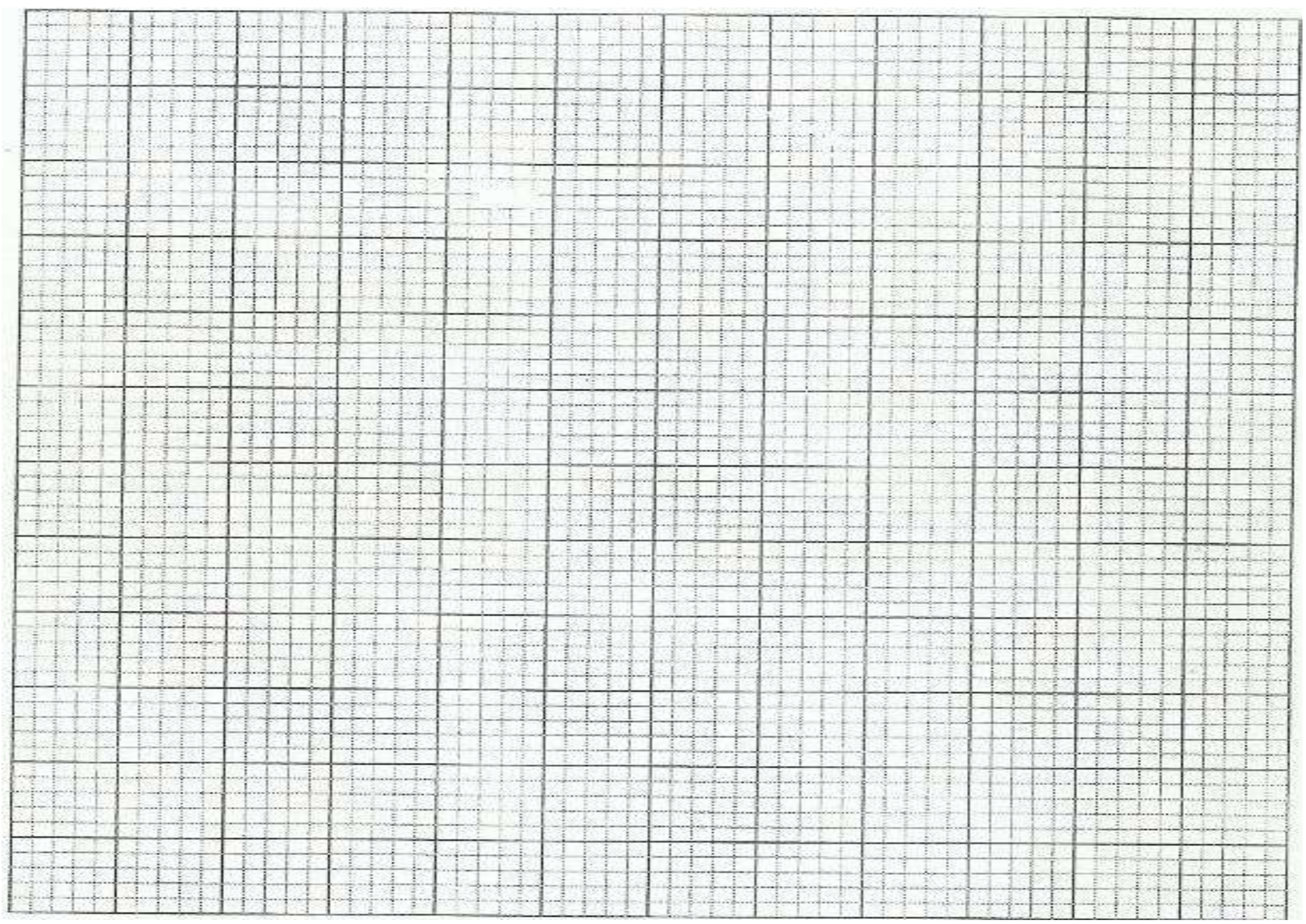
(c) After meeting the Boda boda the matatu stopped for fifteen minutes before resuming the journey. At what speed should it travel then to reach Torongo at the same time when the Boda boda reached Ravine (4mks)

20. The table shows the marks obtained by 40 candidates in an examination

Marks	5-14	15-29	30-34	35-44	45-49
Frequency	2	12	7	15	x

(a) Find the value of x (2mks)

(b) On the grid provided below draw a histogram to represent the data (5mks)



(c) By drawing a straight line on the graph above determine the median mark (3mks)

21. A certain number of people agreed to contribute equally to buy a book worth sh. 1,200 for a school library. Five people pulled out so the others agreed to contribute an extra sh. 10 each. Their contribution enabled them to buy a book worth sh. 200 more than they originally expected.

a) If the original number of people was x, write an expression of how much each was originally to contribute. (1mk)

b) Write down two expressions of how much each contributed after the people pulled out. (2mks)

c) Calculate how many people made the contribution. (5mks)

d) Calculate how much each contributed. (2mks)

22. Three planes P,Q and R left Jomo Kenyatta International Airport at 8.10a.m, 8:40a.m and 9.20a.m respectively. Plane P travelled at 300km/h along N70^0W, plane Q travelled at 240km/h along N67.5^0E and R travelled at 400km/h along 210^0.

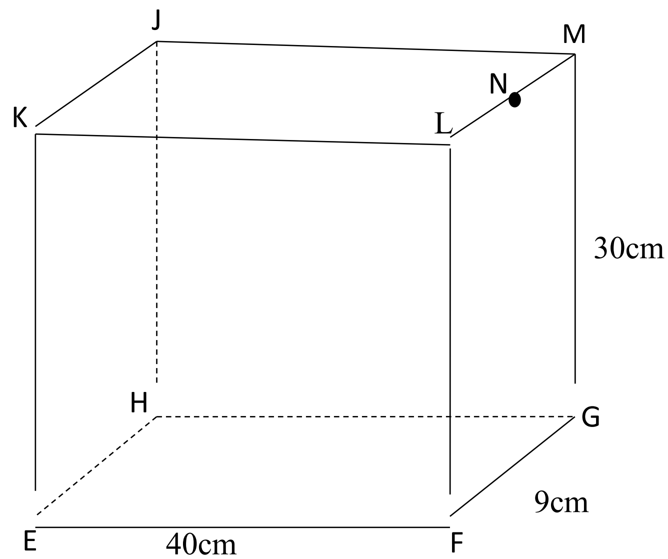
a) Using a scale of 1cm to represent 100km, locate the position of the planes at 10.50a.m. (6mks)

b) Find the distance of plane Q and R at 10.50a.m. (2mks)

c) Find the bearing of plane Q from plane P (1mk)

d) Find the bearing of plane R from plane Q. (1mk)

23. The figure below represents a cuboid EFGHJKLM in which EF =40cm, FG=9cm and GM=30cm. N is the midpoint of LM.



Calculate correct to 4 significant figures:

a) the length of GL; (1mk)

b) the length of FJ; (2mks)

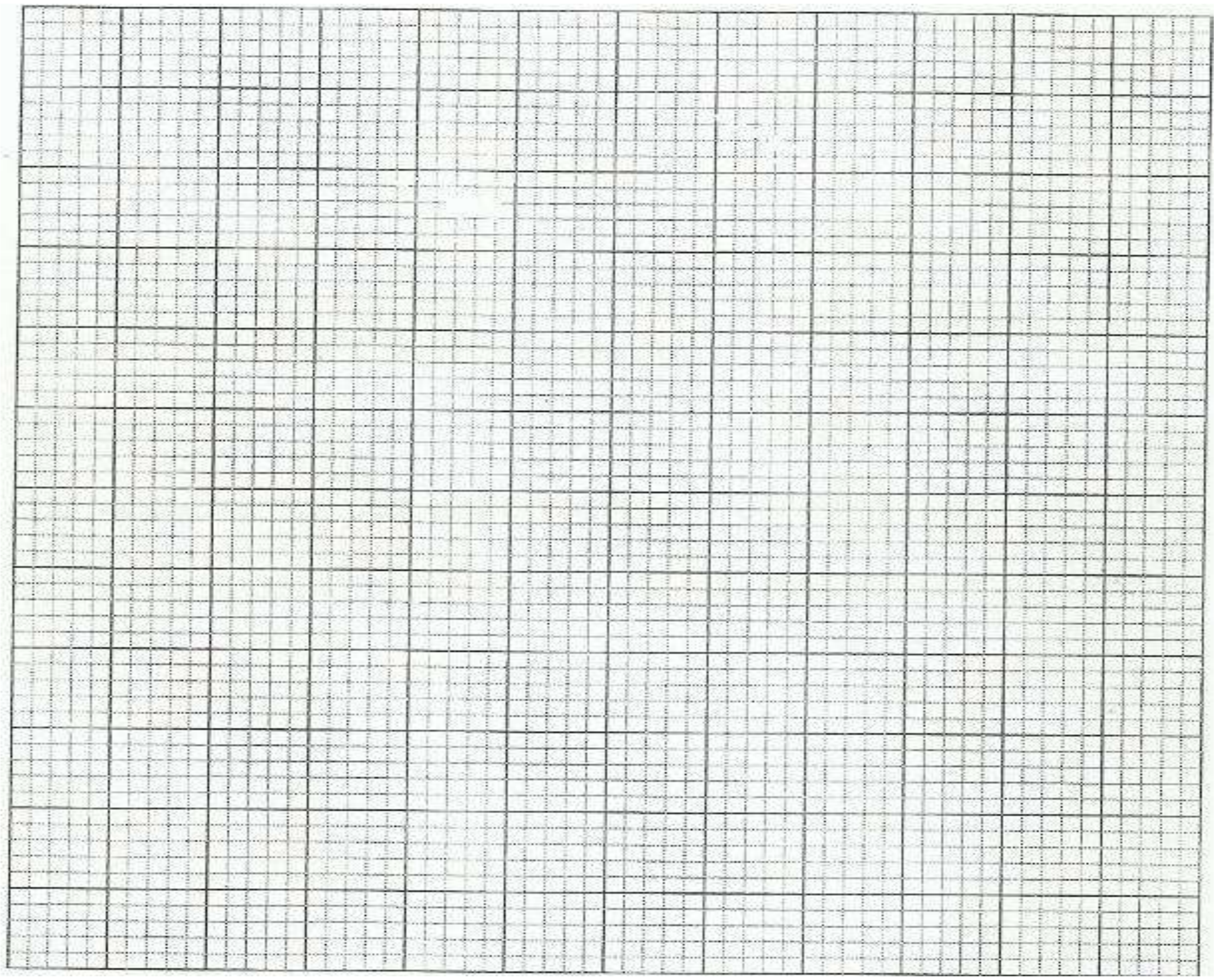
c) the angle between EM and the plane EFGH; (3mks)

d) the angle between the planes EFGH and ENH. (2mks)

e) the angle between the lines EH and GL. (2mks)

24. A triangle ABC with vertices A (-4,2), B(-6,6) and C(-6,2) is enlarged by scale factor -1 and centre (-2,6) to produce triangle $A^1B^1C^1$. Triangle $A^1B^1C^1$ is then reflected in line y=x to give triangle $A^{11}B^{11}C^{11}$.

a) Draw triangle ABC and its successive images on the grid provided. State the co-ordinates of $A^1B^1C^1$ and $A^{11}B^{11}C^{11}$. (6mks)



(b) If triangle $A^{11}B^{11}C^{11}$ is mapped onto a triangle whose co-ordinates are A^{111}(0,-2), B^{111}(4,-4) and C^{111}(0,-4) by a rotation, find the centre and the angle of rotation. (4mks)

KCSE TOP PREDICTION TRIAL 5 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… Adm Number…………………………..

Candidate's Signature ……………………..………………………….Date ……………………………………

INSTRUCTIONS TO CANDIDATES

a) Write your **Name, Index Number** and **School** in the spaces provided at the top of this page.
b) **Sign** and write the **date** of examination in the spaces provided above.
c) This paper contains **TWO** sections: section I and section II
d) Answer **all** the questions in section I and any FIVE questions from section II.
e) **All** answers and **working must** be written on the question paper in the spaces provided below each question.
f) Show **all** the steps in your calculations, giving your answers at each stage in the spaces provided below each question.
g) **Marks** may be given for **correct** working even if the answer is wrong.
h) Non-programmable silent electronic calculators and KNEC mathematical tables **may be** used except where stated otherwise.

FOR EXAMINER'S USE ONLY:

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

17	18	19	20	21	22	23	24	TOTAL

GRAND TOTAL

SECTION I: (50 MARKS)

Answer all the question in this section in the spaces provided:

1. Find the value of χ if $2^{\chi} = 3$. Give your answer correct to 4sf. (3mks)

2. Make χ the subject of the formula:

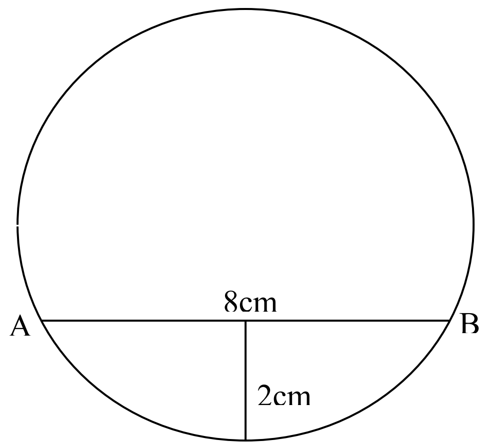
$$A = \sqrt{\frac{3 + 2\chi}{5 - 4\chi}}$$

(3mks)

3. It would take 18 men 12 days to dig a piece of land, if they work for 8 hours a day. How long will it take 24 men if they work 12 a day hours to cultivate three quarters of the same land. (3mks)

4. Kinyua bought soya and millet at sh.65 per kg and sh.40 per kg respectively. He then mixed them and sold the mixture at sh.60 per kg making a profit of 20%. Determine the ratio of soya to millet in mixture. (3mks)

5. Chord AB is of length 8cm and the maximum distance between the chord and lower part of circle is 2cm. Determine the radius of the circle. (3mks)



6. Use the inverse matrix method rule to solve the simultaneous equations. (3mks)
$2\chi + y = 10$
$2\chi + 2y = 14$

7. Solve $\log_2^{(x+7)} - \log_2^{(x-7)} = 3$ (4mks)

8. Construct a circle centre K and radius 2.5cm. Construct a tangent from a point Q which is 6cm from K to touch the circle at M. Measure the length QM. (3mks)

9. Given $4.6 \div 2.0$ find

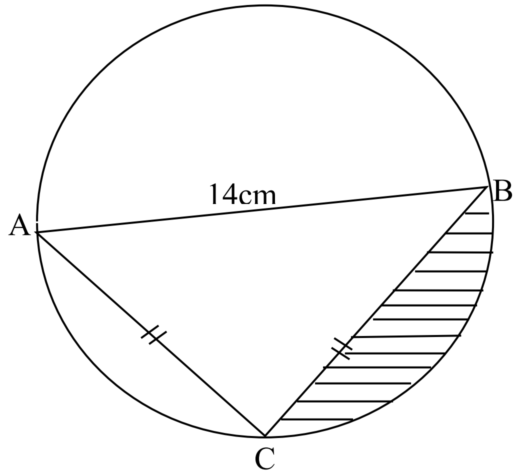
(a) the absolute error in the quotient. (2mks)

(b) the percentage error in the quotient correct to four significant figures. (1mk

10. Find the quartile deviation for the following set of data. (3 marks)
16, 42, 41, 6, 20, 28, 19, 23, 15

.

11. The figure below shows a circle with segments cut off by a triangle whose longest side AB is the largest possible chord of a circle. Determine the area shaded given that AB = 14cm and AC = BC. (3mks)

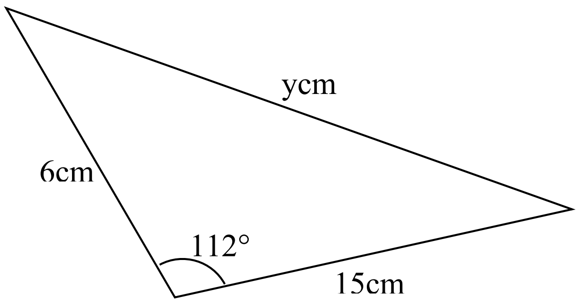


12. A circle of radius 7 units has it's centre at the point of intersection between the lines $\chi + 2y + 1 = 0$ and $2\chi + 3y - 3 = 0$. Find the equation of the circle expressing it in the form $\chi^2 + y^2 + g\chi + fy + c = 0$. (4mks)

13. Without using a Mathematical table or a calculator, evaluate. (2mks)

$$\frac{2.7 \times 2.04}{300 \times 0.054}$$

14. Find the length represented by y in the figure **below**. (3mks)



15. (a) Expand $(1 + 2\chi)^8$ in ascending powers of χ up to and including the term χ^3. (1mk)

(b) Hence evaluate $(1.02)^8$ to 3d.p. (2mks)

16. Under a transformation whose matrix $y = \begin{pmatrix} x-2 & -2 \\ x & x \end{pmatrix}$,
a triangle whose area is $13.5cm^2$ is mapped
on the a triangle whose area is $54cm^2$. Find two possible values of x. (3 marks)

SECTION II: (50 MARKS)

Answer any **five** questions from this section in the spaces provided:

17. (a) Fill the table below for the curves given by $y = 3 \sin (2\chi + 30°)$ and $y = \text{Cos } 2\chi$ for χ values in the range $O \leq \chi \leq 180°$. (2mks)

χ	0°	15°	30°	45°	60	75°	90°	120°	150°	180°
$y = 3 \text{ Sin } (2\chi + 30°)$										
$y = \text{Cos } 2\chi$										

(b) Draw the graphs of $y = 3 \text{ Sin } (2\chi + 30°) = \text{Cos } 2\chi$ on same axes. (2mks)

(c) Use your graph to solve the equation y = 3 Sin (2χ + 30°) and y = Cos 2χ. (2mks)

(d) Determine the following from your graph:

(i) Amplitude of y = 3 Sin (2χ + 30°). (1mk)

(ii) Period of y = 3 Sin (2χ + 30°) . (2mks)

(iii) Phase angle for y = 3 Sin (2χ + 30°). (1mk)

18. OAB is a triangle in which $\underset{\sim}{OA} = \underset{\sim}{a}$ and $\underset{\sim}{OB} = \underset{\sim}{b}$. M is a point on OA such that OM: MA = 2: 3 and N is another point on AB such that AN: NB = 1: 2. Lines ON and MB intersect at X.

(a) Express the following vectors in terms of $\underset{\sim}{a}$ and $\underset{\sim}{b}$.

(i) $\underset{\sim}{AB}$ (1mk)

(ii) $\underset{\sim}{ON}$ (1mk)

(iii) $\underset{\sim}{BM}$ (1mk)

(b) If $\underset{\sim}{OX} = K\underset{\sim}{ON}$ and $\underset{\sim}{BX} = h\underset{\sim}{BM}$ express $\underset{\sim}{OX}$ in two different ways. Hence or otherwise find the values of h and K. (6mks)

(c) Determine the ratio OX: XN. (1mk)

19. A trader bought 8 cows and 12 goats for a total of Ksh.294,000. If he had bought 1 more cows and 3 more goats he would have spend Ksh.337,500.

(a)Form two equations to represent the above information. (2 marks)

(b) Use matrix method to determine the cost of a cow and that of a goat. (4 marks)

(c) The trader sold the animals he had bought making a profit of 40% per cow and 45% per goat.

(i) Calculate the total amount of money he received. (2 marks)

(ii) Determine his profit in Kenya shillings. (2 marks)

20. Income tax is charged on annual income at the rate shown below

Taxable income (K£)	**Rate (shs per K£)**
1 – 1500	2
1501 – 3000	3
3001 – 4500	5
4501 - 6000	7
6001 – 7500	9
Over 7500	10

A civil servant earns a monthly basic salary of Ksh.8570. He is housed by the government and as a result, his taxable income is 15% more than his salary. He is entitled to a tax relief of Ksh. 150 per month.

a) How much tax does he pay in a year? (6mks)

b) From his salary, the following deductions are also made every month.

WCPS	2% basic salary
NHIF	Ksh 20

Calculate the civil servants net salary per month. (4mks)

21. A plane takes off from airport P at (0°, 40°W) and flies 1800 nautical miles due East to Q then 1800 nautical rules due South to R and finally 1800 nautical rules due West before landing at S.

(a) Find to the nearest degree the latitudes and longitudes of Q, R and S. (4mks)

(b) If the total flight time is 16 hours, find the average speed in knots for the whole journey. (3mks)

(c) Find the time taken to fly from R to S, given that this was two hours shorter than the time taken from P to Q to R. (2mks)

22. The 2nd and 5th terms of an arithmetic progression are 8 and 17 respectively. The 2nd, 10th and 42nd terms of the A.P. form the first three terms of a geometric progression. Find

(a) the 1st term and the common difference. (3mks)

(b) the first three terms of the G.P and the 10th term of the G.P. (4mks)

(c) The sum of the first 10 terms of the G.P. (3mks)

23. (a) The acceleration of a particle t seconds after passing a fixed point P is given by $a = 3t - 3$. Given that the velocity of the particle when $t = 2$ is 5m/s, find

(i) its velocity when $t = 4$ seconds. (3mks)

(ii) its displacement at this time. (3mks)

(b) Find the exact area bounded by the graph $\chi = 9y - y^3$ and the Y-axis. (4mks)

24) Two baskets X and Y contain identical balls except for the colours.Basket X contains 6 red balls and 3 black balls. Basket Y contains 2 red balls and 3 black balls.

(a) If a ball is drawn at random from each basket, find the probability that both balls are of the same colour. (4 marks)

(b) If two balls are drawn at random from each basket, one ball at a time without replacement, find the probability that:-

(i) The two balls drawn from basket X or basket Y are red. (4 marks)

(ii) All the four balls drawn are red. (2 marks)

KCSE TOP PREDICTION TRIAL 6 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… **Adm Number**……………………………..

Candidate's Signature ………………………………………………….**Date** ……………………………………

~~INSTRUCTIONS TO CANDIDATES~~

- *Write your **name**, **index number**, **signature** and **date** of the examination in the spaces provided*
- *The paper contains two sections. Section I and section II*
- *Answer **ALL** questions in section I and any five questions in section II*
- *Answers and working **must** be written on the question paper in the spaces provided below each question*

FOR EXAMINERS' USE ONLY

SECTION I

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL
MARKS																	

SECTION II

QUESTION	17	18	19	20	21	22	23	24	TOTAL
MARKS									

GRAND TOTAL

SECTION I (50 MKS) Answer all the questions in this section in the spaces provided.

1. Evaluate without using tables or a calculator the value of $\frac{1.33\times0.15}{0.19\times0.0017}$ **(3marks)**

2. When a certain number is divided by 48, 72 or 100 the remainder is 3 in each case. Find the number. **(3marks)**

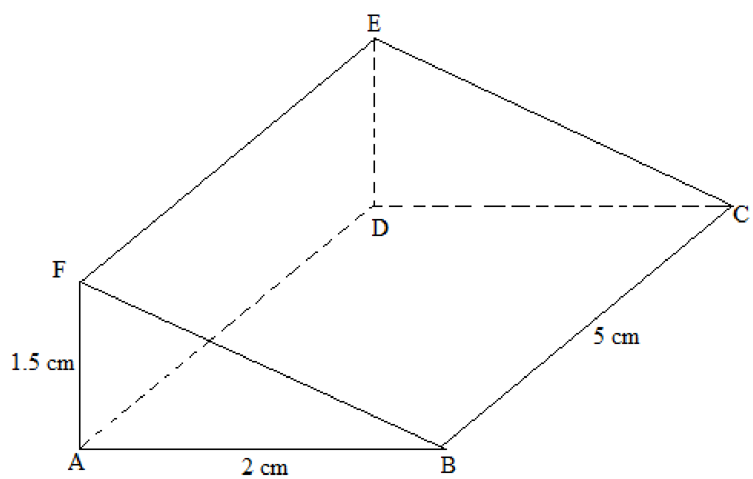
3. Using logarithm table evaluate $\sqrt[3]{\frac{849.6\times2.41}{3941}}$. **(4marks)**

4. A 890kg culvert is made of a hollow cylindrical material with outer radius of 76cm and an inner radius of 62cm. It crosses a road of width 3m. Determine the density of the material used in its construction in Kg/m^3 correct to one decimal place. **(3marks)**

5. Evaluate the value of x in $81^{x+1} + 3^{4x} = 246$. **(3marks)**

6. Simplify the equation $\frac{4x^2-9}{2x^2+5x+3}$. **(3marks)**

7. The diagram below represents a prism whose Cross-section is a right-angled triangle. Draw the net of the solid. **(3marks)**



8. During an annual general meeting at Elimu Mixed Day, goats and chicken were slaughtered. The number of heads for both chicken and goats were 45. The total number of legs were 100. Determine the exact number of goats and chicken slaughtered. **(3marks)**

9. In a mixed school there are 900 students, out of these 600 are girls.
   (a) Find the ratio of boys to girls. **(2marks)**

   (b) What is the percentage of boys in this school? **(1mark)**

10. Find the value of t in the equation $\frac{t-1}{3} - \frac{4+t}{4} = 0.$ **(3marks)**

11. The sum of interior angles of a polygon is 1980^0. Find the number of sides of the polygon and name the polygon. **(3marks)**

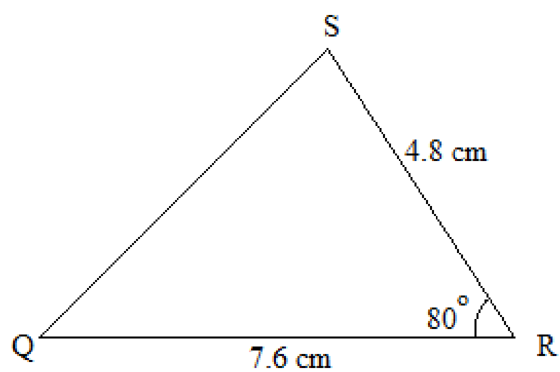
12. Solve the following inequalities and represent the solution on a single number line. **(3mks)**

3-2x<5

4-3x≥-8

13. Line K whose equation is $2y - 6 = 4x$ is perpendicular to another line Q. Find the equation of line Q if it passes through point. (-2, 7). **(3marks)**

14. The figure below is not drawn to scale.

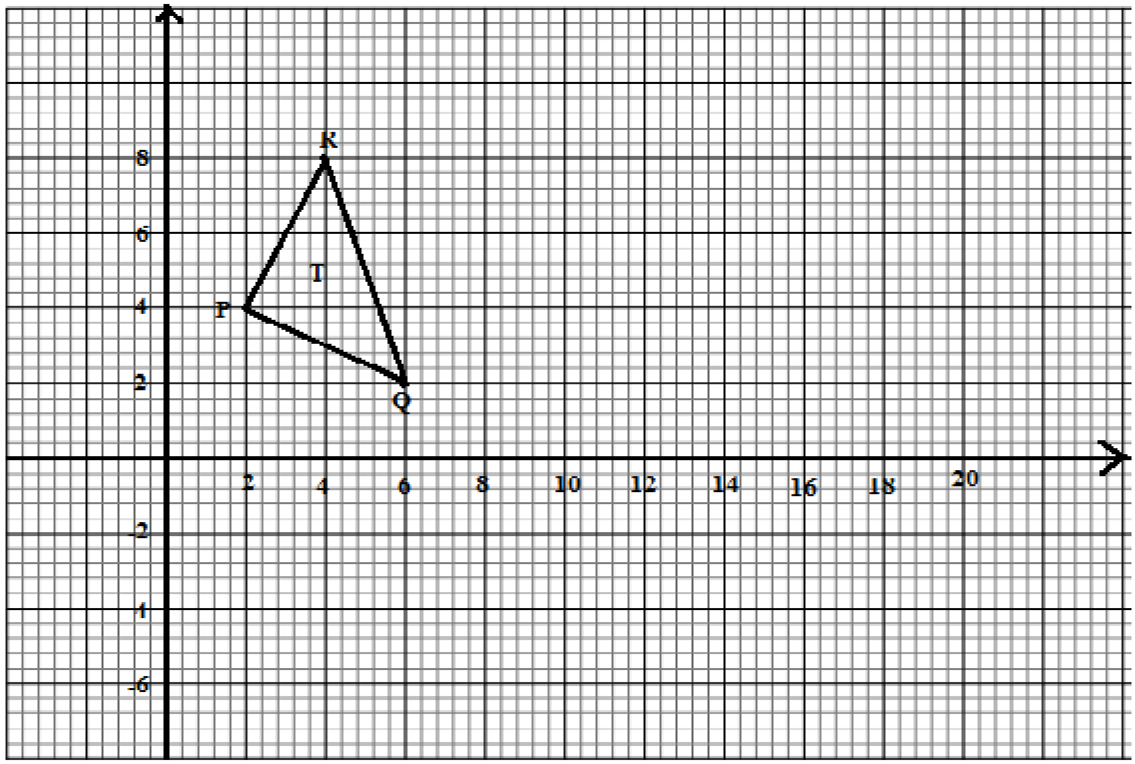


Find correct to 1 decimal place;
(a) Length PQ. **(2 marks)**

(b) Angle ABC **(2 marks)**

15. The marked price of a car ina dealers shop was Ksh.450 000. Kawira bought the car at 7% discount. The dealer still made a profit of 13%. Calculate the amount of money the dealer had paid for the car to the nearest shillings. **(3marks)**

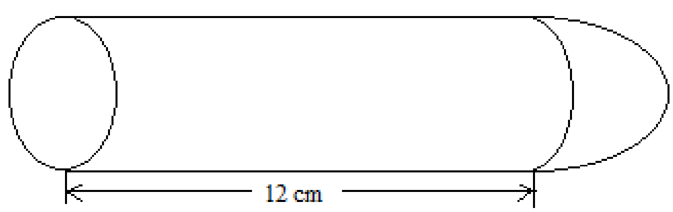
16. The figure below shows triangle T with vertices P(2,4), Q(6,2) and R(4,8). It is mapped onto triangle $T^{/}$ with vertices $P^{/}(10,0)$, $Q^{/}(8,-4)$ and $R^{/}(14, -2)$ by a rotation.
    (a) Draw on the same axis $T^{/}$ the image of triangle T. **(1 mark)**



(b) Determine the center and angle of rotation. (2marks)

SECTION II(50 marks). Answer *only five* questions in this section in the spaces provided.

17. The diagram below shows a solid made of a hemisphere and a cylinder. The radius of both the cylinder and the hemisphere is 3cm. The length of the cylinder is 12cm.



a) i) Calculate the volume of the solid. **(3marks)**

ii) The solid fits in a box in the shape of a cuboid 15 cm by 6cmby 6cm. Calculate thevolume of the box not occupied by the solid correct to four significant figures. **(2marks)**

b) i) Calculate the total surface area of the solid correct to four significant figures. **(3marks)**

ii) The surface of the solid is to be painted. One millilitre of paint covers an area of 8cm^2.The cost of paint is Ksh 900 per litre. Calculate the cost of the paint required. **(2marks)**

18. An expedition has 5 sections AB, BC, CD, DE and EA. B is 200m on a bearing of 050^0 from A. C is 500m from B. The bearing of B from C is 300^0. D is 400m on a bearing 230^0 from C. E is 250m on a bearing 025^0 from D.

(a) Sketch the route **(1 Mark)**

(b) Use the scale of 1cm to 50m to draw the accurate diagram representing the route. **(5 Marks)**

(c) Use your diagram to determine

(i) Distance in metres of A from E **(2 Marks)**

(ii) Bearing of E from A **(2 Marks)**

19. A trader bought 2 cows and 9 goats for a total of Ksh. 98,200. If she had bought 3 cows and 4 goats, she would have spent Ksh. 2, 200 less.

(a) Form two equations to represents the above information. **(2marks)**

(b) Use the matrix method to determine the cost of a cow and that of a goat. **(4marks)**

(c) The trader later sold the animals she had bought making a profit of 30% per cow and 40% per goats.

i) Calculate the total amount of money she received. **(2marks)**

ii) Determine corrects to 4 significant figures the percentage profit the trader made from the sale of animals. **(2marks)**

20. Two towns, Meru and Maua are 80km a part, Kimathi started cycling from Meru to Maua at 10:00 30a.m at an average speed of 40km/h. Mutuma started his journey from Maua to Meru at 10:30 a.m and travelled by car at an average speed of 60km/h.

(a) Calculate:

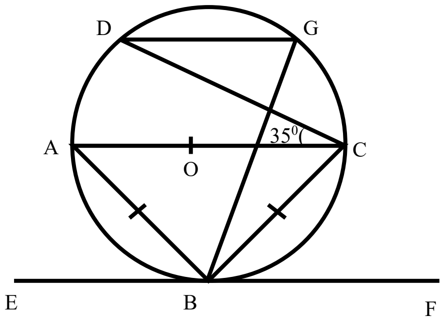
i) The time taken by Kimathi and Mutuma to meet. **(3marks)**

ii) The distance from Meru when Kimathi and Mutuma met. **(2marks)**

iii) The time of the day when the two met. **(2marks)**

(b) Murianki cycled from his home to a school 6km away in 20minutes. He stopped at the school for 5 minutes, before taking a motorbike to a town 40 km away. The motorbike travelled at 75km/h. On the grid provided, draw a distance time graph to represent Murianki's journey. **(3marks)**

21.In the figure below AOC is a diameter of the circle centre O; AB = BC and ∠ACD = 35^0. EBF is a tangent to the circle at B. G is a point on the minor arc CD.



Giving reason

(a) Calculate the size of

(i) ∠BAD **(3 Marks)**

(ii) The obtuse ∠BOD **(2 Marks)**

(iii) ∠BGD **(2 Marks)**

(b) Show that ∠ABE = ∠CBF **(3 Marks)**

22. a) Solve the equation $\frac{(x+3)}{24} = \frac{1}{(x-2)}$ **(4 marks)**

(b). The length of a floor of a rectangular hall is 9m more than its width. The area of the floor is 36m^2.

(i). Calculate the perimeter of the floor. **(4 marks)**

(ii). A rectangular carpet is placed on the floor of the hall leaving an area of 28m^2. If the length of the carpet is twice its width, determine the width of the carpet. **(2 marks)**

23. Using a pair of compass and ruler only construct.

(a) Triangle PQR in which PQ =5cm∠QPR = 30^0 and ∠PQR=105^0. **(3marks)**

(b) A circle that passes through the vertices of the triangle PQR. Measure its radius. **(3marks)**

(c) The height of triangles PQR with PQ as the base. Measure the height. **(2marks)**

(d) Determine the area of the circle that has outsides the triangle correct to 2 decimal places. **(2marks)**

24. (a). The equation of a line L_1 is $7y - 5x - 20 = 0 \cdot$ Find the x-intercept of the equation **(1mark)**

b). Another line L_2 is perpendicular to L_1 and passes through (-5, 3). Find the equation of L_2. **(3marks)**

c). L_3 passes through (0,-3) and parallel to the line L_4 whose equation is $3y - 8x = 3$ find the equation of L_3. **(3marks)** d). Calculate the coordinates of point of intersection between the lines L_1 and L_3. **(3marks)**

KCSE TOP PREDICTION TRIAL 6 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ………………………………………………………………………… Adm Number…………………………..

Candidate's Signature ………………………..……………………….Date ……………………………………

~~INSTRUCTIONS TO CANDIDATES~~

- *Write your **name**, **index number**, **signature** and **date** of the examination in the spaces provided*
- *The paper contains two sections. Section I and section II*
- *Answer **ALL** questions in section I and any five questions in section II*
- *Answers and working **must** be written on the question paper in the spaces provided below each question*
- *Show all steps in your calculation below each question*

FOR EXAMINERS' USE ONLY

SECTION I

QUESTION	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL
MARKS																	

SECTION II

QUESTION	17	18	19	20	21	22	23	24	TOTAL
MARKS									

GRAND TOTAL

SECTION I (50 MKS)

SECTION I (50 MARKS) Answer ALL questions in this section.

1. Make y the subject of the formula : $v = \frac{A\ \sqrt{x^2 + y^2}}{y}$ **(3 marks)**

2. Solve for x in the equation: $Log_8(6 - 2x) - Log_8(x - 2) = {}^{-1}/_{3}$ **(3 marks)**

3. Three quantities P, Q and R are such that P varies directly as the cube of Q and inversely as the square root of R. Find the percentage change in P if Q decreases by 10% and R decreases to 36%.

(3 marks)

4. A circle with its Centre at O (-2, 0) passes through a point (1, 4). Write down the equation of the circle in the form $x^2 + y^2 + ax + by + c = 0$ **(3 marks)**

5. Express: $\frac{COS\ 30^0}{\tan 45^0 + \sqrt{3}}$ in surd form and simplify by rationalizing the denominator

(3 marks)

6. In a transformation, an object with area 9cm² is mapped onto an image whose area is 54cm². Given that the matrix of transformation is $\begin{pmatrix} x & x-1 \\ 2 & 4 \end{pmatrix}$ Find the value of X **(3 marks)**

7 a) Find the binomial expansion of $\left(1+\frac{1}{2x}\right)^7$ up to the 4th term with increasing powers of x. **(2 marks)**

b) Hence estimate the value of $(1.05)^7$ correct to 4 decimal places. **(2 marks)**

8. A computer with a marked price of Kshs. 40000 can also be bought on hire purchase terms. Kibet bought the computer on hire purchase by making a deposit of Ksh 10000 and cleared the balance with equal 12 monthly installments of KShs. 3500 each. Determine the monthly interest rate of hire purchase to 2 significant figures. **(3 marks)**

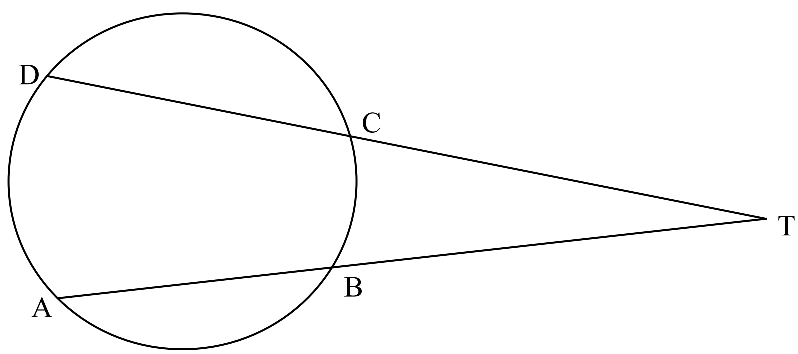
9. A dealer blends Kericho tea that costs sh 300 per 100g packet with Ketepa that costs ksh 160 per 200g packet. In what ratio must the dealer mix the two so that by selling a 100g packet of the blend for ksh 250, a profit of 25% is made? **(4 marks)**

10. Solve for x in the equation: $Sin\ 2\theta = -0.5$ for $0^0 \leq \theta \leq 360^0$ **(3 marks)**

11. Use completing of the square method to solve: $2x^2 - 6x + 4 = 0$ **(3 marks)**

12. A carpenter wishes to cut a 1m long piece of wood into 3 equal parts. He approximates the length of each piece to 0.33m. Calculate the percentage error due to this approximation **(3 marks)**

13. Calculate length DT in the figure below given AT and DT are secants that intersect at T and AB = 5 cm, BT = 4 cm and DC = 9 cm. **(3 marks)**



14. An arc of a circle of diameter 7 cm subtends an angle of 1.2 radians to the Centre of the circle. What is the length of the arc? **(2 marks)**

15. Given matrix $A = \begin{pmatrix} x - 2x & x - 4 \\ x & 2 \end{pmatrix}$. Find the possible values of x if A is a singular matrix.

(3 marks)

16. Working together two taps A and B can fill a tank in 2 hours. By itself tap A can fill the tank in 6 hrs. Tap A and B are opened at the same time and after running for 1 hour, an outlet taps which can drain the full tank by itself in 12 hours is opened and Tap A is closed. Find the total time taken to fill the tank. **(4 marks)**

SECTION II – 50 MARKS **Answer only FIVE questions from this section.**

17. In a form 4 class there are 24 girls and 30 boys. The probability that a girl participates in games is 0.4 whereas that of a boy is 0.6.

a) A student is picked at random from the class. Find the probability that the student picked:

i) Is a boy and will participate in games **(2 marks)**

ii) Will not participate in games **(2 marks)**

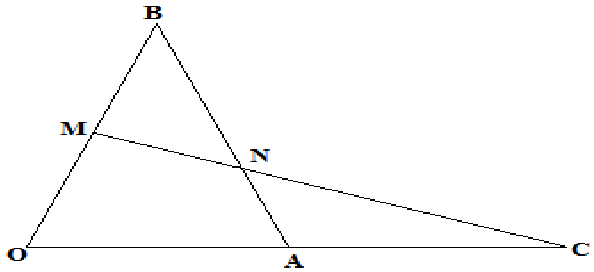
b) Two similar bags A and B are such that, bag A contains 3 blue balls and 7 red balls while bag B contains 4 green balls and 8 blue balls. The balls are similar in shape and size. A ball is drawn from bag A and bag B. Determine the probability that:

i) One of the balls drawn is green. **(2 marks)**

ii) The balls drawn are of the same colour. **(2 marks)**

iii) None of the balls drawn is red **(2 marks)**

18. In the triangle OAB below, **OA** = **a**, **OB** = **b** and 3**OA** = 2**OC**. M divided **OB** in the ratio 1:1 .



a) Express in terms of **a** and **b** only, the vectors

i) **BA** **(1 mark)**

ii) **OC** **(1 mark)**

iii) MC **(1 mark)**

b) Given further that **MN** = h**MC** and BN = k**BA**,

i) Express vector **MN** in two different ways in terms of **a**, **b**, h and k **(2 marks)**

ii) Find the value of h and k. **(4 marks)**

c) State the ratio **MN:MC** **(1 mark)**

19. A group of people planned to contribute equally towards buying a plot valued at Ksh. 270,000. After the contribution, 2 members joined of the group. As a result, a refund of Ksh 1500 was made to the members who had already contributed.

a) Let the original number of people be x, express in terms of x:

i) The original contribution **(1 mark)**

ii) The new contribution **(1 mark)**

b) Find how much each member contributed finally **(5 marks)**

c) Calculate the percentage decrease in the contribution per person caused by the members who joined. **(3 marks)**

20. The table shows income tax rates for a certain year.

Monthly taxable pay K£	Rate of tax in Ksh. Per K£
1- 434	2
435 - 866	3
867- 1298	4
1299 - 1730	5
Over 1730	6

A company employee earns a monthly basic salary and is also given taxable allowances amounting to Ksh 10480. If the employees' tax on the 5th band is Ksh 3420,

a) Calculate the employee's monthly taxable income tax in K£. **(2 marks)**

b) The employee is entitled to a personal tax relief of Ksh. 1056 per month. Determine the net tax. **(4 marks)**

c) In a certain month, the employee received a 25% increment in his basic salary. Calculate his net monthly pay. **(4 marks)**

21.(a) In a geometrical progression the sum of the second and third term is 12 and the sum of the third and fourth terms is -36. Find the first term and the common ratio. **(4 marks)**

(b) In an arithmetic progression the 12^{th} term is 25 and the 7^{th} term is three times the second term, find;

i) The first term and the common difference **(4 marks)**

iii) The sum of the first 10 terms of the arithmetic progression. **(2 marks)**

22 a). The table below shows the frequency distribution of marks scored by students in a test.

Marks	1-10	11-20	21-30	31-40	41-50
Frequency	2	4	8	4	2

On the grid provided, draw a cumulative frequency curve for the data. ***(3 marks)*** b). Use your graph to determine;

(i). The pass mark if only 6 students passed the exam. ***(2 marks)*** (ii). The upper quartile mark *(1 mark)*

c). Find the percentage change if the upper quartile in b(ii) above was found by calculation. **(3 marks)**

23 a) Complete the table below for $y = -2\ sin\ x$ and $y = 3\ cos\ x$. **(2 marks)**

x	0°	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	360°
-2 Sin x	0.00		-1.73			-1.00	0.00			2.00	1.73		
3 Cos x	3.00		1.50			-2.60	-3.00			0.00	1.50		

b,) Draw the graph $y = -2\ Sin\ x$ and $y = 3\ cos\ x$ using 1 cm to represent 30° horizontal axis and 2 cm to represent 1 unit on the vertical axis. **(5 marks)**

c) Use the graph to solve:

i) $3 \text{ Cos } x + 2\sin x = 0$ **(1 mark)**

ii) $1 - 2\sin x = 2$ (2 marks)

24. A triangle **ABC** with vertices at **A (1,-1)** ,**B (3,-1)** and **C (1, 3)** is mapped onto triangle $\mathbf{A^1B^1C^1}$ by a transformation whose matrix is $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.Triangle $\mathbf{A^1B^1C^1}$ is then mapped onto $\mathbf{A^{11}B^{11}C^{11}}$ with vertices at $\mathbf{A^{11}}$ **(2, 2),** $\mathbf{B^{11}}$ **(6, 2)** and $\mathbf{C^{11}}$ **(2,-6)** by a second transformation.

(i) Find the coordinates of $\mathbf{A^1B^1C^1}$ ***(3 marks)***

(ii) Find the matrix which maps $\mathbf{A^1B^1C^1}$ onto $\mathbf{A^{11}B^{11}C^{11}}$. ***(3 marks)***

(iii) Determine the ratio of the area of triangle $\mathbf{A^1B^1C^1}$ to triangle $\mathbf{A^{11}B^{11}C^{11}}$. ***(1 mark)***

(iv) Find the transformation matrix which maps $\mathbf{A^{11}B^{11}C^{11}}$ onto **ABC** ***(3 marks)***

KCSE TOP PREDICTION TRIAL 7 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ……………………………………………….…………………… Adm Number…………………………..

Candidate's Signature ………………………..………………………….Date ……………………………………

~~INSTRUCTIONS TO CANDIDATES~~

1. Write your name, class and admission number in the spaces provided above
2. Sign and write the date of examination in the spaces provided above.
3. This paper consists of **two** sections I and II.
4. Answer all the questions in section **I** and only **five** in questions from section II.
5. All answers and working must be written on the question paper in the spaces provided below question.
6. Show all steps in your calculations, giving your answers at each stage in the spaces provided.

FOR EXAMINER' S USE ONLY

SECTION 1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

SECTION II

17	18	19	20	21	22	23	24	TOTAL

SECTION I(50 MARKS)

ANSWER ALL THE QUESTIONS IN THIS QUESTION

1. Without using a calculator or a mathematical table, evaluate. (3marks)

$$\frac{\frac{1}{2}\ of\ 3\frac{1}{2}+1\frac{1}{2}\left(2\frac{1}{2}-\frac{2}{3}\right)}{\frac{3}{4}\ of\ 2\frac{1}{2}\div\frac{1}{2}}$$

2. Three bulbs are programmed at intervals of 35, 45 and 65 seconds. What is the earliest time they will light up simultaneously if the last time they did this was at 10.17am? (3marks)

3. Two interior angles of an irregular polygon are each 144^0 the rest are 132^0.Find the number of sides of the polygon. (3marks)

4. Use logarithm tables to evaluate. (4marks)

$$\frac{\sqrt{10332\times(31.4)^{\frac{1}{2}}}}{0.235Log8}$$

5. A Kenyan bank buys and sells foreign currencies using the rates shown below.

Buying	Selling
(Ksh)	(Ksh)
1 Euro 86.25	86.97
100 Japanese Yen 66.51	67.26

A Japanese traveling from France arrives in Kenya with 5000 Euros, which he converts to Kenya shillings at the bank. While in Kenya he spent a total of Ksh.289,850 and then converted the remaining Kenya shillings to Japanese Yen at the bank. Calculate the amount of Japanese Yen that he received. (3 mks)

6. Find all the integral values which satisfy the inequality. (3marks)

$$-3-x \leq \frac{2}{3}x+2\angle\frac{1}{3}x+4$$

7. Simplify the expression: (3marks)

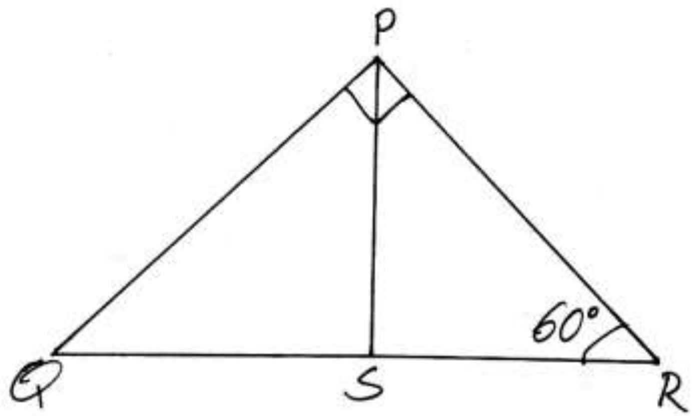
$$\frac{12x^2-16xy+4y^2}{2y^2-18x^2}$$

8. Solve the following equation. (4marks)

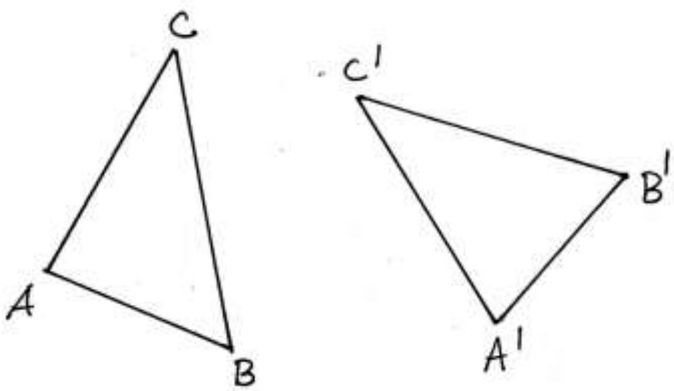
$$\frac{343\times 7^{2x}}{2401\times 7^{x}\div 7^{x-1}}$$

9.Some mangoes are to be distributed among some children. If each receives 8mangoes, there will be 2 mangoes left. If 8 mangoes are added to the original number of the mangoes and redistributed to the children, each will receive 9 mangoes. Find the number of children. (2marks)

10.The figure below represents a right angled triangular plot of land PQR.PS is perpendicular to QR. If angle PQR=60^0 and PS is 6.5cm.Find the area of the plot. (Answer in 1 d.p) (3marks)



11.Given triangle $A^1B^1C^1$ is an image of ABC after rotation .Find the centre and angle of rotation. (3marks)



12. Find the integral values of x for which: (3mks)

$-6 \leq 3x + 3$

$14 – 3x > 2$

13. One diagonal of a rhombus is three quarters the length of the other diagonal, if the rhombus has an area of $96cm^2$ .Find the perimeter of the rhombus. (3marks)

14. A conical flask with radius 10cm and height of 15cm is half full of water. A stone was placed inside and water rose to ¾ of the flask. Find the volume of the stone to 3 s.f. (4marks)

15. It would take 30men, 16days to dig a trench 480m long. Find how many more men would be required to dig 720m long trench in 20 days. (3marks)

16. Given that A (-18, -19),B (-2, - 1) and C (18, 9), express AB and BC as column vectors. Hence show that the points A, B and C are collinear. (3marks)

SECTION B (50MARKS)

ANSWER ANY FIVE QUESTIONS FROM THIS SECTION

17. The line L_1 whose equation is 2x+hy =k cuts x-axis at x=6 and cuts the y-axis at x=12, Find;

(a) The values of the constant h and k. (3marks)

(b) The gradient of the line L_1. (1mark)

(c) Equation of line L_2 which is perpendicular to L_1 and passes through point (-2,-5). Giving your answer in double intercept form. (3marks)

(d) The coordinates of the point of intersection of the two lines. (3marks)

18. A certain matatu ferrying tourists through a National park has a capacity of 43 passangers.It charges ksh 170 per trip. The vehicle however can overload by 5 more passengers but will have to pay an extra charge of ksh 200 in every of five view points along the route .The matatu consumes 1.5litres of petrol per every kilometer and pump price is ksh 40 per litre .The distance covered in each trip is 40 km.

(a)Calculate the amount the matatu owner makes in a round trip when the matatu is not overloaded. (2mks)

(b) How much is lost by overloading the vehicle in each trip. (2marks)

(c) Express as a percentage the profit when overloaded over the profit made at capacity in a single trip (give your answer in1 d.p) (3marks)

(d) If the fuel price is increased by ksh 30 per litre. What is the reduction on profits on normal capacity for a single trip. (3marks)

19. The height of 36 students in a class were recorded to the nearest centimeter as follows.
147, 168, 157,172, 165, 154, 170,157,167
161,160,157,165,165,175,173,172,178
147,168,157,172,165,154,170,157,167
155,159,173,168,160,172,156,167,171

(a) Make a frequency distribution table using a class interval of 5 starting with the class 145-149. (2marks)

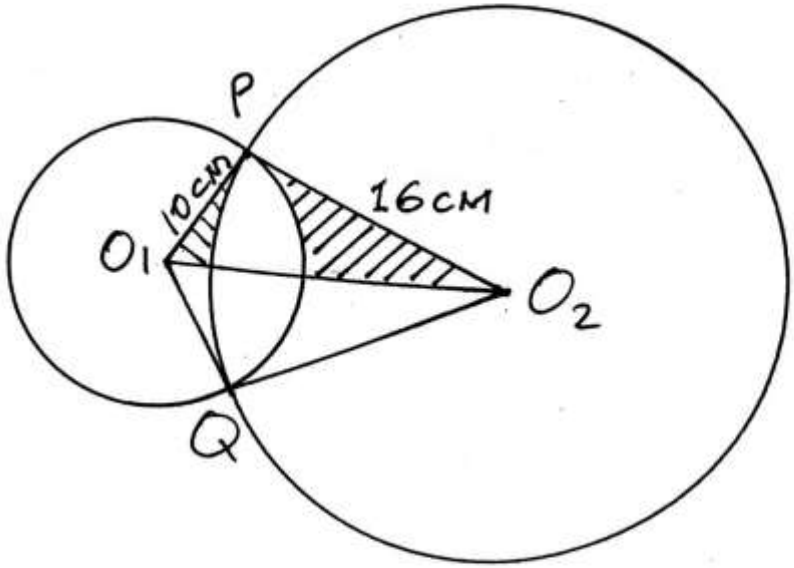
(b) From the table above, calculate;

(i) The mean height (3marks)

(ii) The median height. (3marks)

(c) Draw a frequency polygon to represent the data. (2marks)

20. In the figure below O_1 and O_2 are the centre of the circles whose radii are 10cm and 16cm respectively. The circles intersect at P and Q and angle $PO_1O_2=32^0$ .Calculate the area of the;



(a) Sector

(i) PO_1Q (2marks)

(ii) PO_2Q (2marks)

(b) Calculate the area of intersecting region. (3marks)

(c) Calculate the area of shaded region. (3marks)

21. B is 102km on the bearing of 112° from A. C is 94km on the bearing of 062° from B D is 073° from A and 336° from C.

(a) Using a scale of 1cm to represent 20km,draw a diagram to show the positions of A,B,C and D. (6mks)

b) Using your diagram, determine;

i) The bearing of B from D and A from C. (2mks)

ii) The distance AC and BD (2mks)

22. A bus and a car left Kisumu for Nairobi a distance of 600km at 6.00am.The bus travelled at an average speed of 80km/hr while the car travelled at an average speed of 120km/hr. After 48minutes , the car developed a mechanical problem which took 54minutes to fix.

(a)How long did the car wait before the bus got to where it developed mechanical problem? (2marks)

(b) Find how far from Kisumu the car caught up with the bus. (3marks)

(c) At what time of the day did the car caught up with the bus. (3marks)

(d) How long did the car wait before the bus arrived in Nairobi. (3marks)

23. From points A and B on the same side of abuilding and on a levelground, the angles of elevation of the top of the building are 24^0 and 38^0 respectively .If the distance between A and B is 57m.

(a) Form two expressions for the height of the building. (2marks)

(b) Calculate to the nearest 1 decimal place.

(i) The height of the building. (4marks)

(ii)Find the difference in the distance between the top of the building and the points A and B. (4marks)

24. The displacement S meters of a moving particle after t seconds is given by $s=2t^3-5t^2+4t+2$.
Determine

(a) The velocity of the particle at t=2seconds. (2marks)

(b) The value of t when the particle is momentarily at rest. (3marks)

(c) The displacement when the particle is momentarily at rest. (3marks)

(d) The acceleration of the particle when t=5 seconds. (2marks)

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KCSE TOP PREDICTION TRIAL 7 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name ………………………………………………………………………… **Adm Number**……………………………..

Candidate's Signature ……………………..…………………………….**Date** ………………………………………

INSTRUCTION TO CANDIDATES

a) *Write your name and Admission number in the spaces provided above.*
b) *Sign and write the date of examination in the spaces provided above.*
c) *The paper consists of **two** sections. **Section I** and **Section II.***
d) *Answer **ALL** the questions in Section I and any **FIVE** questions in Section II.*
e) *Show all the steps in your calculations, giving your answer at each stage in the spaces provided below each question.*
f) *Marks may be given for correct working even if the answer is wrong.*
g) *Non-programmable silent electronic calculators and KNEC Mathematical tables may be used except where stated otherwise.*

FOR EXAMINER'S USE

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

SECTION II

17	18	19	20	21	22	23	24	Total

SECTION I (50MARKS)

Answer all questions in this section.

1. The third term of a geometrical progression is 2 and the fifth term is 18. Find the possible values of the common ratio and the second term in each case. (4marks)

2. A plot of land was valued at sh. 1.5 million at the start of 2008. The plot appreciated by 18% during 2008. If in every year after 2008 it appreciated by 12% of its previous year's value. Find

a) The plot's value at the start of 2009. (1mark)

b) The plot's value at the end of 2012. (2marks)

3. A bag contains 3 yellow, 4 black and 5 green beads. Two beads are drawn at random from the bag, one after the other without replacement. Find the probability that at least one is yellow. (2marks)

4. The equation of a circle is $x^2 + y^2 - 2x - 6y + 5 = 0$. Find the centre and radius of the circle. (3marks)

5. In a school trip there were x buses and y luxury vans. Each bus was hired at Ksh. 1000 and could carry 60students. Each van was hired at Ksh. 2000 and could carry 30 students. Express the following statements as inequalities in x and y. (4marks)

i) There must be vans.

ii) There should be at least 3 buses.

iii) The school should not spend more than Ksh. 18,000 on the trip.

iv) Not more than 420 students are to go on the trip.

6. The position of an airport P and Q are (60^0N 45^0W) and Q (60^0N K^0E). It takes a plane 5 hours to travel due east from P to Q at an average speed of 600 knots. Calculate the value of K. (3marks)

7. Without using tables or a calculator, evaluate $\frac{Sin150^o - Cos330^0}{\tan 420}$ leaving your answer in the form $a\sqrt{b} + c$ where a,b and c are constants. (3marks)

8. T in a transformation A represented by the matrix $\begin{pmatrix} 5x & 2 \\ -3 & x \end{pmatrix}$ under T, a square of area $10cm^2$ is mapped onto a square of area $110cm^2$. Find the value of x. (3marks)

9. a) Draw a line MN = 7cm and show the locus of a point P which is such that MPN = 90^0. (2marks)

b) On the locus of P in the diagram in (a) above, locate the locus of T which is such that it is equidistant from M and N. (1mark)

10. Point T divides the line MN internally in the ratio 3:2 taking the position vectors of M and N to be $M=\begin{pmatrix}8\\-12\\15\end{pmatrix}$ and $N=\begin{pmatrix}5\\-6\\0\end{pmatrix}$ respectively. Find the coordinates of T. (2marks)

11.a) Expand $(3 + x)^7$ in ascending power of x up to the term in X^4. (1mark)

b) Use the expansion above (a) to find the value of $(2.97)^7$ correct to 3 d.p. (2marks)

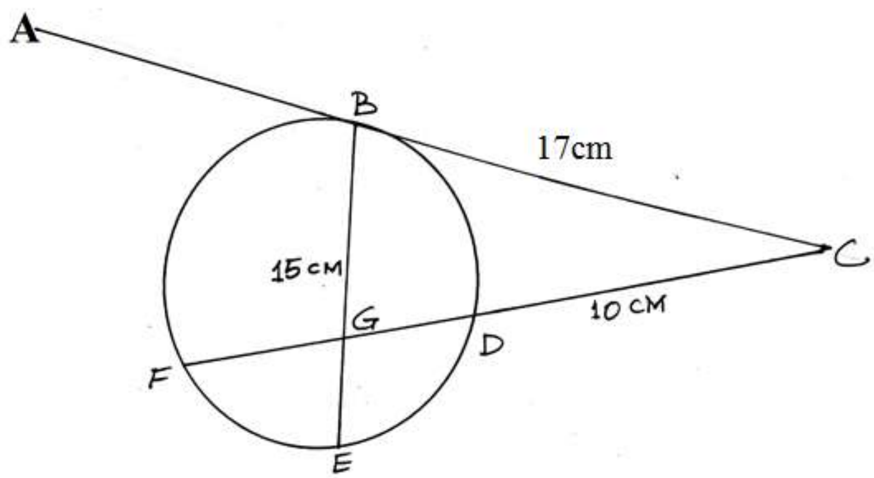
12. Two taps A and B, when opened at the same time can fill a tank in 3 hours and 36minutes. Tap A, working alone takes 3 hours longer than B to fill the tank. How many hours does it take tap A to fill the tank. (3marks)

13.a) ABC is a triangle in which BC = 17cm, CA = 9cm, AB = 10cm. Without using tables or calculators, calculate as fraction in simplest form Cos A. (2marks)

b).Calculate also without using tables or calculators, the area of triangle ABC. (2marks)

14. If x is an angle in the first quadrant such that $18 \cos^2 x + 27 \sin x = 25$. Find tan x without using tables or calculators. (3marks)

15. In the figure ABC is a tangent to circle BDEF at B. BC = 17cm, DC = 10cm and BG = 15cm. Chord FD intersect chord BE at G and FG: GD = 1:5.



Find the length of

a) FD (1mark)

b) GE (2marks)

16. The following table has some values of x and y for the function $y = x^3 + 3$ for the interval $-3 \le x \le 2$

x	-3	-2.5	-2	-1.5	-1	-0.5	0	0.5	1	1.5	2
y	12	9.25	7	5.25	4	3.25	3	3.25	4	5.25	7

Use the values to draw the graph of the function $y = x^3 + 3$ on the grid provided. (4marks)

SECTION II (50MARKS)

Answer ANY FIVE questions only.

17.a) The mass of a cylinder of a given material varies jointly as the square of the radius and as the height. If the radius is increased by 20% and height by 10%. What is the percentage increase in mass? (3marks)

b) Several companies were invited to an inspirational talk by a popular businessman. The charge m, in shillings per person for attending the talk is partly fixed and partly inversely proportional to the total number of people p, in a company that will attend the talk.

i) Express m in terms of p. (1mark)

ii) 60 people from company A attended the talk and their total charge was sh. 30,000. 80 people from company B attended the talk and their total charge was sh. 36,000. What was the charge per person for company C if 75 people from the company attended the talk? (4marks)

iii) When a person that had already paid the full amount decided not to attend and requested for a refund, he/she was only to be refunded the fixed charge. 25% of some people from company D requested for a refund after they decided not to attend the talk. The talk organizers refunded them and were left with 22,800. Determine the number of people from the company D that had initially paid the attendance. (2marks)

18.a) Solve for A in $\frac{1}{\cos A} = 2\cos A \quad for\ 0^0 \leq A \leq 360^0$ (2marks)

b) Complete the table below and use it to answer the questions below. (2marks)

x	-180	-150	-120	-90	-60	-30	0	30	60	90	120	150	180
2cosx						1.73				0			
Sin2x						-0.87				0			

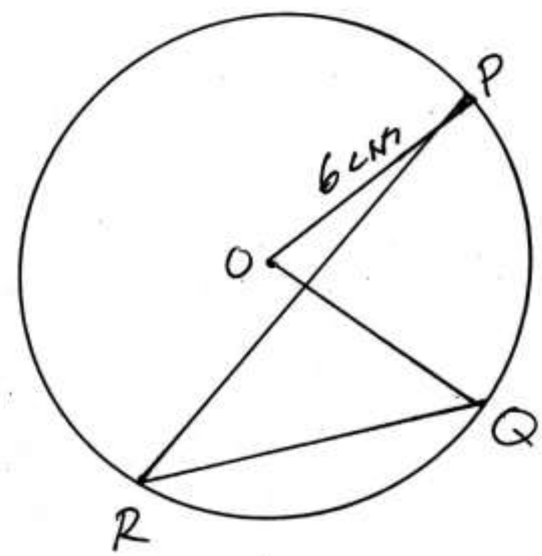
i) Draw the graph of y = sin 2x and y = cos x on the same set of axes using a scale of 1cm for 30^0 on x-axis and 2cm for 1unit on y-axes.

ii) Estimate the values of x for which

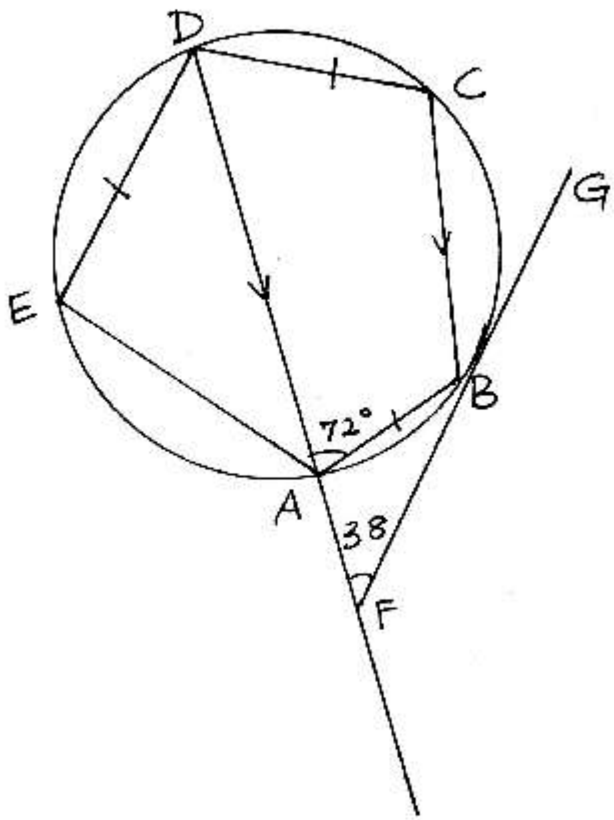
a) $2 \cos x - \sin 2x = 0$ (1mark)

b) $2 \cos x = - 0.6$ (1mark)

19.a) The figure below is a circle centre O and radius 6cm. if the area of the sector OPQ = 4π cm^2. Calculate the size of angle PRQ. (2marks)



b) In the figure below ABCDE is a pentagon inscribed in a circle centre B. CB is parallel to DAF and AB = CD = DE. Angle BAD = 72^0 and BFA = 38^0. Giving reasons in each case, determine the values of the following angles.



i) ABF (2marks)

ii) BDA (2marks)

iii) BCD (2marks)

iv) DAE (2marks)

20.a)A particle starting from rest at a point O moves in a straight line such that its velocity V m/s after t,seconds is given by $V = 6t - 2t^2$. Calculate the distance travelled by the particle between t = 2 and t = 5. (3mks)

b) The acceleration of a particle a m/s^2 of an object that starts moving from rest at time t seconds is given by a = 2t – 6. If the initial velocity is 15m/s.

i) Give an expression for the objects velocity. (2marks)

ii) Find the velocity at t = 3sec. (1mark)

iii) Give an expression in terms of t for the object's displacement. (1mark)

iv) Determine its total displacement in 2^{nd} and 3^{rd} seconds. (2marks)

c) Find the minimum velocity

y that the object attained. (1mark)

21. The following table shows the ages in a years of 70 people that attended a conference.

Age in years	30-34	35-39	40-44	45-49	50-54	5-59	60-64	65-70
Frequency	3	5	7	9	11	15	14	6

a) Calculate

i) Mean (2marks)

ii) Standard deviation (2marks)

b) Draw an ogive for the data. (3marks)

c) Use your graph to find

i) Median (1mark)

ii) The inter quartile range. (1mark)

iii) The 35^{th} percentile (1mark)

22. A tetrahedron has equilateral triangular base ABC of side 10cm. The vertex V is such that VA= VB = VC= 8cm. Calculate

a) The height of the pyramid. (2marks)

b) The angle between the planes ABC and BCV. (2marks)

c) The angle between AV and the base ABC. (2marks)

d) The angle between planes ACV and BCV. (3marks)

e) The volume of the pyramid. (1mark)

23. The table below shows income tax rules that were used in 2023.

Income (sh per month)	Rate %
1-10164	10
10165 – 19740	15
19741 – 29316	20
29317 – 38892	25
Over 38,892	30

From January to June that year Janet's monthly taxable earnings amounted to sh. 32,500.

a) Find the gross tax on her monthly earnings. (3marks)

b) Janet was entitled to the following tax reliefs.

Monthly personal relief of sh. 1,162

Monthly insurance relief at the rate of 15% of the premium paid subject to a maximum relief amount of sh. 5,000 per month.

How much tax did she pay in a month if she paid a monthly premium of sh. 2,900 towards her life insurance policy? (2marks)

c) In June 2023, she received additional earnings that were taxed at 25% in each shilling. If she paid 31.55% more tax that month. Find the percentage increase in her earnings. (5marks)

24. The coordinates of the vertices of a rhombus ABCD are A (3, 4), B (5, 1),C(3,-2),D(1, 1).

a.(i) Find the co-ordinates of vertices of its image $A'B'C'D'$ under the transformation defined by the matrix $\begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$ (2marks)

ii) Draw the object and its image on a grid. (2marks)

iii) On the same grid draw the image A''B''C''D'' of A'B'C'D' under the transformation given by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ (2marks)

b(i) Describe the transformation that map A''B''C''D'' back onto A'B'C'D'. (1mark)

ii) Find the matrix of transformation that map A''B''C''D'' onto A'B'C'D' (1mark)

iii) Find a single matrix which will map ABCD onto A''B''C''D''. (2marks)

KCSE TOP PREDICTION TRIAL 8 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ………………………………………………………..…………… Adm Number…………………………..

Candidate's Signature ………………………..………………………….Date ……………………………………

INSTRUCTIONS TO CANDIDATES

1. *Write your name and index number in the spaces provided at the top of this page.*
2. *This paper consists of **Two** sections: **Section I** and **Section II***
3. *Answer **ALL** questions in Section I and **any five questions** from Section II*
4. *Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.*
5. *Marks may be given for correct working even if the answer is wrong.*
6. *Non-Programmable silent electronic calculators and KNEC Mathematical Tables may be used.*
7. *This paper consists of 16 printed pages. Candidates should check the question paper to ensure that all the pages are printed as indicated and no questions are missing.*

For Examiners' Use Only

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

SECTION II

17	18	19	20	21	22	23	24	Total

Grand Total	

SECTION I (50 MARKS)

Answer all questions in this section

1. Evaluate without using a calculator or mathematical table. (3 marks)

$$\frac{(2\frac{2}{7} - 1\frac{5}{6}) \div \frac{5}{6}}{\frac{2}{3} of\ 2\frac{1}{4} - 1\frac{1}{7}}$$

2. A perpendicular line is drawn from a point (4, 6) to the line 5y+4x = 20. Find its equation in the form of, ay+bx = c where a, b, c are integers. (3 marks)

3. Simplify completely by factorization $\frac{20-45x^2}{6x^2-x-2}$ (3 marks)

4. Solve for x in the equation $8^{(2x-1)} \times 32^{x} = 16^{x+1}$ (3 marks)

5. A bank in Kenya buys and sells foreign currencies as follows.

	Buying (Ksh)	Selling (kshs.)
1 Us dollar	85.86	86.06
1 sterling pound	142.41	142.73

A tourist from United States of America converted 43521 US dollars into Kenya shillings.

i) Calculate the amount in Kenya shillings that she received (1 mark)

ii) While in Kenya, the tourist spent sh. 2437821 and converted the balance into sterling pounds. How much in Sterling pound did the tourist receive? (3 marks)

6. Wasike bought 3 bags of DAP fertilizer and 4 bags of CAN fertilizer at sh. 21500. Had he bought 2 bags of DAP and 5 bags of CAN, he would have spent sh. 400 less. Find the cost of one bag of each type of fertilizer. (3 marks)

7. Use square roots and reciprocals tables to evaluate to 4 significant figures the expression.

$$(0.06458)^{\frac{1}{2}} + \frac{2}{0.4327}$$ (3 marks)

8. Express 128 and 200 as a product of their prime factors in power form hence evaluate

$$\sqrt{\frac{128}{200}}$$ (3 marks)

9. Given that $\sin\theta = \frac{2}{3}$ and θ is acute angle, find $\tan(90 - \theta)$. Give the answer to 4 significant figures. (3 marks)

10. Given the column vector $\underset{\sim}{P} = \begin{pmatrix} -5 \\ 3 \end{pmatrix}$, $\underset{\sim}{q} = \begin{pmatrix} 4 \\ -8 \end{pmatrix}$ and $\underset{\sim}{r} = \begin{pmatrix} 6 \\ -9 \end{pmatrix}$, $\underset{\sim}{t} = 2\underset{\sim}{p} - \frac{1}{2}\underset{\sim}{q} + \frac{1}{3}\underset{\sim}{r}$

    i) Express t as a column vector (2 marks)

    ii) Calculate the magnitude of vector t in (i) above correct to 2 d.p. (1 mark)

11. Four machines give out signals at intervals of 24 seconds, 27 seconds, 30 seconds and 50 seconds. At 5.00 p.m. all the four machines give out a signal simultaneously. Find the time this will happen again. (3 marks)

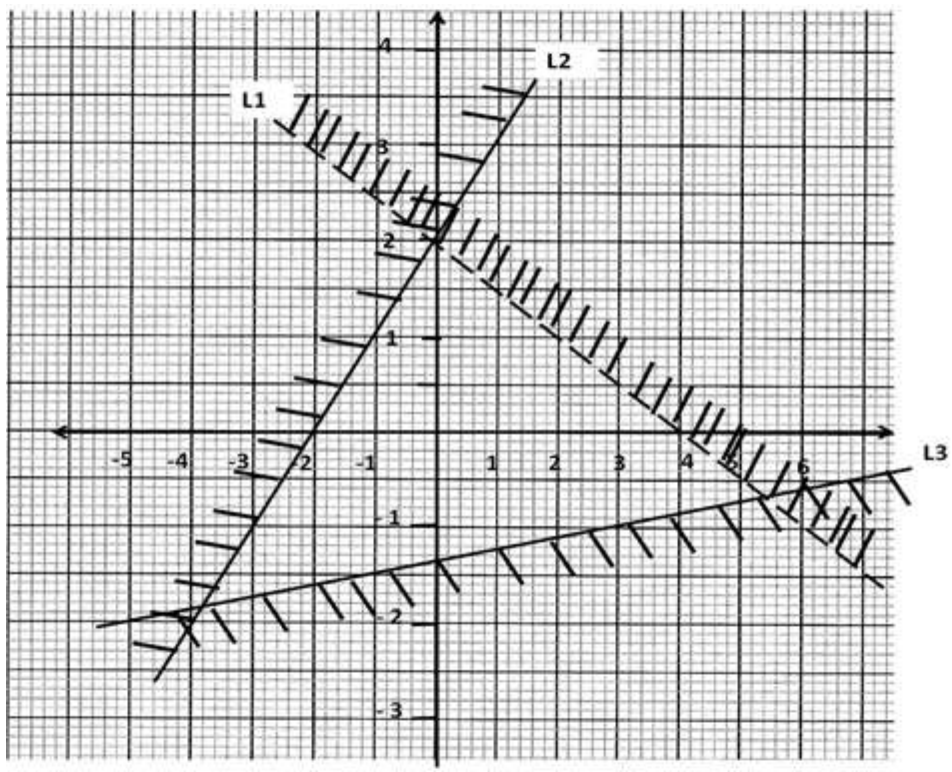
12. A wire in form of an arc with radius 12 cm and center angle 150^0 is formed into a circle. Find the radius of the circle. (3 marks)

13. The exterior angle of a regular polygon in equal to one – third of interior angle. Calculate the number of sides of the polygon and give its name. (3 marks)

14. The area of a parallelogram is 121 cm^2 and its sides are 11 cm and 16 cm long. Find the sizes of all the angles of the parallelogram. (3 marks)

15. Use logarithms table to evaluate $\sqrt[3]{\frac{0.006549}{81.06 tan 43^0}}$ (4 marks)

16. a) Form the three inequalities that satisfy the given region R. (3 marks)



SECTION II (50 MARKS)

Answer any five questions from this section in the spaces provided below in each question.

17. (a) Using a ruler and a pair of compasses only, construct triangle ABC in which BC = 6cm, AB = 8.8 cm and < ABC = 22 ½ ^{0}, (4 marks)

b) Measure AC and <ACB (2 marks)

c) Construct a circle that passes through A, B and C. (3 marks)

d) What is the radius of this circle (1 mark)

18) Forty students in a form 2 class were weighed and their masses recorded to the nearest kilogram as shown below.

45	48	56	39	47	36	45
37	46	33	43	51	42	47
47	40	46	41	45	43	46
54	42	51	39	42	45	44

49	50	46
39	42	48
50	38	45
35	52	46

a) Using class intervals of 5kg tabulate this data in a frequency table (4 marks)

b) Find the modal frequency (1 mark)

c) Modify the table and use it to calculate the mean mass of the students (5 marks)

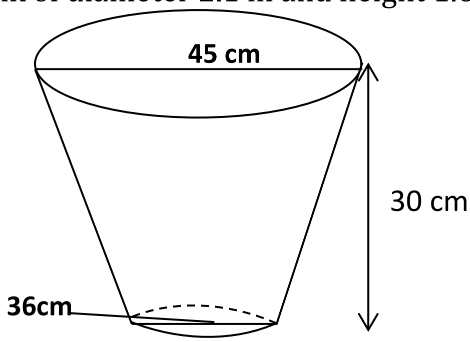
19) A bus left town A at 11.45 a.m. and travelled towards town B at an average speed of 80km/h. A Matatu left town B at 1.15 p.m. and travelled towards town A along the same road at an average speed of 120km/hr. The distance between the two towns is 840km. Determine;

a) The time of day when the two vehicle met. (4 marks)

b) How far from town A they met? (2 marks)

c) How far outside town B the bus was when the Matatu reached town A. (4 marks)

20) The diagram below shows a frustrum which represents a bucket with an open end diameter of 30 cm and a button diameter of 24 cm. The bucket is 30 cm deep as shown and it is used to fill an empty cylindrical tank of diameter 2.1 m and height 1.6 m.



a. Leaving π in your answer, calculate

(i) The capacity of the bucket (6 marks)

b) The capacity of the tank (2 marks)

b) Determine the number of buckets that must be drawn in order to fill the tank. (2 marks)

21) Four business partners Duncan, Dennis, David and Derrick decided to buy a minibus. They agreed to pay for the cost of the minibus in the ratio 3:4:5:6. Basing their contributions on the marked price of the minibus, they found that Derrick would pay sh. 360000 more than Duncan. The sales agent however allowed them a 10% discount on cash payment.

a) Calculate

i) The marked price of the minibus (2 marks)

ii) How much more Duncan and David paid than Derrick. (5 marks)

b) The four partners agreed to share monthly profits from the minibus in the ratio of their contributions after setting a side 12% of the profit for emergencies. If the minibus realized sh.72, 000 in profits during one month, calculate David's share of the profits for that month. (3 marks)

22) a) The points A (2, 6), B (1, 1) C (3, 4) and D (5, 3) are vertices of a quadrilateral ABCD. Plot ABCD on the grid below to form quadrilateral ABCD. (2 mark)

b) ABCD undergoes a rotation of positive 90^0, about the origin. on the same grid, draw the image A'B'C'D' and state the coordinates of A'B'C'D'. (3 marks)

c) A'B'C'D' undergoes a reflection in the X – axis to give A''B''C''D''. On same grid, draw A''B''C''D'' and state the coordinates of A''B''C''D'' (3 marks)

d) A''B''C''D'' is the image of ABCD under a reflection. On the grid, mark the mirror line and state its equation. (2 marks)

23 . (a) Find the inverse of the matrix (2 marks)

$$\begin{bmatrix} 2 & 5 \\ 4 & 3 \end{bmatrix}$$

(b) A transport company has two types of vehicles for hire: Lorries and buses. The vehicles are hired per day. The cost of hiring two lorries and five buses is Sh. 156,000 and that of hiring 4 lorries and three buses is Sh. 137,000.

(i) Form two equations to represent the above information. (2 marks)

(ii) Use matrix method to determine the cost of hiring a lorry and that of hiring a bus. (3 marks)

(c) Find the value of x given that $\begin{bmatrix} 2x-1 & 1 \\ x^2 & 1 \end{bmatrix}$ is a singular matrix (3 marks)

24) Four points A, B, C, and D are situated on a horizontal plane such that B is 200 m on a bearing of 065^0 from A. C is 300m on a bearing of 120^0 from B and D is 150m due west of C.

a. Using a suitable scale, draw an accurate scale drawing representing the positions of A, B, C, and D.

(6 marks)

c) By measuring use your scale drawing to find the distance and bearing of:-

i) D from A (2 marks)

ii) B from D (2 marks)

KCSE TOP PREDICTION TRIAL 8 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… Adm Number…………………………..

Candidate's Signature …………………………………………………Date ……………………………………

~~INSTRUCTION TO CANDIDATE'S~~:

- ✓ *Write your name, stream and index number in the spaces provided at the top of this page*
- ✓ *This paper contains two sections:* ***Section I and Section II***
- ✓ ***Answer all questions in section I and any five in section II***
- ✓ ***Show all the steps in your calculations giving your answer at each stage in the spaces provided below each question.***

For examiner's use only

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

17	18	19	20	21	22	23	24	TOTAL

Grand Total	

SECTION I (50 MARKS).

(Attempt all the questions in this section)

1. Find the selling price of 5 kg of a mixture of maize flour and millet flour if 4 kg of maize flour costing sh. 75 per kg is mixed with 6 kg of millet flour costing sh. 50 per kg and a profit of 20% is realized. (3 marks)

2. The radius of a cylinder is given as 3.50 cm while its height is exactly 12 cm. Find the percentage error in the volume of the cylinder, giving your answer to three significant figures. (take $\pi = \frac{22}{7}$) (3 marks)

3. Solve the equation $\log_2(2x^2 - 8) - \log_2(x + 2) = 3$ (3 marks)

4. Solve the equation $2x^2 + 10x + 12 = 0$ by completing the square method. (3 marks)

5. Make b the subject of the formula. (3 marks)

$$x = \frac{a}{\sqrt{(a-b)(a+b)}}$$

6. Simplify the expression: $\frac{5}{7-2\sqrt{3}} - \frac{2}{7+2\sqrt{3}}$ (3 marks)

7. Triangle ABC is such that AB = 10 cm, BC = 12 cm and AC = 20 cm. calculate correct to 2 decimal places the;
   a) Angle ABC (2 marks)

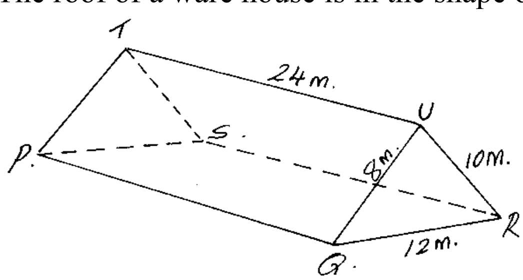
   b) A circle passes through the vertices of triangle ABC. Calculate the radius of the circle.(2 marks)

8. Karanu bought a computer whose marked price was 25,000 on hire purchase terms. She paid a deposit of 25% of the cash price and cleared the balance by paying 12 equal monthly instalments of sh2000 each. Calculate the rate of compound interest charged per month. (3 marks)

9. Solve the following equation $8\cos x = sin^2x + 6$ for $0^0 \leq x \leq 360^0$ (4 marks)

10. Given that A (4, -2, 6) and B (-2, 1, -3) and that a point N divides AB in the ratio -2:5. Find the vector **ON** in terms of **i** **j** and **k**. (3 marks)

11. The roof of a ware house is in the shape of a triangular prism as shown below



Calculate the angle between faces RSTU and PQRS (3 marks)

12. The equation of a circle is given by $2x^2 + 2y^2 - 12x + 8y - 6 = 0$. Determine the centre and the radius of the circle. (3 marks)

13. Expand $\left(1 - \frac{x}{2}\right)^6$ in ascending powers of x up to the fourth term. Hence use your expansion to evaluate $(0.95)^6$ to three significant figures. (3 marks)

14. The gradient function of a curve is given as $\frac{dy}{dx} = 3x^2 + 4x + 4$. Find the equation of the curve given that the curve passes through the point (1, 2). (3 marks)

15. The first, second and fifth terms of an arithmetic sequence are the first three consecutive terms of a geometric sequence. Find the common ratio of the geometric sequence. (3 marks)

16. In a transformation, an object with area $12cm^2$ is mapped onto an image whose area is $24cm^2$. Given that the matrix of transformation is $\begin{bmatrix} x & 4 \\ 1 & (x-1) \end{bmatrix}$ find the value of x, hence state two possible matrices (3 marks)

SECTION II (50 marks)

(Answer ONLY FIVE questions in this section in the spaces provided)

17. Three quantities x, y and z are such that x varies directly as the square root of y and inversely as the square of z.

a) (i). Given that $x = 12$, and y= 36, and $z = 3$, find an equation connecting x, y and z (3 marks)

(ii) Find x when $y = 121$ and $z = 5$ (1 mark)

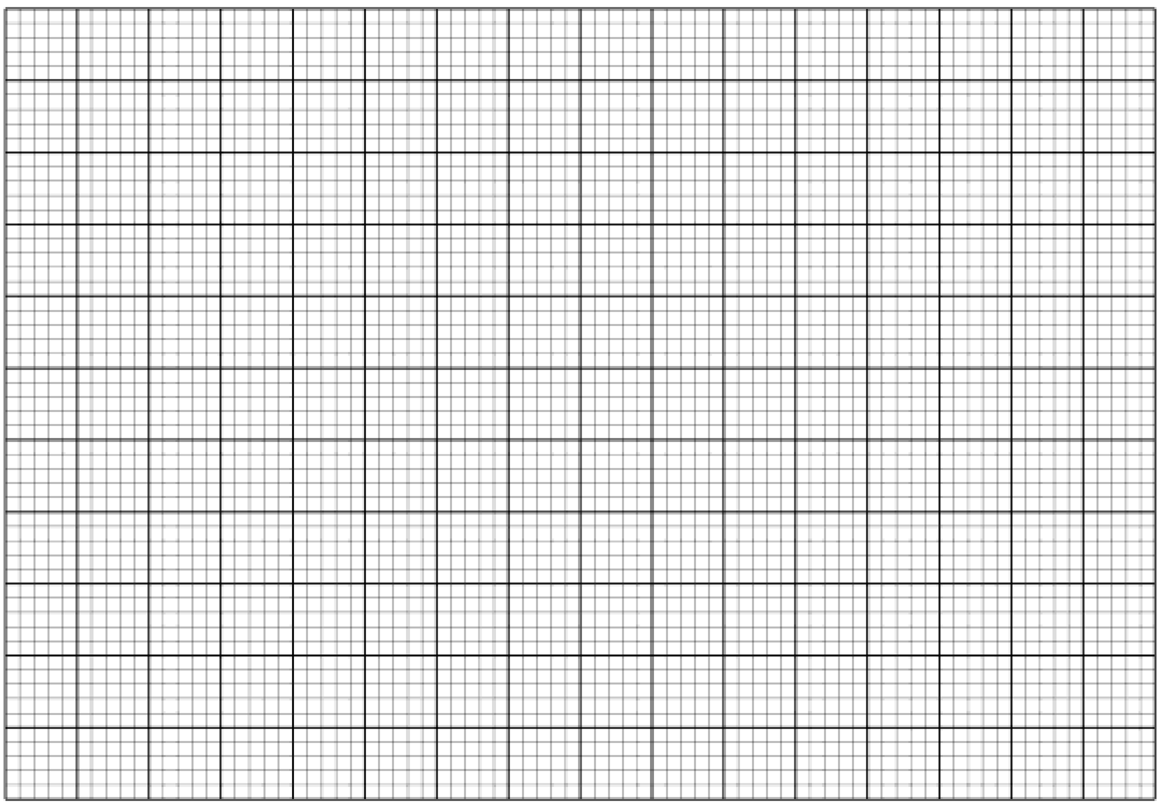
b) i) Express y in terms of x and z (2 marks)

ii) If x increases by 21% and z decreases by 10%, find the percentage increase in y (4 marks)

18. Complete the table below correct to 2 decimal places. (2 marks)

X	-180^0	-150^0	-120^0	-90^0	-60^0	-30^0	0^0	30^0	60^0	90^0	120^0	150^0	180^0
Y=2Cos x		-1.73			1		2		1	0			
Y=Cos (x-60)	-0.5			-0.9		0			1				-0.5

(b) On the same axes plot the graphs of y= Cos $(x-60^0)$ and y = 2 Cos x. (use a scale of 1 unit for 30^0 on the x axis and 1 unit for 0.5 units on the y axis) (4 marks)



(c) Using the graph above determine the values of x for which $\cos(x-60^0) - 2\cos x = 0$ (1 mark)

(d) State the period and amplitude of each of the waves above. (2 marks)

	amplitude	period
Y = 2Cos x		
Y = Cos(x - 60)		

(e) Describe the transformation which maps y= $\cos(x-60^0)$ to y = 2cosx. (1 mark)

19. Mrs Mutiso has two cupboards A and B in her kitchen. Cupboard A has 7 white cups and 5 brown cups while cupboard B has 4 white cups and 6 brown ones. All the cups are identical in size and shape.
    a) There was a blackout in the town and Mrs Mutiso had to select two cups, one after the other without replacing the previous one, from any of the cupboards.
        i. Draw a tree diagram for the information. (2 marks)

        ii. Calculate the probability that she chooses two cups of the same colour. (3 marks)

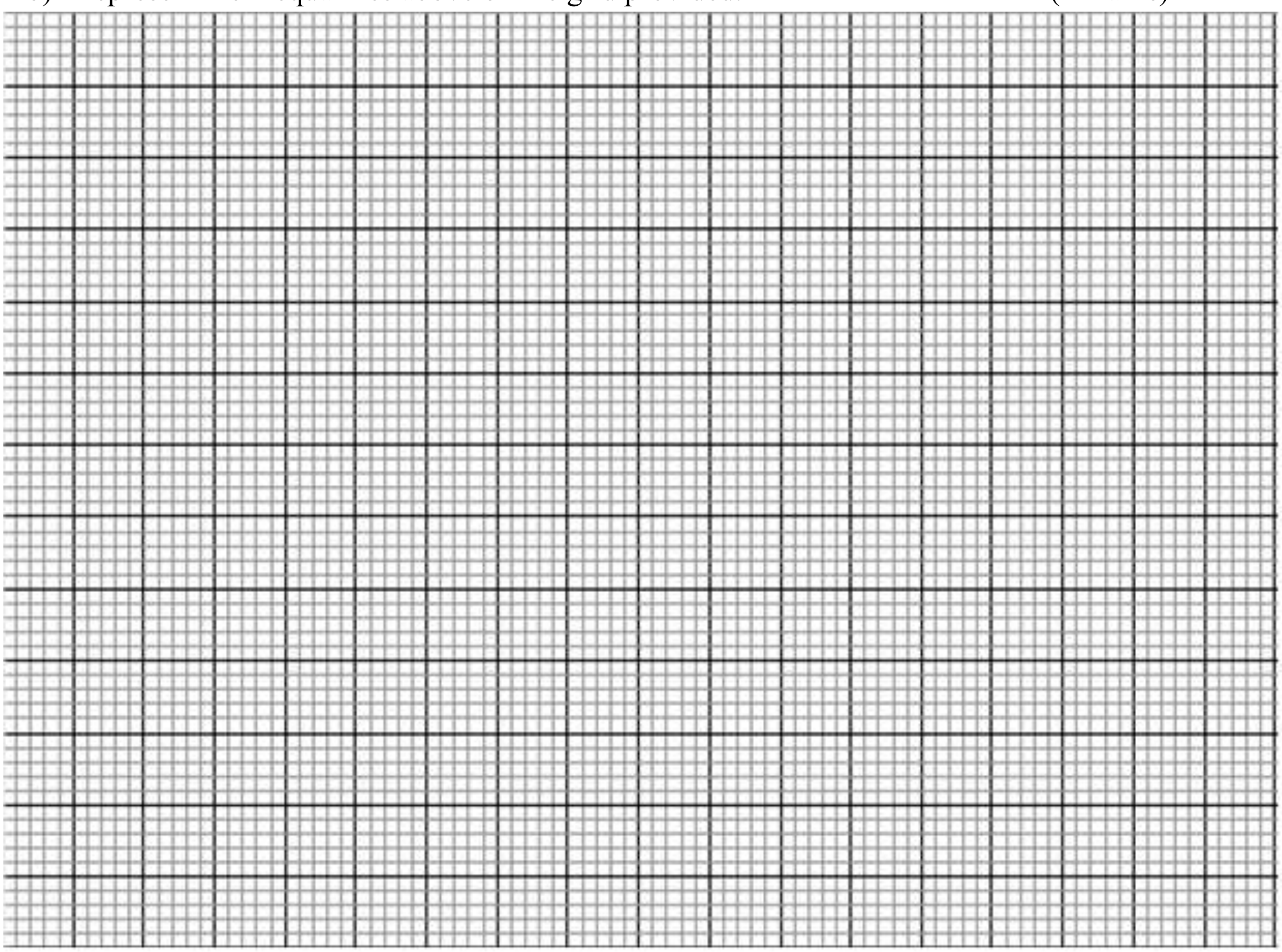
        iii. Calculate the probability that she chooses two cups of different colours (1 mark)

    b) On another day, she selected two cups from cupboard B. After using them she placed them in cupboard A. Later in the day, she selected one cup from cupboard A. Find the probability that she chose a white cup. (4 marks)

20. A contractor intends to transport 800 bags of cement using a lorry and a pick up. The lorry can carry a maximum of 40 bags while a pick up can carry a maximum of 20 bags. The pickup has to make more than twice the number of trips the lorry makes. The total number of trips made by the pickup has to be less than 40 while those made by the lorry have to be at least 5. By letting x and y to be the trips made by the lorry and pickup respectively:

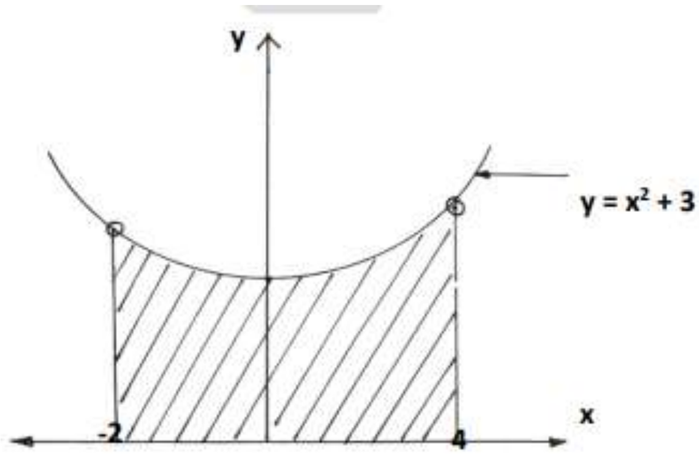
    a) Write down all the inequalities to represent the above situation. (4 marks)

    b) Represent the inequalities above on the grid provided. (4 marks)



    c) Find the number of trips to be made by the lorry and the pickup respectively in order to minimise cost if the cost per trip is Ksh 2000 for the lorry and Ksh 1000 for the pickup. (2 marks)

21. Below is a sketch of the area bounded by the curve $y = x^2 + 3$ and the lines x=-2, x=4 and y=0.



a) Use the trapezium rule with six trapezia to estimate the area of the shaded region (4 marks)

b) Find the exact area bounded by the region in (a) above (3 marks)

c) Find the percentage error in the area for the approximation using trapezium rule (3 marks)

22. An aero plane left town P $(60^{O}N, 55^{0}E)$ at 7:00 am local time on Monday and travelled towards another town Q $(60^{0}N, 125^{0}W)$ at a speed of 500 knots using the **shortest route**. (Take $\pi = \frac{22}{7}$ and radius of the earth R= 6370 km)

a) (i) Calculate the distance travelled in nautical miles. (2 marks)

(ii) Calculate the local time at Q when the plane arrived at Q (3 marks)

b) Another plane left P at 1.30 pm local time and travelled to T $(60^{0}N, 90^{0}E)$ along the parallel of latitude. Calculate the:

(i) Distance between P and T to the nearest km (2 marks)

(ii) Local time of arrival at town T if the plane flew at the speed of 470 km/h (3 marks)

23. The data below shows the heights of students in a class.

Height	130 -139	140 -149	150-159	160-169	170-179	180-189	190-199
Frequency	3	12	32	22	7	3	1

a) Draw a cumulative frequency curve on the grid provided (4 marks)

b) Use the graph to find:

i) The median height. (2 marks)

ii) The interquartile range (2 marks)

iii) The percentage of students with a height greater than 172cm (2 marks)

24. Using a ruler and a pair of compasses only, construct a parallelogram ABCD such that AB=10 cm, AD=6 cm and angle BAD=60º. (3 marks)

(b) On the same diagram, construct:

(i) The locus of a point P such that P is equidistant from AB and AD; (1 mark)

(ii) The locus of a point Q such that Q is equidistant from A and B; (1 mark)

(iii) The locus of a point T such that ATB = 120^0; (3 marks).

(c) Shade the region R bounded by the locus of P, the locus of Q and the locus of T within the trapezium such that AR ≤ RB, angle BAR ≤ angle DAR and angle ARB ≥ 120. (2 marks)

KCSE TOP PREDICTION TRIAL 9 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… Adm Number…………………………..

Candidate's Signature ………………………..…………………….Date ……………………………………

Instructions to Candidates

1. Write your **name, Admission number and class** in the spaces provided.
2. Sign and write date of the examination in the spaces provided.
3. The paper contains **TWO sections**: **Section I and II**
4. Answer **ALL** questions in section I and **STRICTLY ANY FIVE** questions from section II.

For Examiner's use only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	TOTAL

Section II

17	18	19	20	21	22	23	24	TOTAL

GRAND TOTAL

SECTION I(50MARKS)

Answer all the questions in this section

1. Without using a calculator or mathematical tables evaluate. (3 marks)

$$\frac{0.32 \times 9}{\sqrt{0.0256 \times 9}}$$

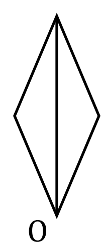
.

2. Odour has a driver, a gardener and a house girl. The driver's daily wage is thrice that of the gardener. That of the gardener is Sh. 50 less than that of the house girl. Their total daily earnings are Sh. 1550. Express the ratio of their earnings; driver: house girl: gardener in the simplest form. (3 marks)

3. If $4^{3y-4x} = 64$ and $3^y \div 9^x = 1$, solve for x and y. (3 marks)

4. Given that $log3 = 0.4771$ and $log5 = 0.6990$ find, without using mathematical tables or calculator, the value of $log3375$. (3 marks)

5. The diagram below shows one section of a design that has **rotational symmetry of order 4** about point O. Complete the diagram (3 Marks)



6. The sum of interior angles of two regular polygons of sides; n and n+2 are in the ratio 3:4. Calculate the sum of the interior angles of the polygon with n sides. (3marks)

7. An arc subtends an angle of 0.9 radians at the centre of a circle whose radius is 13 cm. Find the length of the arc. (3 marks)

8. A hospitality company has three hotels, A, B and C. Hotel B is $60km$ on a bearing of 100^0 from A while hotel C is $80km$ from A and on a bearing of $S30^0W$ from A.
   (a) Using a scale of 1cm to represent 10km, draw a diagram that represents the situation above. (2 marks)

   (b) Use your diagram in (a) above, find the distance between B and C. (1 mark)

9. In a construction site there are 3 painters and 4 masons. The sum of their weekly wages is sh. 33,600. When one of the painters resigns and one mason is employed, the sum of their weekly wages of the 7 employees is sh. 32,900. Given their salaries remained the same, calculate using matrices, the weekly wage of a mason (4 marks)

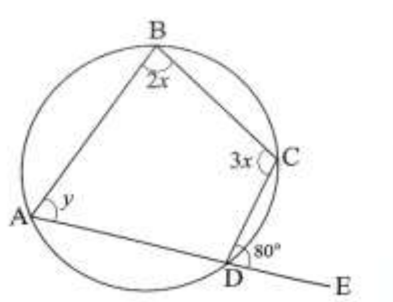
10. Without using a calculator or mathematical tables , find the values of Θ in the range $0^0 \leq \Theta \leq 90^0$ given that $\sin{(3\Theta - 45)^0} = \cos 60^0$. (3 marks)

11. A tourist is at a point A at the top of a $60m$ high cliff. B is a point on the surface of water directly below A. From his position, the angles of depression of two boats at points P and Q are $32^0 45'$and $54^0 30'$ respectively. If B,P and Q are in a straight line, find correct to 3 significant figures the distance between the two boats. (4 marks)

12. Solve the following inequalities and find all integral values that satisfy the inequality.

$5 - 3x > -7$ (3 marks)
$x - 6 \leq 3x - 4$

13. In the figure below ABCD is a cyclic quadrilateral and ADE is a straight line. Given that $< CDE = 80^0, < ABC = 2x^0, < BCD = 3x^0$ and $< DAB = y^0$, find the values of x and y. (3 marks)



14. Find the equation of the perpendicular bisector of the line AB where the coordinates of A and B are (-4, 2) and (6, 4) respectively. (3 marks)

15. One basket contains 195 bananas while another has 250 bananas. The bananas are to be distributed equally among a group of children. Find the largest number of bananas that can be given to each child so that 3 bananas remain in the first basket and 2 in the second. (3marks)

16. Find the exact value of $0.\dot{3} + 0.2\dot{5}$ (3marks)

SECTION II(5MARKS)

Answer only 5 questions in this section

17. Two trains, Train A and Train B, are moving on parallel tracks in the same direction. Train A is 150 meters long and travels at 60 km/h. Train B is 100 meters long and travels at 90 km/h.

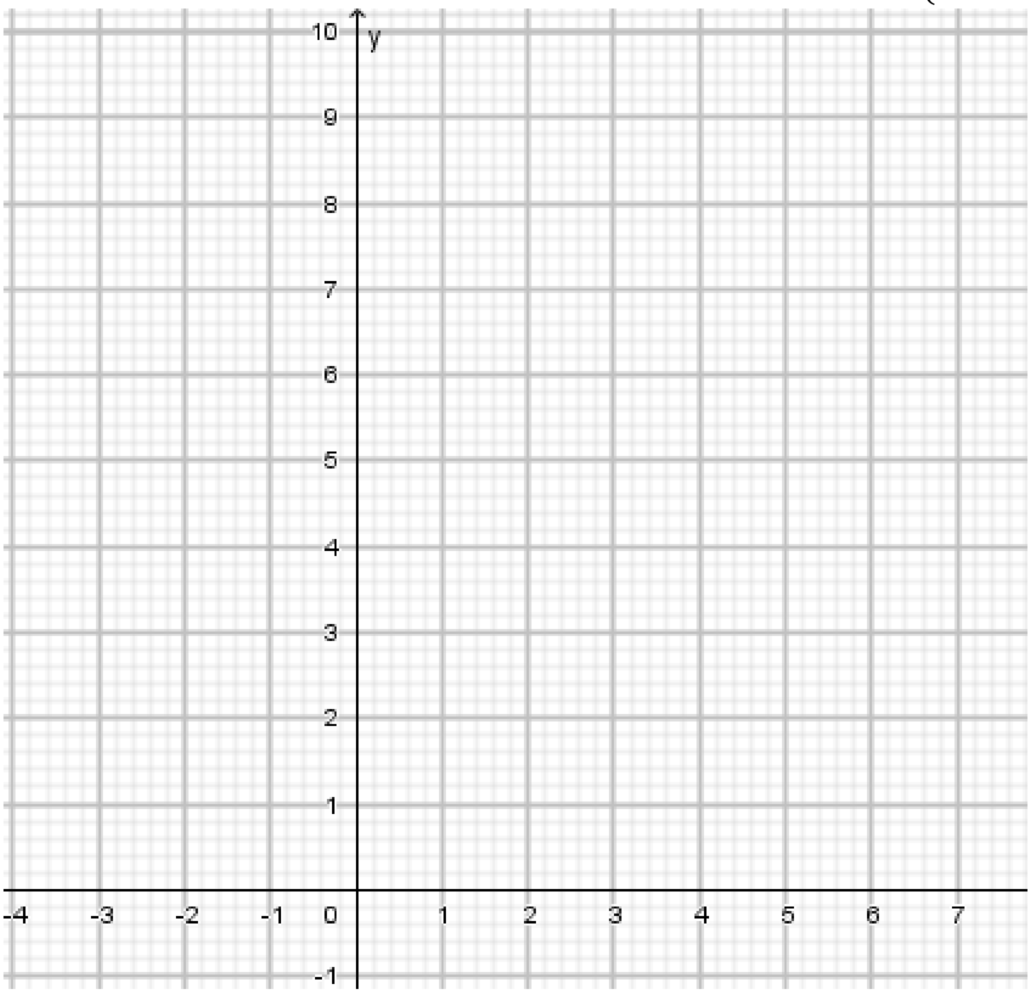
(a) Convert the speeds of both trains into meters per second. (2 marks)

(b) Calculate the relative speed of Train B with respect to Train A. (2 marks)

(c) How long will it take for Train B to completely pass Train A? (3 marks)

(d) If the trains were moving in opposite directions, how long would it take for them to completely pass each other? (3 marks)

18. (a) On the grid below, plot the points A(1,2), B(1,1), C(2,1), D(3,3) and join them to form quadrilateral ABCD. (2 marks)



(b) (i) On the same grid draw quadrilateral A'B'C'D' which is the image of ABCD under an enlargement centre P(1,1) and scale factor 3. (3 marks)

(ii) State the coordinates of vertices of A'B'C'D'. (1 mark)

(c) On the same axes draw quadrilateral A"B"C"D" with vertices A"(-2,1), B"(-1,1), C"(-1,2) and D"(-3,3). (2 marks)

(d) Describe fully transformation T which maps ABCD onto A"B"C"D". (2 marks)

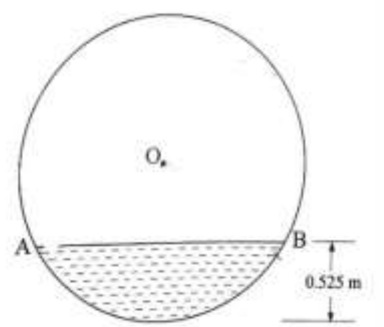
19. The heights of tree seedlings in a nursery were measured and recorded as in the table below.

$Height\ x\ cm$	$0 \le x < 5$	$5 \le x < 15$	$15 \le x < 25$	$25 \le x < 45$	$45 \le x < 75$
$no. of\ seedlings$	7	46	72	64	11

a) Estimate the mean height (4marks)

b) (i) Using a scale of 2cm to represent 10 units along the horizontal axis, and a scale of 1 cm to represent 1 unit along the vertical axis, draw a histogram to represent the distribution. 3 marks (ii)Use the histogram to estimate the median height (3marks)

20. The figure below shows the vertical cross section of cylindrical fuel tanker when the depth of the fuel it contain is 0.525m. AB is the horizontal level of the fuel. Its internal length is 7m and its internal diameter is 2.1m.



Calculate

(a) Angle AOB, where O is the centre of the circular section. (3 marks)

(b) The area of sector AOB correct to 2 decimal places. (Use $\pi = \frac{22}{7}$) (2 marks)

(c) The shaded area (3 marks)

(d) The mass of the fuel in the tanker given that the density of the fuel is $700kg/m^3$ (2 marks)

21. Mohamed is a second hand car dealer. A customer wanted to confirm the efficiency of a car tracking systems in two cars, car M and car N he wanted to buy. The dealer drove car M while the customer drove car N. They both leave point P at 10.50a.m. The dealer heads due East at an average speed of 90Km/h while the customer heads due North at an average speed of 120km/h. The trackers in the two cars detect each other as long as they are at most 270km apart. Car M moved $h$km and car N moved $t$km before the trackers stopped detecting each other.

    (a) (i) Form two simultaneous equations in h and t (3 marks)

    (ii) find the values of h and t (5 marks)

    b) Determine the time when the trackers stopped detecting each other. (2 marks)

22. Members of an technology investment group were to contribute some money equally to start a software development company. The total cost of starting the company was Sh. 4,950,000. Some days before the contributions were made, 4 members withdrew from the group. The remaining members were to increase their contributions by Sh.120,000 per member if they had to go on and start the company.

(a) Calculate the original number of members in the group. (4 marks)

(b) Due to financial difficulties, the group applied for a loan of $Sh. x$ from Hustler Funds. They received the loan, which was 18% of the total cost. They were then to contribute towards the balance equally. Find the amount that each member contributed. (3 marks)

(c) The ratio of the actual cost of starting the company to all the government taxes and levies was 7:2. Express correct to 1 decimal place a member's contribution as a percentage of the actual cost of the company. 3marks

23. The position vectors of point P with respect to the origin O is $\begin{pmatrix} 8 \\ -3 \end{pmatrix}$ and that of Q is $\begin{pmatrix} -4 \\ 5 \end{pmatrix}$.
Points A and B are the mid points of PQ and OQ respectively.
(a) Determine the:
(i) co-ordinates of A and B. (3 marks)

(ii) magnitude of AB correct to 2 decimal places. (3 marks)

(b) Express vector **AB** in terms of **OP**. (1 mark)

(c) A point M maps onto M' by a translation $\begin{pmatrix} -2 \\ 7 \end{pmatrix}$. Given that: $\mathbf{OM} = 3\mathbf{PQ} - 2\mathbf{OP}$,
find the co-ordinates of M' (3 marks)

24. The equation of a curve is given as $y = x^3 + x^2 - x$. Determine:

(a) The value of y when $x = -3$ (2 marks)

(b) The stationary points (4 marks)

(c) the nature of the stationary points (4 marks)

KCSE TOP PREDICTION TRIAL 9 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… **Adm Number**…………………………..

Candidate's Signature …………………………………………………**Date** ……………………………………

INSTRUCTIONS TO CANDIDATES:

- *Write your name, index number, admission and class in the spaces provided above.*
- *Sign and write the date of examination in the spaces provided above.*
- *This paper contains **TWO** sections: Section **I** and Section **II**.*
- *Answer **ALL** the questions in Section **I** and **ONLY FIVE** questions from section **II**.*
- *All answers and working **MUST** be written on the question paper in the spaces provided below each question.*
- *Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.*
- *Marks may be given for correct working even if the answer is wrong.*
- *Non-programmable silent electronic calculators and KNEC Mathematical tables may be used, except where stated otherwise.*
- *Candidates should answer the questions in English.*

FOR EXAMINE'S USE ONLY

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

SECTION II

17	18	19	20	21	22	23	24	Total

Grand Total

SECTION 1 (50 Marks)

Answer ALL the questions in the spaces provided after each.

1. Rationalise the denominator and simplify leaving your answer in the form $\sqrt{a} + b$ (3marks)

$$\frac{\sqrt{2} + 2\sqrt{5}}{\sqrt{5} - \sqrt{2}}$$

2. The radius of a sphere was corrected to 1 decimal place as 3.5cm. Taking $\pi = \frac{22}{7}$, find :

(a) the surface area of the sphere. (1mark)

(b) the percentage error in calculating the surface area of the sphere. (3marks)

3. Find the value of p if $px^2 - \frac{3}{2}x + \frac{1}{16}$ is a perfect square and p is a constant. (2marks)

4. A circle of radius 3cm has its centre at (3, 2).Express the equation of the circle in the form $x^2 + y^2 + mx + ny + c = 0$ where m, n and c are constants. (3marks)

5. A transformation matrix $\begin{pmatrix} p & 2 \\ 4p-3 & 3p \end{pmatrix}$ maps an object of area $21cm^2$ to an image of area $42cm^2$. Determine the value of p. (3Marks)

6. Solve for x in the equation.

   $6Cos^2x - Sin\ x - 4 = 0$ in the range $0^0 \leq \ \leq 180^0$ (4Marks)

7. Given the function ; $y = \frac{\sqrt[3]{x}}{z - r^2}$

(a) find the value of y truncated to 2 decimal places if $x = 200$, $z = 50$ and r = 6 (1mark)

(b) Make r the subject of the function above. (2marks)

8. (a) Using the binomial expansion, expand $\left(2 + \frac{2}{x}\right)^5$ up to the term in x^{-3} (2marks)

(b) Use the expansion above to evaluate $(3.5)^5$ (2marks)

9. Five men working 8 hours a day take 2 days to cultivate an acre of land. How many days would four men working 10 hours a day at double rate to cultivate 3 acres of land? (3marks)

10. The following table shows monthly income tax rates for a certain year

Monthly income in kshs	Tax % in each shilling
0 – 24, 000	10
24, 001 – 32, 333	25
32, 334 – 50, 000	30
Above 50, 000	32.5

In June of that year, an employee earnings were as follow:

Basic salary sh.92, 000

House allowance sh.45, 000

Travel allowance sh.12, 000

He was entitled to a monthly tax relief of sh. 2, 400.

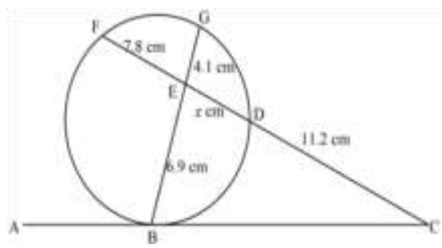
Calculate his monthly tax (4 Marks)

11. Draw a line $B = 9cm$. On the upper side of line AB, construct the locus of a point P such that the area of $\Delta APB = 13.5cm^2$. On the same locus of P locate two points labeled P_1 and P_2 such that $< AP_1B = < AP_2B = 90^0$. (3Marks)

12. Calculate the quartile deviation for the following scores
    9, 12, 14, 16, 18, 20, 23, 24 (3Marks)

13. The position vectors of **A** and **B** are given as $= \mathbf{2i - 3j + 4k}$ and $\mathbf{b = -2i - j + 2k}$ respectively. Find to 2 decimal places, the length of the vector **AB.** (3Marks)

14. In the figure below ABC is a tangent to the circle at point B. Given that BE = 6.9cm, FE = 7.8 cm, GE = 4.1 cm, DC=11.2cm and ED = x cm. Determine the length BC, give your answer in four significant figures. (4 marks)



15. Find m given that:
$\int_2^m (5x - 12)dx = 16.5$(4Marks)

16. In an experiment involving two variables, time (t) hours and height (h)cm, the following results were obtained.

$Time\ (t) hours$	0	0.7	1.5	2.1	3.2	3.6
$Height\ (h) cm$	97	82	64	46	32	18

(a) On the grid provided, plot height, against time, t (h) (1mark)

(b) Use the plotted points to draw the line of best of fit for the data and hence determine the rate of change of height with time. (3Marks)

SECTION II (50 MARKS)

Answer only five questions in this section in the spaces provided.

17. Three similar bags A, B and C each contains grey and pink balls. A has 2 grey and 7 pink balls, bag B has 4 grey and 3 pink balls while bag C has 5 grey and 5 pink balls. The balls are identical except of colour. A bag is selected at random and two balls randomly picked from it, one at a time without being returned.

a) using a tree diagram or otherwise, find the probability that;

i) The two balls are from bag A and are both pink. (2Marks)

ii) The two balls are of different colours and picked from bag B or C. (3Marks)

b) All balls initially in the three bags are gathered in one bag and 3 balls randomly picked from it, one at a time and placed in a second bag which is empty. Find the probability that all the three

balls picked are grey. (2Marks)

c) A bag contains 15 red balls and x blue balls, two balls are selected at random one at a time, the probability of selecting balls with different colours is $\frac{1}{2}$. Find the number of balls in the bag. (3Marks)

18. (a) Complete the table below for the functions $y = -3\cos 2x$ and $y = 2\sin\left(\frac{3}{2}x + 30\right)^0$ for $0^0 \leq x \leq 180^0$ (2marks)

x^0	0	20	40	60	80	100	120	140	160	180
$-3\cos 2x$	-3.00			1.50	2.82	2.82		-0.52	-2.30	
$2\sin\left(\frac{3}{2}x + 30\right)^0$	1.00		2.00	1.73		0.00	-1.00			-1.73

(b) Using the grid provided, draw the graphs of $y = -3\cos 2x$ and $y = 2\sin\left(\frac{3}{2}x + 30\right)^0$ for $0^0 \leq x \leq 180^0$ on the same pair of axes. Take 1cm to represent 20^0 on x – axis and 2cm to represent 1 unit on the y – axis. (5Marks)

© From the graphs in (b) above, find:

(i) The period of $y = 2\sin\left(\frac{3}{2}x + 30\right)^0$ (1Mark)

(ii) The values of x that $2\sin\left(\frac{3}{2}x + 30\right)^0 + 3\cos 2x = 0$ (2Marks)

19. A triangle ABC with vertices at $(1, -1)$, $B(3, -1)$and $C(1, 3)$ is mapped onto triangle $A^1B^1C^1$ by a transformation whose matrix $\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$. Triangle $A^1B^1C^1$ is mapped onto $A^{11}B^{11}C^{11}$ with vertices at $A^{I}(2, 2)$, $B^{II}(6, 2)$ and $C^{II}(2, -6)$by a second transformation.

(a) Find the coordinates of $A^1B^1C^1$. (2Marks)

(b) Find the matrix which maps $A^1B^1C^1$ on to $A^{11}B^{11}C^{11}$. (3Marks)

(c) Determine the ratio of the area of triangle $A^1B^1C^1$ to triangle $A^{11}B^{11}C^{11}$. (2Marks)

(d) Find the transformation matrix which maps $A^{11}B^{11}C^{11}$ onto ABC. (3Marks)

20. An aircraft leaves town $(30^0S, 17^0E)$ and flies due North to $Q(60^0N, 17^0E)$. It then flies at an average speed of 300 knots for 8hours due West to town R. Determine;

(a) The distance PQ in nautical miles. (2Marks)

(b) The position of town R. (3Marks)

(c) The local time at R if the local time at Q is 3.12pm on Tuesday. (2Marks)

(d) The shortest distance between Q and R to the nearest kilometers.$(R = 6370km, \pi = \frac{22}{7})$

(3Marks)

21. The 2^{nd} and 5^{th}terms of an arithmetic progression are 8 and 17 respectively. The 2^{nd}, 10^{th} and 42^{nd} terms of the A.P form the first three terms of a geometric progression.

Find :

(a) The 1^{st} term and the common difference of the A.P. (3Marks)

(b) The first three terms and the 10^{th} term of the G.P. (4Marks)

(c) The sum of the first 10 terms of the G.P. (3Marks)

22. The following are marks scored by form 4 students in a Mathematics test.

Marks	10 - 19	20 -29	30 - 39	40 -49	50 - 59	60 - 69	70 - 79	80 - 89	90 - 99
Number of students	2	6	10	16	24	20	x	8	2

(a) If the assumed mean was 54.5 and $\sum fd^2 = 31400$ where $d = (x - A)$, determine;

(i) The value of x. (5Marks)

(ii) The mean. (3Marks)

(b) Calculate the standard deviation. (2Marks)

23. (a) The speed V m/s of a moving particle is partly constant and partly varies as time t seconds. It is given $V = 28m/s$ when $t = 2s$ and $V = 53\ m/s$ when $t = 7s$.

Find the speed of the particle when $t = 11s$. (4Marks)

(c) A quantity R varies directly as T and inversely as the cube root of Q. Given that $Q = 64$ when $T = 6$ and $R = 30$;

(i) Find the equation connecting Q , R and T. (3Marks)

(ii) Find the percentage change in R when T is decreased by 10% and Q increased by 25%. (3Marks)

24. A sculptor takes 2.5 hours to mould a teapot and 5 hours to carve a dining table. It takes the sculptor at least 100 hours to mould x teapots and carve y dining tables. The labor cost of molding a teapot is Ksh 100 and that of carving a table is Ksh 150. The total labor cost must not exceed Ksh 6000. The dining tables the sculptor's carved must not be more than twice the teapots moulded.

i) Write down in terms of x and y, all the inequalities representing the above information. (4Marks)

ii) On the grid provided, draw the inequalities and shade the unwanted region. (4Marks)

c) The sculptor makes a profit of Ksh 160 on a teapot and Ksh 270 on a dining table. Use the graph to determine the maximum profit that the sculptor can make. (2Marks)

KCSE TOP PREDICTION TRIAL 10 - 2026

Kenya Certificate of Secondary Education

121/1 - MATHEMATICS - Paper 1

(Alt A)

Time 2 ½ hours

Name ………………………………………………………………………… **Adm Number**…………………………..

Candidate's Signature ……………………………………………………….**Date** ……………………………………

Instructions to candidates

- Write your Name, Admission Number, Class, Signature and Date of the Examination on the spaces provided at the top of this page.
- This paper consists of two sections: **Section I** and **Section II**. Answer **All** questions in **section I** and **ONLY FIVE** questions in **section II**.
- Show **All** the steps in your calculations, giving your answers at each stage on the spaces below each question.
- Marks may be given for correct working even if the answer is wrong.
- KNEC Mathematical tables & Silent Non-Programmable Calculators may be used except where stated otherwise.

For Examiner's Use Only

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

SECTION II

17	18	19	20	21	22	23	24	TOTAL

SECTION 1(50MKS)

Answer all questions in this section

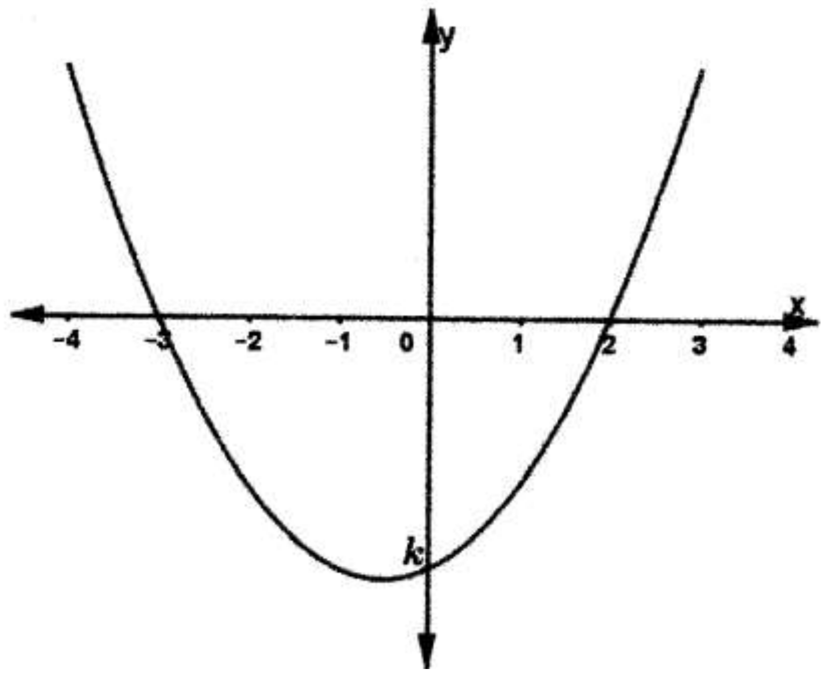
1. Simplify the expression (3mks)

$$\frac{8x^2 - 18}{4x^2 + 12x + 9}$$

2. Dr. Lexine needs to import a car from Japan that costs US dollars (USD) 5000 outside Kenya. He intends to buy the car through an agent who deals in Japanese Yen (JPY). The agent charges a 20% commission on the price of the car and a further 80 325 JPY for shipping the car to Kenya. Find the amount in Kenya shillings that Dr. Lexine will need to send to the agent to get the car given that: (3mks)
   1USD = Ksh. 120
   1 USD = 135 JPY

3. The distance from **x** to **y** is **dkm** and that from y to z is **hkm.** If a bus maintains an average speed of **50km/h** between x and y and **60km/h** between **y** and **z**, it takes 3 hours to travel from **x** and **z**. if it maintains **60km/h** between **x** and **y** and **50km/h** between y and z the journey takes 8 minutes less. What is the distance between **x** and **z**? (4mks)

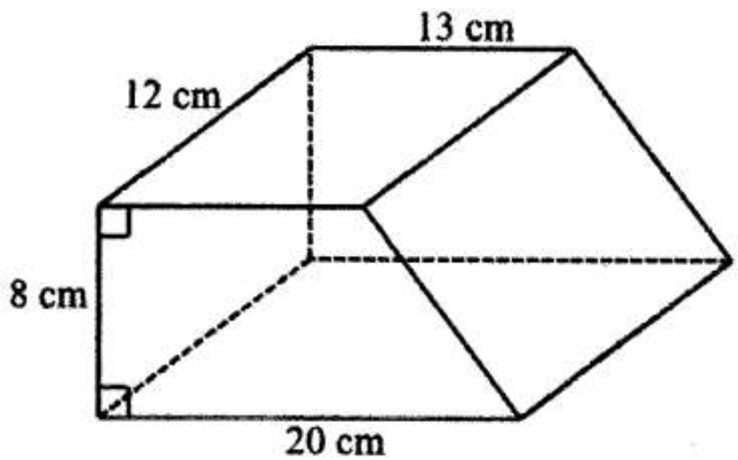
4. The graph below (not drawn to scale) is a plot for the function $y = ax^2 + bx + k$ where a, b and k are constants.



Determine the values of a, b and k. (3mks)

5. An arc of a circle centre O. subtends an angle of 72^0 at the centre. Given that the perimeter of the sector is 34.2cm. calculate the area of the sector (Take $\pi = \frac{22}{7}$) (3mks)

6. The figure below shows a solid with the given dimensions.



Calculate the mass of the solid in kilograms given that its density is $3.5 g/cm^3$. (3mks)

7. Solve for **k** in the equation $\frac{\sin(2k+10)}{\cos(3k-20)^0} = 1$ (3mks)

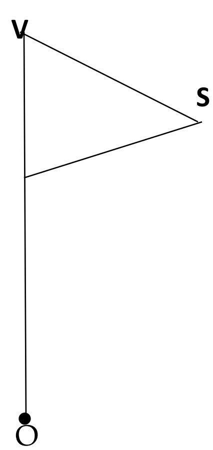
8. PQ and QR are two sides of a regular polygon. The size of angle RPQ is 15^0. Calculate the sum of the interior angles of the polygon. (3mks)

9. Use logarithms only to 4 decimal places to evaluate (4mks)

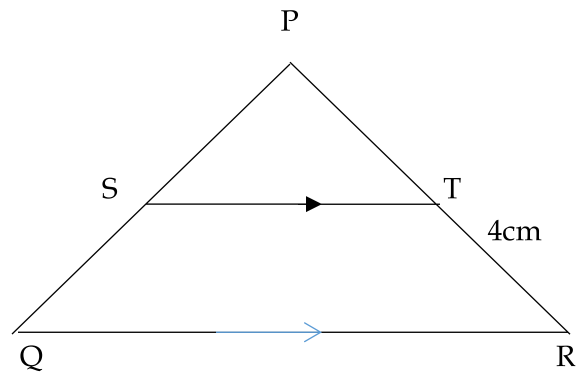
$$\sqrt{\frac{0.8008 \sin 67.12}{34.59^2}}$$

10. From the top of a watch tower 10m high an observer see the foot of a post with an angle of depression of 30^0 and the top of the post with an angle of elevation of 20^0. Calculate to 1 decimal place the height of the post. (3mks)

11. Complete the figure below to show a rotational symmetry of order 3 about O. (3 mks)



12. The figure below shows a triangle PQR. PR = 12CM, TR = 4cm and ST is parallel to QR.



If the area of triangle PQR is 336cm². Find the area of the quadrilateral QRTS. (3mks)

13. The position vectors of M and N are $\overrightarrow{OM} = \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $\overrightarrow{ON} = \begin{pmatrix} x \\ -6 \end{pmatrix}$ respectively. Given that the magnitude of $\overrightarrow{MN}$ is 26. Find the possible values of x. (3mks)

14. The base ABCDEF of a right pyramid is a regular hexagon of sides 2.5cm. Point V is the vertex of the pyramid and the length of the slanting edges is 4cm. draw a labelled net of the pyramid. (3mks)

15. A sales lady earns a basic salary of ksh 12 400 per month. In addition, she gets a 2% commission on the first ksh 80 000 worth of good and an additional 13% commission on goods worth over ksh 80 000. In a certain month, she earned a total of ksh 20 000. Calculate the value of the goods sold that month. (3mks)

16. Using mid-ordinate rule with six strips estimate the area between the curve $y = 3x^2 + 4x + 10$, x-axis and the line $x = -2$ and $x = 4$. (3mks)

SECTION II (50MARKS)

Answer any five questions in this section

17. Three villages P, Q and R are such that P is 2.1 km from Q on a bearing of $S30^0W$, R is on a bearing of 120^0 from Q and 080^0 from P.
    (a) Using a scale of 1cm to represent 300m, draw a diagram to show the relative positions of the villages. (4 mks)

    (b) Find the distance of between

    i. R and P (1mk)
    ii. R and Q (1mk)

    (c) Locate a point S on the diagram which is equidistant from P, Q and R. Find the distance between S and R. (4mks)

18. A swimming pool is 20m by 12m and it slopes gently from a depth of 1m at the shallow end to a depth of 3m at the deep end.
    (a) Calculate the volume of water in the swimming pool (in m^3) when it is full. (3 mks)

    (b) If the swimming pool is to be drained by a pump which removes water at the rate of $2.5m^3$ per minute, how long will it take this pump to drain the swimming pool if it was full. Give your answer in hours. (3 mks)

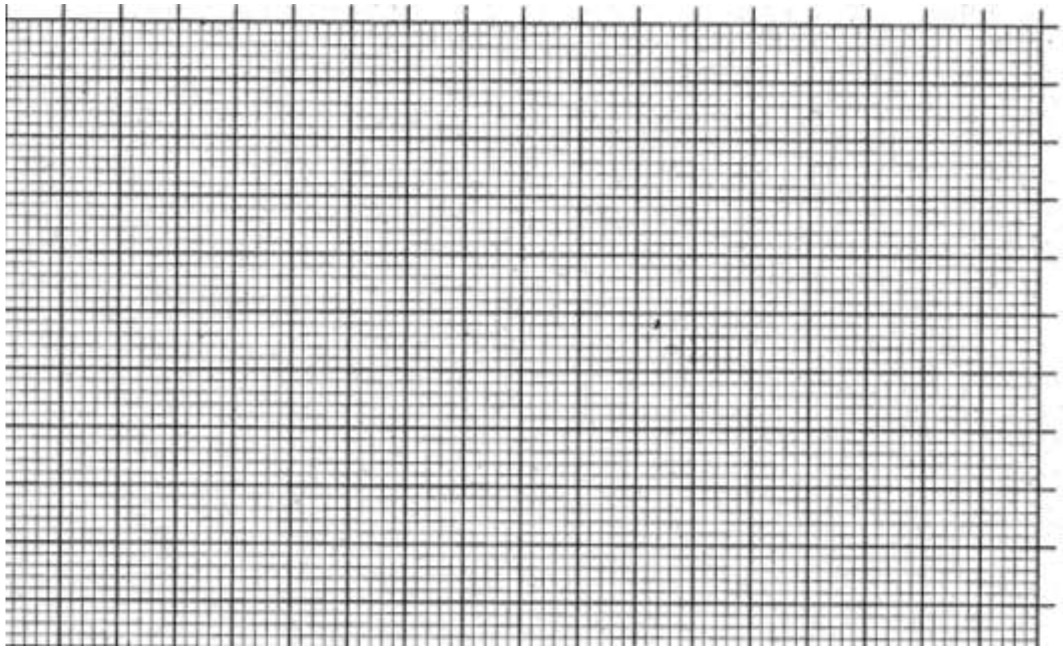
    (c) If the sides of the swimming pool and the floor are to be covered with square tiles of side 20cm, find to the nearest 100, the number of tiles required. (4 mks)

19. The heights of tree seedlings in a nursery were measured and recorded as below;

Height xcm	$0 \leq x < 5$	$5 \leq x < 15$	$15 \leq x < 25$	$25 \leq x < 45$	$45 \leq x < 75$
No. of seedlings	7	46	71	64	11

(a) Estimate the mean height (5mks)

(b) Using a scale of 1cm represent 5 units along the horizontal axis and 2 cm to represent 5 units along the vertical axis draw a histogram to represent the distribution. (3mks)



(c) Draw a vertical line on the histogram above to estimate the median value. (2mks)

20. The vertices of a triangle ABC are A (-5, -3), B(-3,-3) and C(-5,-1)

(a) On the grid provided below, draw the triangles ABC and $A^1B^1C^1$ under reflection in the line x = -1. (2mks)

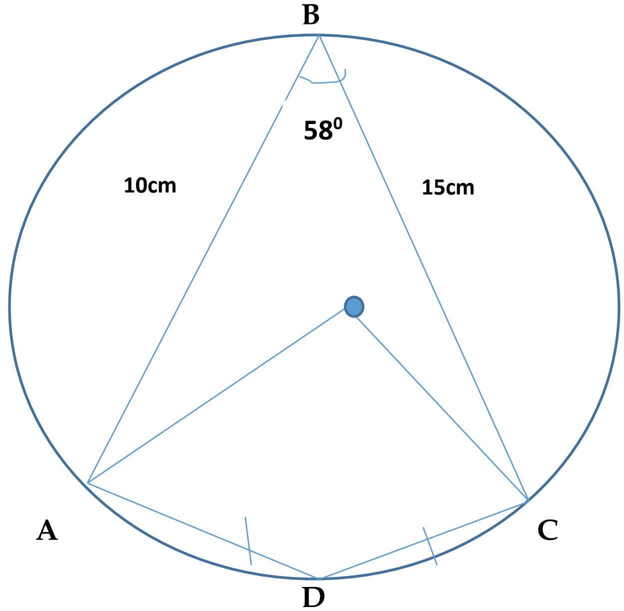
(b) Triangle $A^{11}B^{11}C^{11}$ is the image of triangle $A^1B^1C^1$ under a rotation. The coordinates A′′ and B′′ are (1,1) and (1,3) respectively.

i. Determine the angle and the centre of rotation. (3mks)

ii. Find the co-ordinates of C′′ hence draw triangle A′′B′′C′′. (2mks)

(c) Triangle A′′′B′′′C′′′ is the image of A′′B′′C′′ under an enlargement centre (0,2) and scale factor -2. On the same axes, draw the triangle A′′′B′′′C′′′. (3mmks)

21. In the figure below, **ABC** and **D** are points on a circumference of a circle centre **O**. **AB = 10cm**, **BC = 15cm** angle **ABC = 58⁰** and **AD = DC**.



(a) Calculate the size of angle;

(i) Angle **AOC** reflex. (1 mk)

(ii) Angle **ADC** (1mk)

(b) Calculate correct to 1 decimal place;

(i) The length **AC** (2mks)

(ii) The radius of the circle. (2mks)

(c) Calculate the area in the circle not covered by quadrilateral **ABCD.**

(Take π = 3.142) (4 mks)

22. Masomo bookshop sells 120 calculator at ksh x. Talanta at ksh. 120 less. With a sum of ksh. 54,000, a school is able to buy 5 more calculators from Talanta bookshop than from Masomo bookshop.
    a. Write down an expression in terms of x to represent the number of calculators that a school can buy from
        i. Masomo bookshop (1mk)
        ii. Talanta bookshop. (1mk)
    b. Form an equation in x and hence determine the number of calculators a school can buy from Talanta Bookshop. (4mks)
    c. The two bookshops offer special discounts to students. On a certain day a student purchased 2 calculators from Masomo Bookshop and 3 calculators from Talanta bookshop for ksh 5 118. Another student purchased 3 calculators from Masomo Bookshop and 2 calculators from Talanta Bookshop for ksh 5 112. Calculate the percentage discount offered to the student by each bookshop. (4mks)

23. A salesman dealing in mattresses earns a basic salary and commission as follows:

	Commission
For sales up to ksh 150 000	0%
For sales above ksh 150 000	
First ksh. 85 000	3%
Next ksh. 85 000	4%
Any amount above ksh 320 000	5%

(a) In the month of November 2020, the salesman earned a basic salary of ksh. 35,000 and he sold 90 mattresses at ksh. 4000 each. Calculate

i. His total sales in the month of November 2020. (1mk)

ii. His total earnings that month. (3mks)

(b) In the month that followed his basic salary was decreased by 10%. If he recovered a total earning of ksh 36 970 in that month, calculate

i. Total salts that month (4mks)

ii. The number of mattresses sold in that month. (2mks)

24. Distance moved by a particle is given as $S = t^3 - 3t^2 - 9t + 100$ metres. Determine;

(a) The displacement of the particle after 3 seconds. (1mks)

(b) The velocity of the particle after 3 seconds. (2mks)

(c) The acceleration when $t = 5$ seconds (1mk)

(d) Distance moved during the 3rd second. (2mks)

(e) The time when the particle is momentarily at rest. (2mks)

(f) The minimum velocity of the particle. (1mk)

(g) Initial velocity (1mk)

KCSE TOP PREDICTION TRIAL 10 - 2026

Kenya Certificate of Secondary Education

121/2 - MATHEMATICS - Paper 2

(Alt A)

Time 2 ½ hours

Name …………………………………………………………………… Adm Number…………………………..

Candidate's Signature ………………………...……………………….Date ………………………………………

Instructions to candidates

(a) Write your name and admission number in the spaces provided above.
(b) Sign and write the date of examination in the spaces provided.
(c) This paper consists of two sections: **Section I** and **Section II**.
(d) Answer all questions in **section I** and **only five** questions from section **II**.
(e) Show all the steps in your calculations, giving the answers at each stage in the spaces provided below each question.
(f) Marks may be given for correct working even if the answer is wrong.
(g) **Non-programmable** silent electronic calculators and KNEC mathematical tables may be used, except where stated otherwise.

For Examiner's Use Only

Section I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

Section II

17	18	19	20	21	22	23	24	Total

Grand Total

SECTION I (50 marks)

Answer all questions in this section

1. The dimensions of a rectangle are 40 cm and 45 cm. If there is an error of 5% in the measurement of it's dimensions, find the percentage error in calculation of it's the area of the rectangle. (3 Marks)

2. Solve the following pairs of simultaneous equations. (4 marks)

$$x + y = 1$$

$$x^2 - 2xy + y^2 = 4$$

3. Without using a calculator or mathematical tables, evaluate leaving your answer in the form $a\sqrt{b}+c$ where a, b and c are integers. (3 Marks)

$$\frac{tan\ 60^0}{1-\cos 30^0}$$

4. A quantity y varies partly as x and partly as the square of x. When $x = 20$, $y = 45$ and when $x = 24$, $y = 60$. Find y when $x = 4$. (3 Marks)

5. Solve for X in $\log_2 X + 3 = \log_2 X^4$. (3 Marks)

6. Make x the subject of the formula in: (3 Marks)

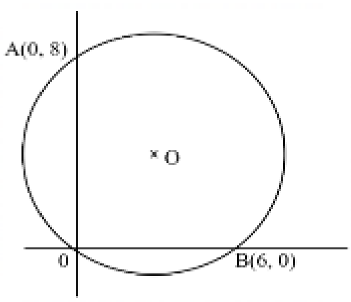
$$y + x^2 = (x + t)(x + y)$$

7. The position vectors of points P and Q are $5\underset{\sim}{i} + 4\underset{\sim}{j} - 6\underset{\sim}{k}$ and $2\underset{\sim}{i} - 2\underset{\sim}{j}$ respectively. A point X divides PQ in the ratio 5:-3, find the coordinates of X, hence magnitude of OX. (3 Marks)

8. A jewelry room is guarded by three policemen X, Y and Z. The probability of a thief being caught by X is $\frac{1}{3}$, Y is $\frac{1}{5}$ and Z is $\frac{1}{4}$. Find the probability that the thief is caught by policeman Z. (3 Marks)

9. Solve the equation $\cos^2 x - 2\sin x = 2$ for the range $0^0 \leq x \leq 360^0$. (3 Marks)

10. In the figure below, O is the centre of the circle and A if joined to B, passes through the centre of the circle O.



(a) Determine the centre and radius of the circle. (2 marks)

(b) Express the equation of the circle in the form $x^2 + y^2 + ax + by = 0$ where a *and* b are constants. (2 marks)

11. The table below shows the monthly income tax rates for the year 2024

Monthly taxable income	Tax rates (percentages)
$1 - 9{,}880$	10%
$9{,}881 - 19{,}480$	15%
$19{,}481 - 29{,}080$	20%
$29{,}081 - 38{,}680$	25%

Mr. Michael's monthly earning comprises of a basic salary of ksh. X and allowances amounting to sh.5,200. When Mr. Michael claims a monthly tax relief of ksh. 1, 056, his employer deducts from his earnings a tax of ksh. 1,282. Calculate the value of X. (3 marks)

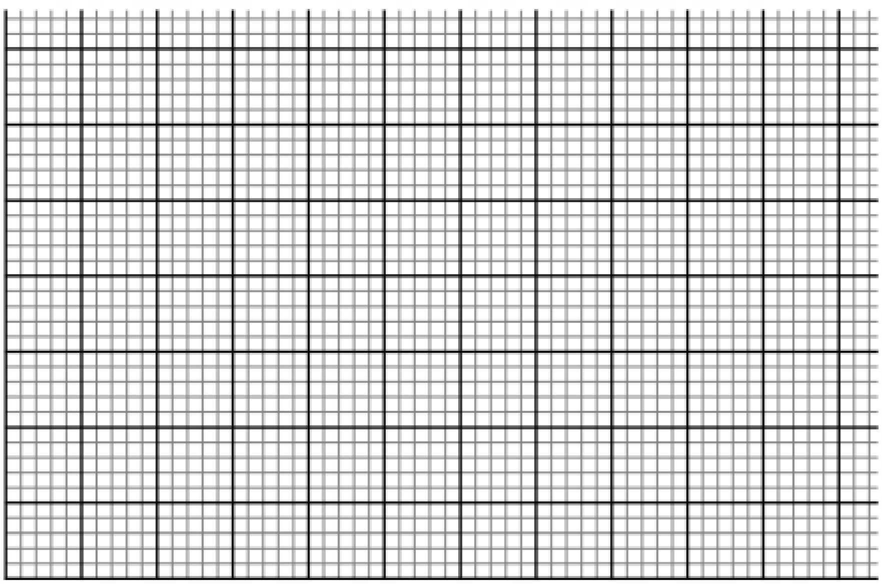
12. A matrix M$\begin{pmatrix} 2 & 1 \\ 2 & 2 \end{pmatrix}$ maps an object point A(x,y) to A'(x_1,y_1). A matrix T $\begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix}$ maps A' to A''(X_2,Y_2). Find the matrix which maps A'' to A. (3 Marks)

13. Three grades of coffee A, B and C were mixed in the ratio 1:2:3. The cost per kg of each of the grades A, B and C were sh 700, sh 490 and sh 630 respectively. Calculate the selling price per kg if a profit of 20% is to be realized. (3 Marks)

14. The gradient function of a curve is given by $4x^3 + 2x - 1$. If the curve passes through a point (-2,1), find the equation of the curve. (3 Marks)

15. Complete the table below for curve $y = x^2 – 2x$, draw the graph. Using the graph, find the gradient when x=2. (3 Marks)

X	1	2	3	4
Y				



16. The difference between the fourth and the seventh terms of an increasing arithmetic progression is 12. Calculate the sum of the first fifteen terms of the progression if the first term is 9. (3 Marks)

SECTION II (50 MARKS)

Answer any five questions from this section

17. The table below shows the heights (in metres) of seedlings in Mr. Wanjala's seedbed

(a) Use the information above to complete the table below. (3 Marks)

Height (m)	F	$mid-point(x)$	$t=\left(\frac{1000x-105}{40}\right)$	ft	t^2	ft^2
$0.01-0.04$	12			-24		48
$0.05-0.08$	17		-1		1	
$0.09-0.12$	31		0	0	0	0
$0.13-0.16$	28		1			
$0.17-0.20$	23		2	46	4	
$0.21-0.24$	9					

(b) Using the completed table above, determine;

(i) The mean height of the seedlings. (2 Marks)

(ii) The standard deviation of height of the seedlings. (3 Marks)

(c) Determine the probability of the number of seedlings whose height exceeds 0.142 m. (2 Marks)

18. Taking the radius of the earth, $R = 6370$km and $\pi = {}^{22}/_{7}$.
    (a) Calculate the shortest distance in km between the two cities, P(60ºN, 80ºW) and Q (60ºN, 10ºE). (2 Marks)

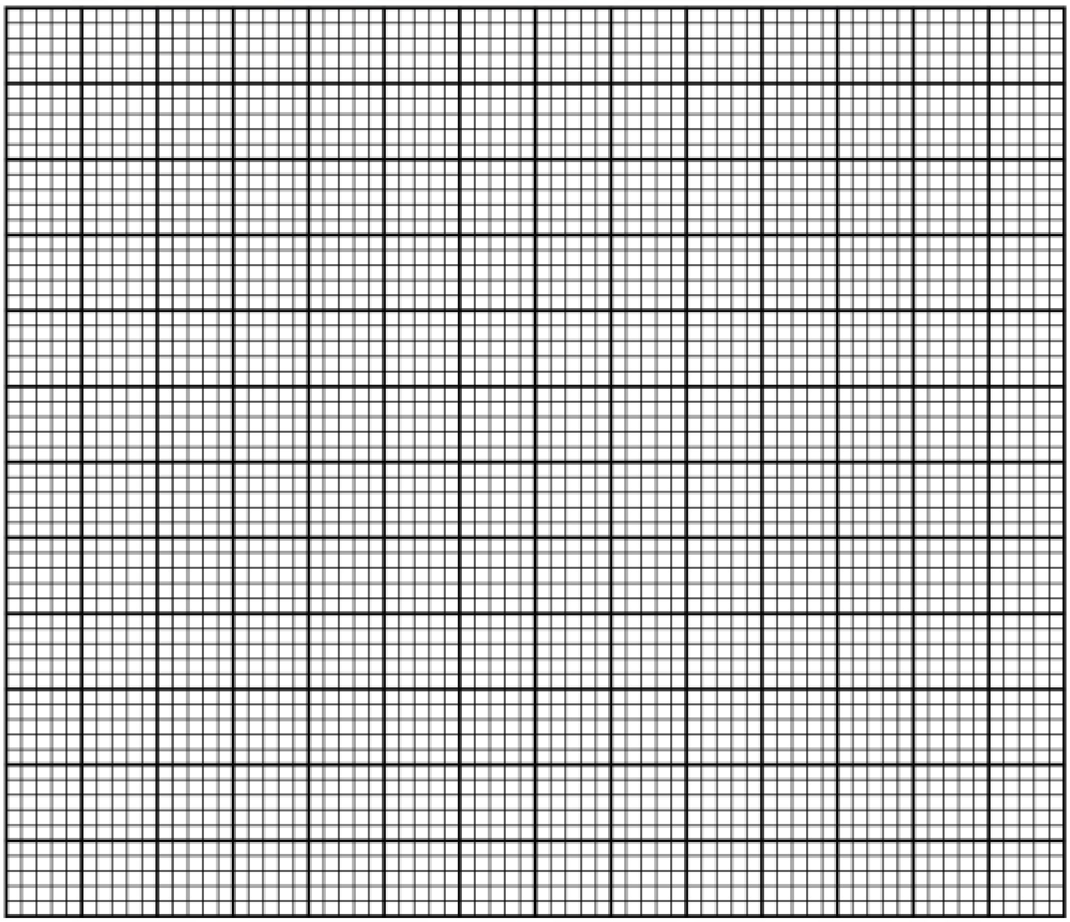
    (b) An airplane flew from P at 1200hrs local time to Q along the parallel of latitude at a speed of 1001km/h. What was the local time at Q when it arrived? (4 Marks)

(c) The airplane then flew due south from point Q(60ºN, 10ºE) to point B at speed of 194.4 knots. The distance covered by the airplane was 3888nm. Find the position of B and the local time it arrived at B if it rested for 30mins at Q in twelve hour system. (4 Marks)

19. (a) Complete the table below correct to 2 decimal places. (2 Marks)

$x°$	0	15	30	45	60	75	90	105	120	135	150	165	180
$-2\cos 2x$		-1.7	-1.00				2.0		1.0		-1.0	-1.7	-2.0
$3\sin(2x+30)$	1.5					0.0				-2.6	-1.5		

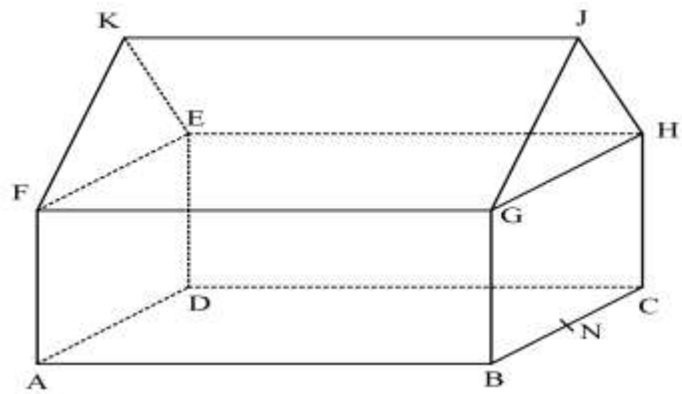
(b) Using a scale of 1 cm for 15° on the x-axis and 2 cm for 1 unit on the y-axis, and on the same Axes draw the graphs of $y = -2\cos 2x$ and $y = 3\sin(2x+30)$ for $0° \leq x \leq 180°$. (5 Marks)



c) Using the graph in (b) above, solve the equation $3\sin(2x+30) + 2\cos 2x = 0$ (2 Marks)

d) Find the range of values of x for which $3\sin(2x+30) \leq -2\cos 2x$ (1 Mark)

20. In the figure below a triangular prism is placed on top of a cuboid. Point K and J lie vertically above line AD and BC respectively. AB = KJ = 18 m, BC = 12 m and CH = 10 m. FK=KE=JH=JG=8m . N is the mid-point of BC.



Calculate,

(a) The height of ridge KJ above the base ABCD. (2 Marks)

(b) The angle between line FJ and plane EFGH. (2 Marks)

(c) The angle between plane FEJ and plane ABCD. (2 Marks)

(d) The angle between plane FEN and plane ABCD. (2 Marks)

(e) The angle between GC and AD. (2 Marks)

21. A piece of land can be bought on cash at ksh.1,000,000 or on hire purchase terms, by making a down payment of ksh. 200,000 and a 24 monthly installment of sh.100,000 each.

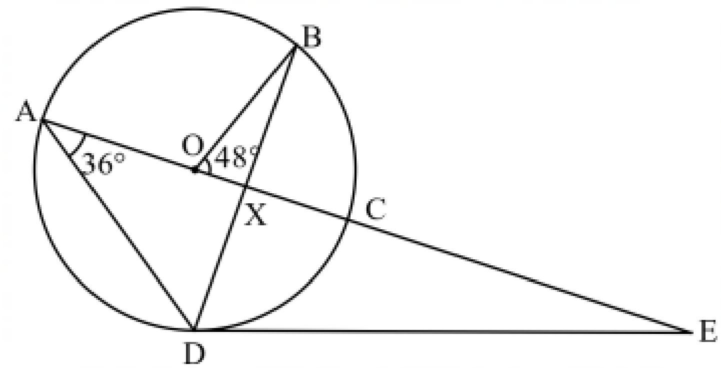
(i) Calculate the carrying charge (2 Marks)

(ii) Calculate the rate of interest charged per month on hire purchase terms. (2 Marks)

(iii) If the piece of land appreciated at a rate of 5% p.a for the first two years compounded annually, in the next 3 years it appreciated at a rate of 12% p.a compounded quarterly. Calculate the value of the land after 5 years. (4 Marks)

(iv) After five years, due to the environmental degradation the value of the land depreciated at a rate of 6% p.a. How long will it take for the value to depreciate to 1,153,625. (2 Marks)

22. In the figure below O is the centre of the circle. A, B, C and D are points on the circumference of the circle. A, O, X, C and E are points on a straight line. DE is a tangent to the circle at D. Angle BOC= 48° and angle CAD= 36°.



(a) Giving reasons in each case, find the value of each of the following angles:

(i) DBC (2 Marks)

(ii) DEC (2 Marks)

(iii) BCD (2 Marks)

(b) Given that AD = 15 cm, CE = 12cm. Calculate correct to 2 decimal places the:

(i) radius of the circle. (2 Marks)

(ii) length DE (2 Marks)

23. Using a ruler and a pair of compasses only for all constructions in this question;

(a) Construct triangle ABC in which $AB = 6cm$, $BC = 7cm$ and $\angle ABC = 75°$. (3 Marks)

(b) Find locus x such that $AX = 3$ cm. (1 Mark)

(c) On the same side of BC as A, construct the locus of P such that $\angle BPC = 120°$. (3 Marks)

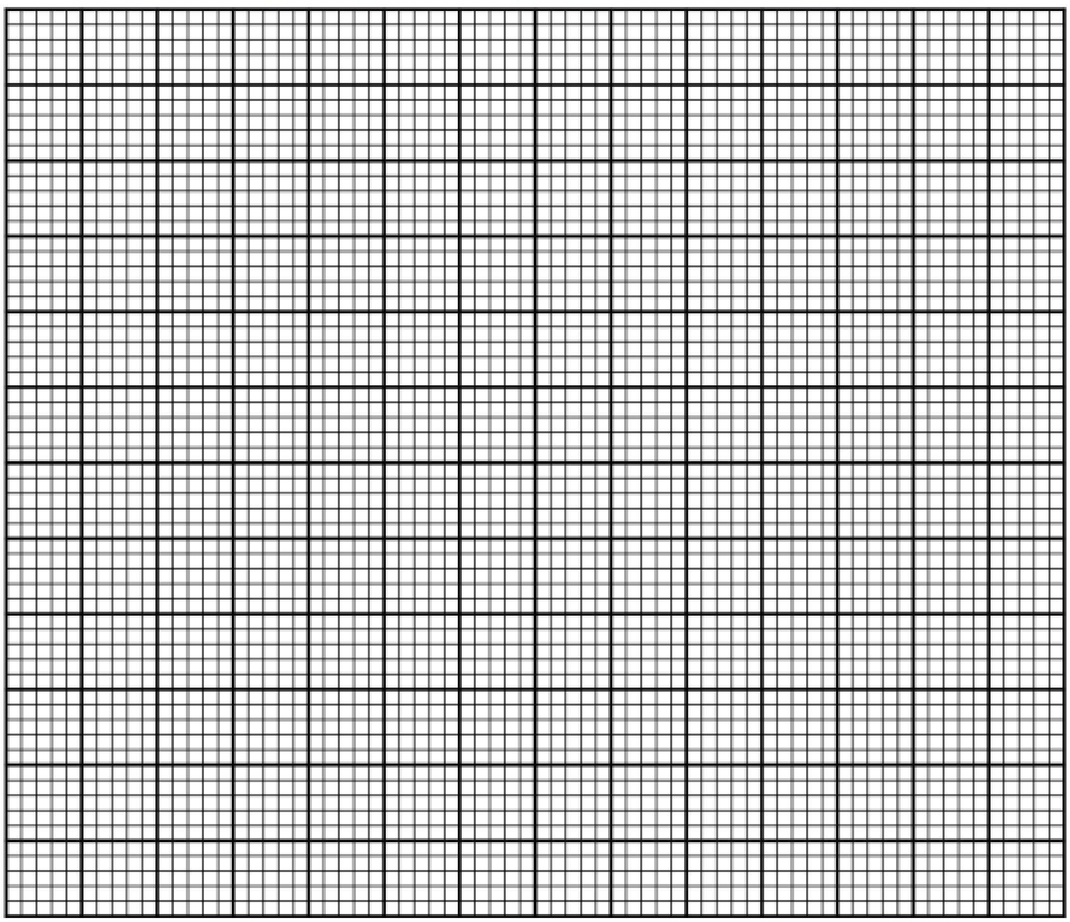
(d) Show by the shading, the locus of Q inside triangle ABC such that BPC $\leq 120^0$ and AQ $\geq$ 3 cm. (1 Mark)

(e) On the opposite of C, construct the locus of T such that the area of triangle ATB=6 cm^2 (2 Marks)

24. A farmer has 50 hectares of land in which he plants potatoes and carrots. Each hectare of potatoes require 6 men while carrots require 2 men. The farmer has 240 men available. He must plant at least 10 hectares of potatoes. The profit on potatoes is sh.1,000 per hectare and carrots is sh.1,200 per hectare. If he plants x hectares of potatoes and y hectares of carrots.

(a) Write down four inequalities to represent the given information. (4 Marks)

(b) On the grid provided draw the inequalities. (3 Marks)



(c) Using your graph determine the number of hectares for each vegetable which will give maximum profit. (3 Marks)